

Department of Computer Science

COMP4300 - Graduation Project

Fall 2025/2026

Crime Scene Virtual Reality Simulation for Forensic Science Education.

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June 25, 2026

Section – B

Title of Project: Crime Scene Virtual Reality Simulation for Forensic Science Education.
Project No: 26
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Key Areas: VR, Education, Unity

Section – C

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Abstract

Virtual Reality (VR) is increasingly recognized as a powerful tool in forensic science education, offering immersive and interactive environments that enhance learning and practical skill development. This project leverages VR technology to create a realistic and controlled virtual crime scene simulation designed for undergraduate forensic science students. The system allows learners to navigate crime scenes, examine evidence, and interact with three-dimensional objects using VR tracking and intuitive controls. Key elements such as scene layout and evidence items are modeled with high fidelity to support authentic forensic investigation practices.

The system records user actions, provides immediate procedural feedback, and generates performance reports to support both training and assessment. By offering a safe, repeatable, and resource-efficient environment, the VR simulation enables students to practice evidence collection, scene analysis, and decision-making without the limitations of physical mock crime scenes. Upon evaluation, the project aims to demonstrate that VR-based training improves engagement, comprehension, and procedural accuracy compared to traditional instructional methods. Ultimately, this VR system is intended to serve as an effective, scalable, and innovative platform for enhancing forensic science education.

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Chapter 1

Introduction

1.1 General overview

Forensic science plays a fundamental role in the criminal justice system by analyzing physical evidence to reconstruct events and identify those involved in criminal activities. Traditional forensic training, however, faces challenges such as limited access to real crime scenes, safety concerns, and high resource requirements. These limitations often prevent students from gaining hands-on experience in realistic investigative environments. Virtual Reality (VR) has emerged as a powerful educational and investigative tool that enables the creation of immersive, interactive, and risk-free simulations. By integrating VR technology with forensic science education, learners can explore complex crime scenes, practice evidence collection, and enhance their analytical and observational skills in a controlled virtual setting. This project focuses on the integration of Virtual Reality (VR) and forensic science to create an immersive crime scene simulation for educational use. The aim is to enhance students' learning experiences by providing a realistic environment where they can practice and experiment with crime scene investigation equipment and techniques safely and interactively. Such a simulation can improve observational skills and evidence handling in forensic science education. This report aims to present the scope, and main features of the VR and Forensic Science project.

The **scope** of this project covers the design, development, and evaluation of a Virtual Reality-based crime scene investigation training tool intended for undergraduate forensic science students. The system is designed to support multiple crime scene environments,featuring distinct layouts, realistic evidence items, and interactive elements that support essential investigative tasks such as observing, identifying, collecting, tagging, and recording evidence. To accommodate different learning stages and skill levels, the system implements multiple interaction modes. The base mode provides an observational learning experience where students can explore crime scenes through movement and observation only, without interactive evidence collection. This mode supports the integration of 360-degree recorded crime scenes, allowing students to familiarize themselves with scene assessment and spatial awareness in a passive learning environment. The advanced interactive mode introduces full forensic interaction, including evidence identification, collection, tagging, photography, fingerprint recovery, and all investigative procedures, providing comprehensive hands-on training as described throughout this report. To enhance learning, the project incorporates immediate procedural feedback, a gamified scoring system, and a report-generation feature. Additionally, it implements an optional NPC-based hint system where students can interact with a virtual forensic

advisor to ask forensics-related questions or request procedural hints, assisting them during challenging investigative steps. The project also introduces limited randomization of evidence items and their placements to create varied investigative scenarios and increase replay value for each scene without extending into automatically generated scenes or multi-path investigative narratives. While the simulation focuses on core crime scene investigation processes, it does not address advanced forensic laboratory analysis or large-scale multi-case narratives.

1.2 Motivation

Training undergraduate forensic science students currently relies heavily on lectures, photographs and other visual materials, which limits the students' ability to learn and experiment with forensic equipment and techniques. Setting up physical mock crime scenes can be very costly, time-consuming and requires extensive supervision. Furthermore, the inability to make mistakes in a safe environment means students enter the field without experiencing the consequences of procedural errors, which can compromise real investigations.

VR provides an opportunity to bridge this gap, by providing an environment that allows students to explore, experiment and learn in a controlled, safe, maintanable space. By simulating realistic crime scenes, VR can significantly improve engagement, observation, procedural understanding, and hands-on experience without the constraints of physical labs.

Additionally, the Science Faculty at Birzeit University was formally consulted and has identified a significant gap in available VR tools. Current systems fail to deliver realistic crime-scene simulations, intuitive interaction, and the procedural accuracy required for effective forensic training. They also lack alignment with the local forensic context, where regional practices, environmental settings, and resource constraints differ from the assumptions built into most international VR solutions.

This project aims to address these challenges for undergraduate forensic science programs, by providing students with a realistic, safe, scalable and immersive hands-on learning experience through Virtual Reality Technology.

1.3 Method

This project follows the Agile iterative software development methodology. The work begins with a detailed review of existing literature and commercial solutions related to Virtual Reality, forensic science education, and immersive training systems. This review helps identify the teaching limitations in current approaches. Based on this analysis, system requirements are derived and translated into a structured system design, including architectural models, interaction flows, and UML diagrams. The project is then implemented incrementally, starting with core navigation and scene visualization features, followed by evidence interaction, tagging, scoring, and feedback mechanisms.

1.4 Aim and Objectives

The aim of this project is to utilize VR technology to design and develop a training system that addresses the limitations of traditional crime scene investigation teaching methods. The system provides an immersive and interactive environment in which undergraduate forensic science students can practice essential investigative procedures, supporting both hands-on training and performance evaluation.

To achieve that overall aim the project pursues the following objectives:

- Design multiple immersive virtual crime scene environments featuring distinct layouts and realistic evidence items.
- Identify and define the system requirements, and design a structured system model that includes the system architecture, interaction flows, and UML diagrams.
- Implement multiple interaction modes to support progressive learning, including a base observational mode for scene familiarization and an advanced interactive mode for comprehensive forensic training.
- Develop interactive VR mechanics that allow students to identify, collect, tag and record virtual evidence naturally using VR controllers.
- Implement comprehensive evidence tagging and documentation procedures following professional forensic standards, including evidence marker placement, labeling, and chain of custody tracking.
- Implement limited randomization of evidence items and their placements to create diverse investigative scenarios within each crime scene.
- Incorporate a scoring system to introduce gamification and motivate students through performance-based feedback.
- Construct a report generation tool that summarizes collected evidence for instructor evaluation and student review.
- Provide immediate feedback on investigative decisions and actions to guide learning following standard forensic protocols.
- Implement an NPC-based hint system to guide students during investigations.
- Evaluate the VR system's impact on student engagement, procedural understanding, and evidence-handling accuracy using structured user testing.

1.5 Overview of the technical area

This project is a Virtual Reality system, an interactive simulation, and an educational technology. Achieving these project objectives involves the development of an immersive VR application that utilizes motion tracking, controller-based interaction, and realistic 3D rendering.

The system integrates VR interaction techniques such as scene navigation, object manipulation, and spatial evidence placement, alongside gamification mechanisms including

scoring and feedback. These mechanisms are used to guide the user through the investigative process and to provide structured evaluation of procedural correctness.

From a technical perspective, the system is developed using the Unity 3D game engine and targets the OpenXR framework to ensure compatibility across different VR devices. VR-specific interactions are implemented using standard interaction toolkits, allowing users to naturally explore crime scenes and interact with forensic evidence.

The system also incorporates data management capabilities to store scene configurations, investigation data, and generated reports. This enables persistence of user progress and supports performance evaluation while keeping the system modular and extensible.

Overall, the technical area of this project combines real-time VR rendering, interactive simulation design, and structured data handling to support immersive and repeatable forensic training scenarios.

1.6 Overview of the report

The remainder of this report is organized as follows. Chapter 2 presents the theoretical background, covering Virtual Reality technologies, gamification in education, forensic science fundamentals, and a critical analysis of existing forensic VR applications. Chapter 3 details the system analysis and design, including the overall architecture, functional decomposition, use case modeling, and system diagrams. Chapter 4 describes the system implementation, detailing how the design was turned into a functional application. Chapter 5 concludes the report by reviewing the project outcomes and outlining potential future work and extensions. The appendices provide supplementary materials such as detailed use case specifications.

Chapter 2

Background

2.1 Details of Relevant Theory

2.1.1 Background Concepts

Extended Reality (XR)

Extended Reality (XR) is an umbrella term encompassing all forms of immersive technology that combine or mirror the physical world with a twin digital world. This domain includes Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR)[1], [2].

The conceptual relationships between these technologies are best understood through the reality-virtuality continuum, introduced by Milgram and Kishino in 1994. This spectrum defines the extent to which the physical and digital worlds are blended[3]:

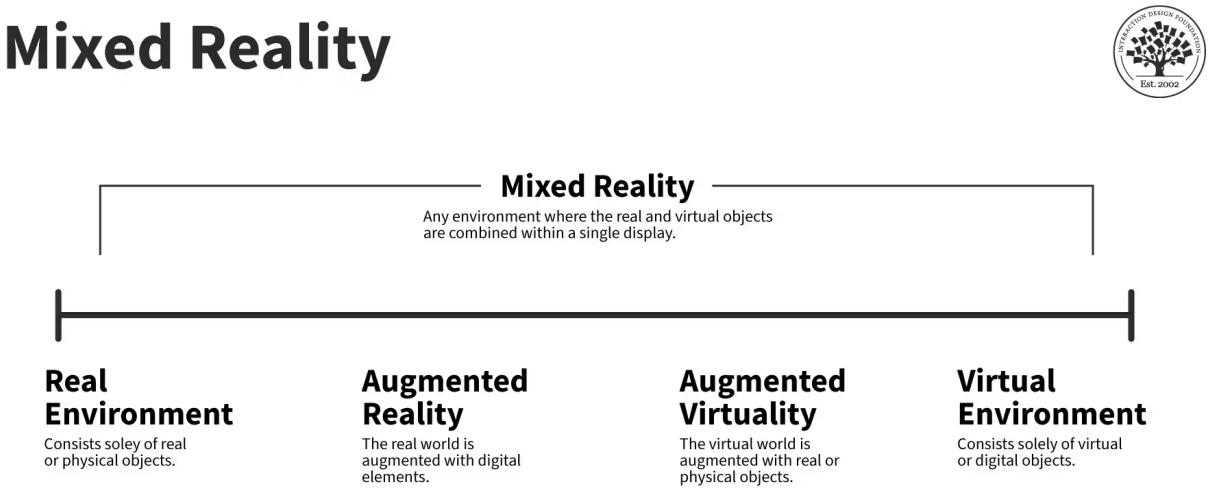


Figure 2.1: Reality-Virtuality Continuum showing the spectrum from real environment through AR and MR to virtual environment [3].

1. **Virtual Reality (VR):** At the virtual end of the continuum, VR is characterized by complete immersion. The technology utilizes computer images to create simulated three-dimensional environments that entirely block out the physical world, providing users with an interactive, synthetic experience via display devices such as headsets or visors.

2. **Augmented Reality (AR):** Located closer to the real environment, AR superimposes digital elements onto the physical world, typically through handheld devices or headsets. However, in AR, the digital and physical elements generally do not interact.

3. **Mixed Reality (MR):** MR occupies the crucial space between AR and VR, facilitating an environment where the user can interact meaningfully with both digital and physical elements simultaneously.

XR technologies are utilized across diverse fields such as medicine, architecture, and most importantly, education. [2]

2.1.2 Virtual Reality

Overview

Virtual Reality (VR) is a computer-generated environment that users can interact with in real time, simulating physical presence through spatial audio, haptic feedback, and precise motion tracking. By immersing the user's senses, VR offers unlimited, customizable environments that can be visualized, explored, and manipulated safely and repeatedly. This makes it a highly maintainable and scalable alternative to physical setups. Modern VR headsets balance visual fidelity with accessibility, enabling immersive learning experiences in both classroom and professional contexts. Overall, VR has progressed from a conceptual technology into a practical, effective tool for delivering engaging and interactive simulations across numerous fields including forensic science education.[4]

Components

VR systems are composed of hardware and software elements that work together to generate an interactive virtual environment. The head-mounted display (HMD) forms the core of the system, providing the visuals with a wide field of view, creating a sense of realism. Meanwhile input devices, including handheld controllers, gesture-tracking systems, and haptic gloves enable users to manipulate objects and interact with virtual elements.

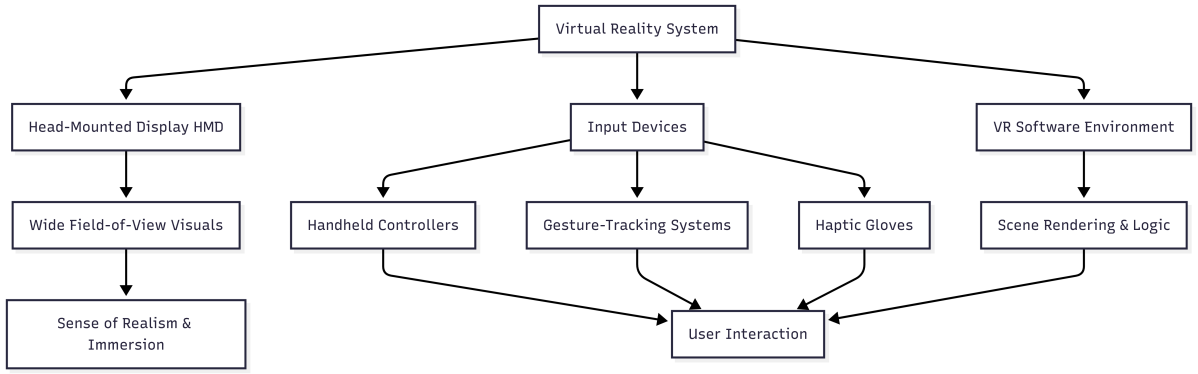


Figure 2.2: Key components of a VR system including HMD, controllers, and tracking sensors.

Types of VR systems

VR systems fall into three major categories, differentiated by the degree of immersion and hardware complexity. [5]

1. **Immersive VR:** Uses head-mounted displays (HMDs), motion tracking, and spatial audio to fully isolate the user from the physical world. High-end systems support six degrees of freedom (6DoF) motion tracking [6], enabling natural exploration and realistic interaction. Immersive VR is the dominant format in training and simulation applications because it maximizes spatial presence and situational awareness.



Figure 2.3: Head-mounted display (HMD) used in immersive VR systems [7].

2. **Semi-Immersive VR:** Often implemented through large projection screens, CAVE environments (Cave Automatic Virtual Environment), or panoramic displays. Although users are not entirely isolated from the real world, these environments support collaborative work and large-scale visualization. One common example of this is pilot training, where the user is in a cockpit-like setting, where instead of the windshield they are surrounded by large displays.

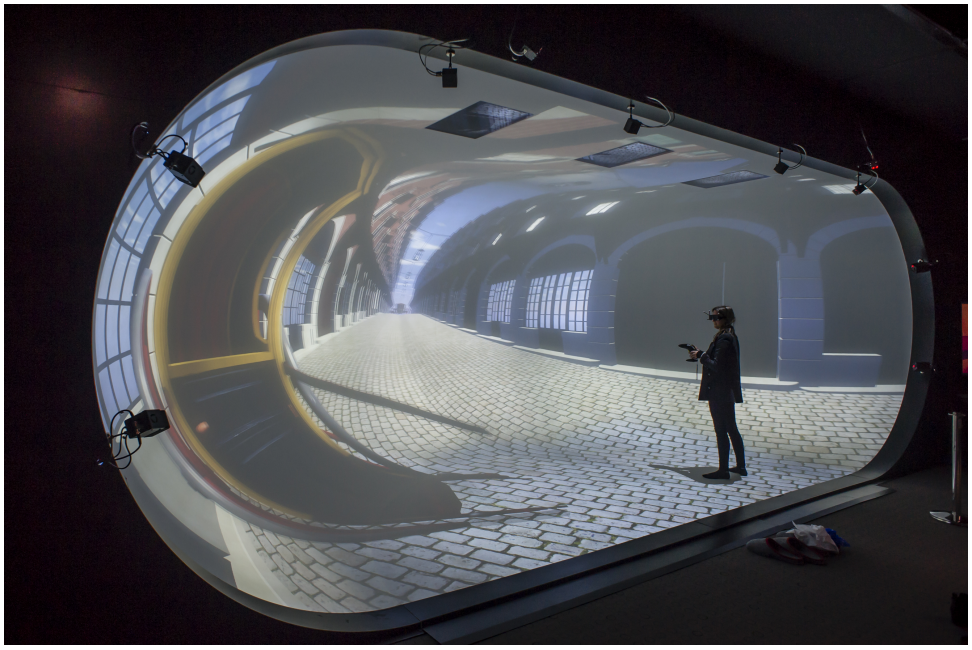


Figure 2.4: CAVE (Cave Automatic Virtual Environment) system providing semi-immersive VR experience [8].

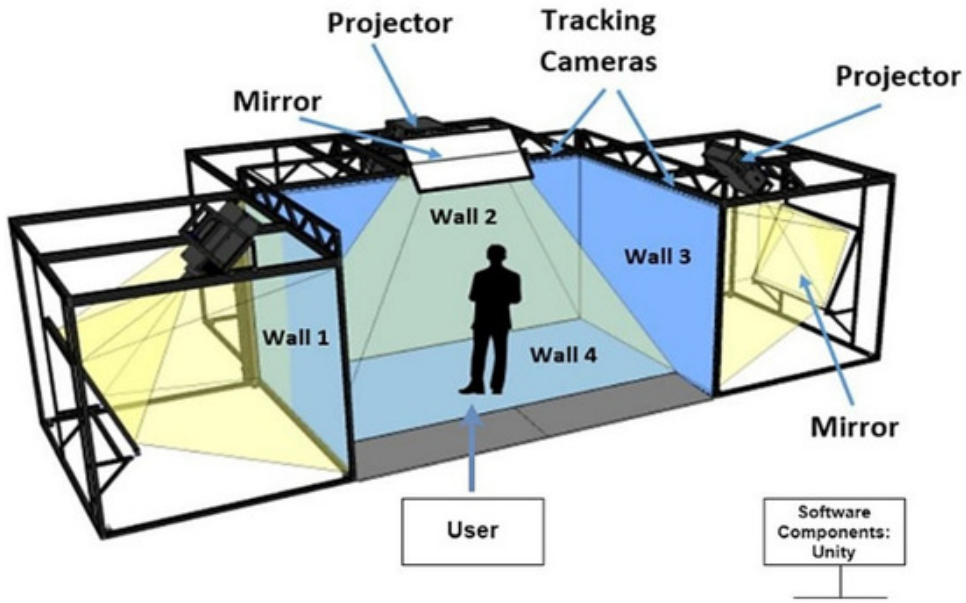


Figure 2.5: Technical diagram of CAVE system architecture showing projection setup [9].

3. **Non-Immersive VR:** Delivered through traditional displays with mouse/keyboard or gamepad input. Desktop VR, or non-immersive VR, is cheap and accessible but offers limited immersion.

Interaction & locomotion techniques

The field of Virtual Reality (VR) locomotion, which is the ability to navigate in VR environments, has garnered significant research interest. Various locomotion techniques

have been proposed to acccommodate user preferences, task demands, and system capabilities.[10]

To characterize this space, a typology of VR locomotion techniques was developed, which calssifies them based on three key interaction-related characteristics:

- **Interaction Type:** Describes how the user triggers VR navigation.
  - **Physical:** Navigation is enabled by using physical motion cues, translating natural movement to VR motion through body tracking.
  - **Artifical:** Navigation is directed by utilizing input devices (eg., controllers).
- **VR Motion Type:** Assesses the nature of the user's movement in the virtual environment.
  - **Continuous:** Supports smooth, uninterrupted movement.
  - **Non-continuous:** Provides instantaneous movement transitions, often described as visual "jumps".
- **VR Interaction Space:** Defines the limits of the virtual navigation area.
  - **Open:** Supports navigation in a virtual environment that surpasses the limits of the real physical environment.
  - **Limited:** The size of the virtual environment is restricted by the dimensions of the physical environment.

Based on these characteristics, the typology organizes VR locomotion into the following five distinct types:

1. **Motion-based:** These techniques utilize physical movement to enable interaction and support continuous motion in open VR spaces.
   - Examples: Walking-in-place, redirected walking, arm swinging, gesture-based locomotion, and reorientation.
2. **Room scale-based:** This type also utilizes physical movement for interaction and supports continuous motion, but the interaction takes place in VR environments whose size is limited by the real environment's size.
   - Example: Real-walking.
3. **Controller-based:** For this type, controllers are used to move the user artificially in the VR environment, with continuous motion in an open VR interaction space.
   - Example: Joystick-based.
4. **Teleportation-based:** These techniques utilize artificial interactions in open VR spaces with non-continuous movement, where the user is instantaneously teleported to a new position.
   - Example: Point and teleport.

5. **Motion-based Teleporting:** This is a new locomotion technique introduced recently, combining physical motion cues (like gestures) with teleportation.

Application to Our System: For the VR forensic training system, we have chosen to implement **teleportation-based locomotion**. This interaction type is most suitable for our application for several reasons. First, teleportation provides an artificial, non-continuous movement approach that significantly reduces motion sickness and simulator discomfort, which is critical for extended training sessions in educational environments. Second, it supports navigation in open VR spaces that exceed the physical boundaries of typical classroom or training laboratory settings, allowing us to create larger, more realistic crime scenes without spatial constraints. Third, teleportation enables precise positioning near evidence items, which is essential for detailed forensic examination and documentation procedures. Finally, this technique has low cognitive overhead compared to continuous locomotion methods, allowing students to focus their attention on forensic tasks rather than navigation mechanics. The point-and-teleport implementation will provide intuitive, comfortable, and task-appropriate movement throughout crime scene investigations.

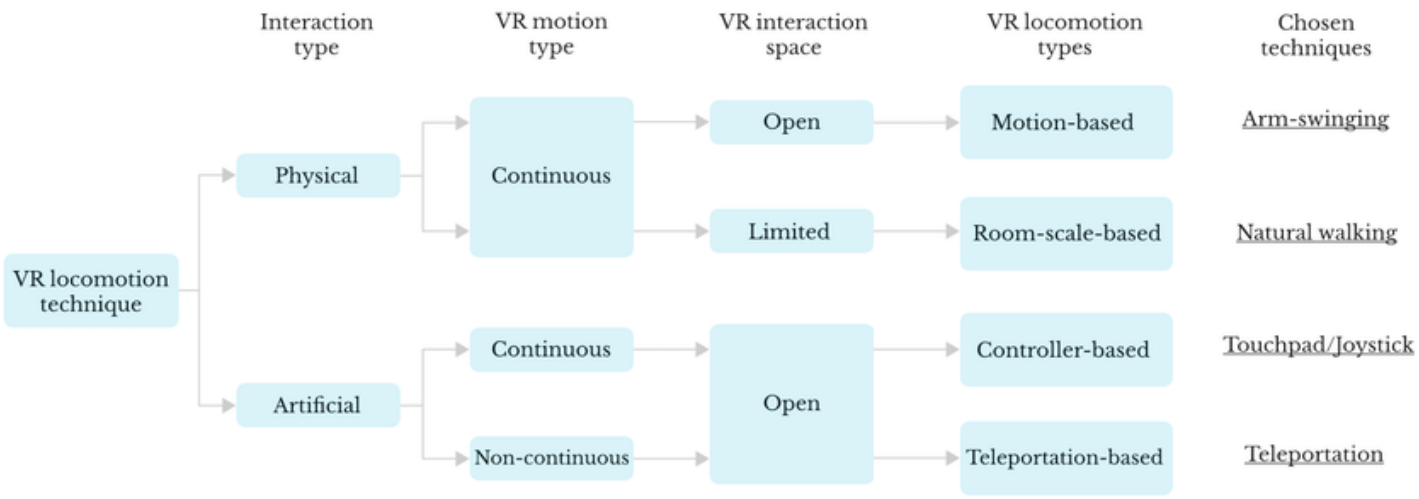


Figure 2.6: Typology of VR locomotion techniques showing the five distinct locomotion types and their characteristics [11].

Beyond locomotion, VR systems rely on a variety of interaction techniques that enable users to select, manipulate, and control virtual objects and interface elements. According to recent surveys of VR interaction design, these techniques can be broadly categorized into the following groups:[12], [13], [14]

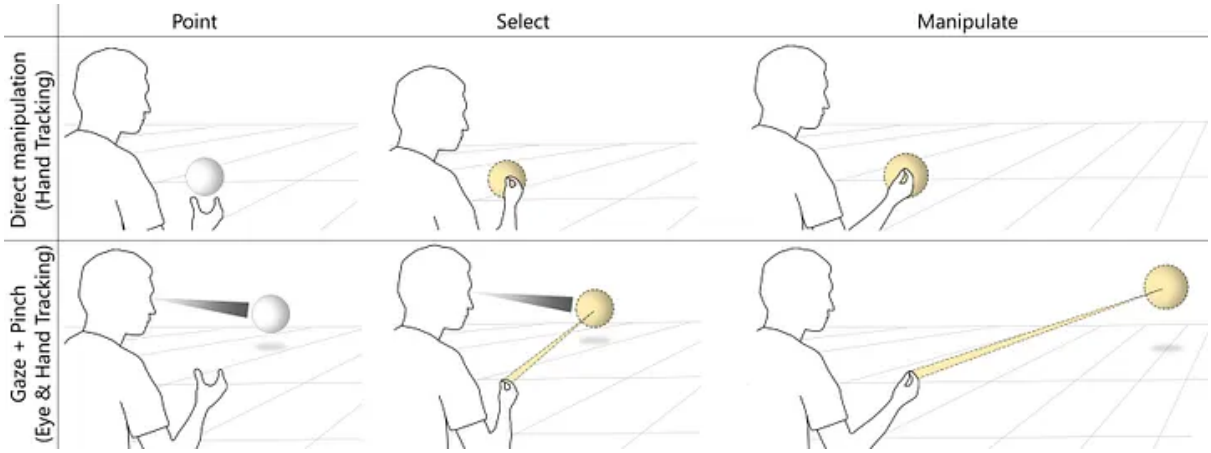


Figure 2.7: Comparison of direct manipulation and gaze-based interaction techniques in VR environments [15].

1. **Direct Manipulation Techniques:** These allow users to interact with virtual objects through hand movements or tracked controllers, closely mimicking real-world manipulation.
    - Examples: Grabbing, moving, rotating, scaling objects with tracked hands or controllers.
2. **Ray-Based Interaction Techniques:** Ray-casting extends a virtual pointer into the environment, enabling selection and interaction at a distance, particularly useful for UI elements and far-away objects.
    - Examples: Laser-pointer selection, pointing-based menu interaction.
3. **Gesture-Based Interaction Techniques:** These rely on hand poses, motions, or symbolic gestures captured through hand tracking or sensors, enabling more natural and expressive interaction.
    - Examples: Pinching to select, swiping to activate commands, mid-air gesture controls.
4. **Gaze-Based Interaction Techniques:** Eye tracking is used to direct attention, select targets, or support multimodal input, often combined with other modalities to reduce manual effort.
    - Examples: Dwell-based selection, gaze-assisted pointing, gaze + gesture hybrid interaction.
5. **Voice-Based Interaction Techniques:** Speech input enables hands-free interaction and complements manual or gaze-based input, especially for system-level commands.
    - Examples: Voice-activated menus, spoken commands for system control.
6. **Haptic and Feedback-Based Techniques:** These techniques incorporate tactile or force feedback to enhance realism, precision, and embodiment.

- Examples: Controller vibration, wearable haptics, force-feedback gloves.

7. **Adaptive and Space-Efficient Techniques:** Emerging techniques modify the mapping between the user's real movements and virtual actions to improve usability in constrained environments.[16]

    - Examples: Amplifying small finger or hand motions to large virtual arm movements, interaction remapping.

VR in education and training

Virtual Reality (VR) is rapidly transforming education and training by offering immersive, experiential learning environments that extend beyond the limitations of traditional classrooms. VR enables learners to explore locations, scenarios, and concepts that would otherwise be inaccessible, impractical, or too costly to experience in real life.

Recent work by Aufenanger et al.[17] demonstrates the effectiveness of immersive Virtual Reality Learning Environments (VRLEs) designed within the EU-funded REVEALING project. Their study implemented VR modules covering topics such as ancient Greek technology, historical expeditions, sea-urchin measurements, and linear algebra. Students reported high levels of engagement, motivation, and positive emotional responses, especially within interactive VR experiences that allowed them to manipulate objects, solve problems, or complete tasks in fully immersive environments.

Research on VR use in post-primary education further highlights its pedagogical value. A systematic narrative review by Hamash et al.[18] shows that VR is most widely applied in STEM fields, where its capacity for visualization and simulation supports deeper conceptual understanding. Their review identifies numerous examples of VR-based learning activities, including virtual science laboratories, 3D physics simulations, and immersive explorations of biological systems. Across the studies surveyed, VR-enhanced learning was frequently associated with improved knowledge retention, increased motivation, and stronger learner engagement when compared to conventional methods.

Beyond conceptual learning, VR is also well suited for specialized training contexts. By simulating complex or hazardous real-world environments, VR provides a safe and controlled space for learners to practice skills, rehearse procedures, or apply theoretical knowledge prior to performing tasks in authentic settings. Such training-oriented VR applications are especially common in fields such as engineering, health sciences, and laboratory-based disciplines, where hands-on practice is crucial but often constrained by cost, equipment availability, or safety concerns [18]. In these domains, VR enables repeated, risk-free practice and supports competency development before transitioning to real-world scenarios.

Overall, the evidence from current research indicates that VR can significantly enrich both educational and training environments when thoughtfully integrated into instructional design. Although challenges such as motion sickness, technical limitations, and varying levels of user acceptance remain, VR shows strong potential to enhance experiential learning and support more effective skill acquisition across diverse academic and professional fields.

2.1.3 Gamification in Education

Overview of Gamification

Gamification is the application of game-design elements and game principles in non-game contexts to enhance user engagement, motivation, and learning outcomes[19]. In educational settings, gamification transforms learning activities by incorporating mechanics commonly found in games—such as points, badges, leaderboards, challenges, and narrative structures—to make educational experiences more engaging and effective[20].

The theoretical foundation of gamification draws from multiple disciplines, including behavioral psychology, motivational theory, and game design. Self-Determination Theory (SDT), developed by Deci and Ryan, provides a key framework for understanding gamification's effectiveness[21]. SDT identifies three fundamental psychological needs that drive intrinsic motivation: *autonomy* (feeling of control over one's actions), *competence* (sense of mastery and achievement), and *relatedness* (connection to others). Well-designed gamification addresses all three needs, supporting sustained engagement and deeper learning[22].

Core Gamification Elements

Effective gamification in educational applications typically incorporates several core elements[23], [24]:

1. **Points and Scoring Systems:** Quantifiable metrics that provide immediate feedback on performance. Points reward correct actions, efficient procedures, and task completion, offering learners clear indication of their progress and competency development.

2. **Badges and Achievements:** Visual representations of accomplishments that recognize specific milestones or mastery of particular skills. Badges serve both as extrinsic rewards and as markers of competency that learners can display.

3. **Levels and Progressive Difficulty:** Structured progression through increasingly challenging content. This scaffolded approach prevents learner frustration (by not overwhelming beginners) while maintaining engagement (by providing appropriate challenges for advancing learners).

4. **Leaderboards and Social Comparison:** Rankings that enable learners to compare their performance against peers. While potentially motivating for competitive learners, leaderboards must be carefully implemented to avoid demotivating lower-performing students.

5. **Challenges and Quests:** Goal-oriented tasks that provide clear objectives and purpose. Challenges give structure to learning activities and create a sense of mission that can increase engagement.

6. **Feedback and Progress Visualization:** Immediate, clear information about performance and advancement. Progress bars, completion percentages, and skill trees help learners understand their current standing and what remains to be accomplished.

7. **Narrative and Storytelling:** Contextual frameworks that give meaning to learning activities. Stories create emotional engagement and help learners understand the relevance and application of skills being developed.

8. **Unlockable Content:** Resources, tools, or scenarios that become available as learners progress, creating anticipation and rewarding advancement with new experiences.

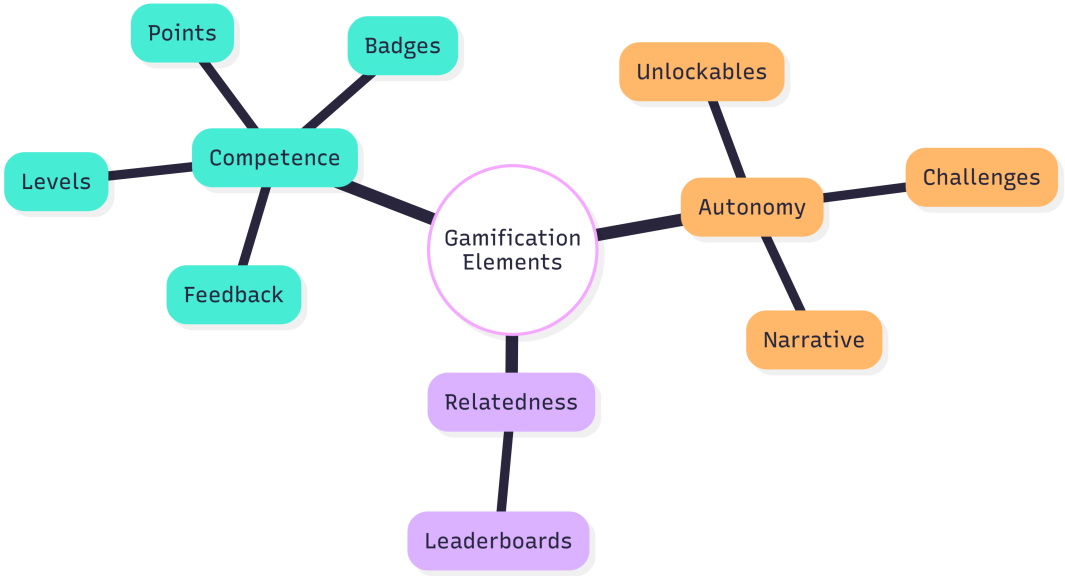


Figure 2.8: Core gamification elements used in educational applications.

Gamification in VR Training Environments

Virtual Reality provides an ideal platform for gamification due to its inherently immersive and interactive nature. VR environments naturally support many game-like qualities: spatial presence, embodied interaction, real-time feedback, and consequence-free experimentation that align with effective gamification principles[25].

The combination of VR and gamification offers several advantages for training applications[26], [27]:

- **Enhanced Immersion:** VR's sensory immersion amplifies the impact of gamification elements. Earning an achievement or unlocking new tools feels more meaningful when experienced in an embodied, three-dimensional space.

- **Natural Feedback Integration:** VR enables feedback to be seamlessly integrated into the environment itself. Rather than abstract point notifications, feedback can be presented through realistic environmental responses, character reactions, or spatial indicators.

- **Embodied Learning:** Physical actions in VR (reaching, grasping, manipulating) combined with gamification mechanics create stronger memory encoding through embodied cognition principles.

- **Safe Failure Environment:** Gamification's emphasis on iterative attempts and learning from mistakes aligns perfectly with VR's ability to provide risk-free practice environments where failure has no real-world consequences.

2.1.4 AI-Powered Learning Support in Virtual Reality

Intelligent Assistance Systems for Procedural Training

The integration of AI-powered assistance in educational VR environments is grounded in established learning theories that emphasize adaptive scaffolding and learner-centered support. In the context of VR training, intelligent assistance systems can dynamically adjust the level of support based on learner performance, providing guidance when needed while promoting independent problem-solving.

Recent advances in Large Language Models (LLMs) have enabled more natural and contextually aware conversational interfaces that can serve as virtual tutors or advisors. When integrated with real-time system state information, LLMs can provide situation-specific guidance that adapts to the learner's current context, making them particularly suitable for complex procedural training environments like forensic investigation[28].

NPC-Based Pedagogical Agents

The use of Non-Player Characters (NPCs) as interface agents for intelligent assistance systems offers several advantages over traditional help interfaces. NPCs provide a more immersive and socially engaging interaction paradigm compared to menu-based help systems, supporting the sense of presence that is central to effective VR learning. The embodied nature of NPC advisors can also reduce cognitive load by integrating assistance naturally into the virtual environment rather than requiring learners to exit the immersive experience to seek help[29].

NPCs in procedural training environments can serve multiple roles: demonstrating correct procedures, providing real-time feedback on learner actions, answering context-specific questions, and offering strategic guidance for complex tasks. When powered by AI systems, these agents can maintain conversation history, understand domain-specific terminology, and adapt their explanations to match the learner's demonstrated competency level. This combination of embodied presence, contextual awareness, and adaptive dialogue makes NPC-based assistance particularly well-suited for immersive training simulations where maintaining engagement and spatial presence is critical to learning effectiveness.

2.1.5 Forensic Science

Overview

Forensic science is the application of scientific principles and methods to matters of law, criminal and civil. It is an interdisciplinary field encompassing numerous sub-disciplines in the natural, physical, and computational sciences. Nearly any scientific field can become "forensic" when its techniques are used to solve crimes or resolve legal questions.

Crime Scene Investigation (CSI)

Forensic science comprises a wide range of specialized branches, each focused on the analysis of specific types of evidence or the application of particular scientific principles. One of the major branches of forensic science is Crime Scene Investigation (CSI), this branch involves the scientific methods of recognizing, documenting, collecting, and analyzing physical evidence from crime scenes. CSI specialists (forensic investigators or criminalists) gather trace materials, fingerprints, bloodstains, and other evidence, and reconstruct events based on the physical clues present.[30]

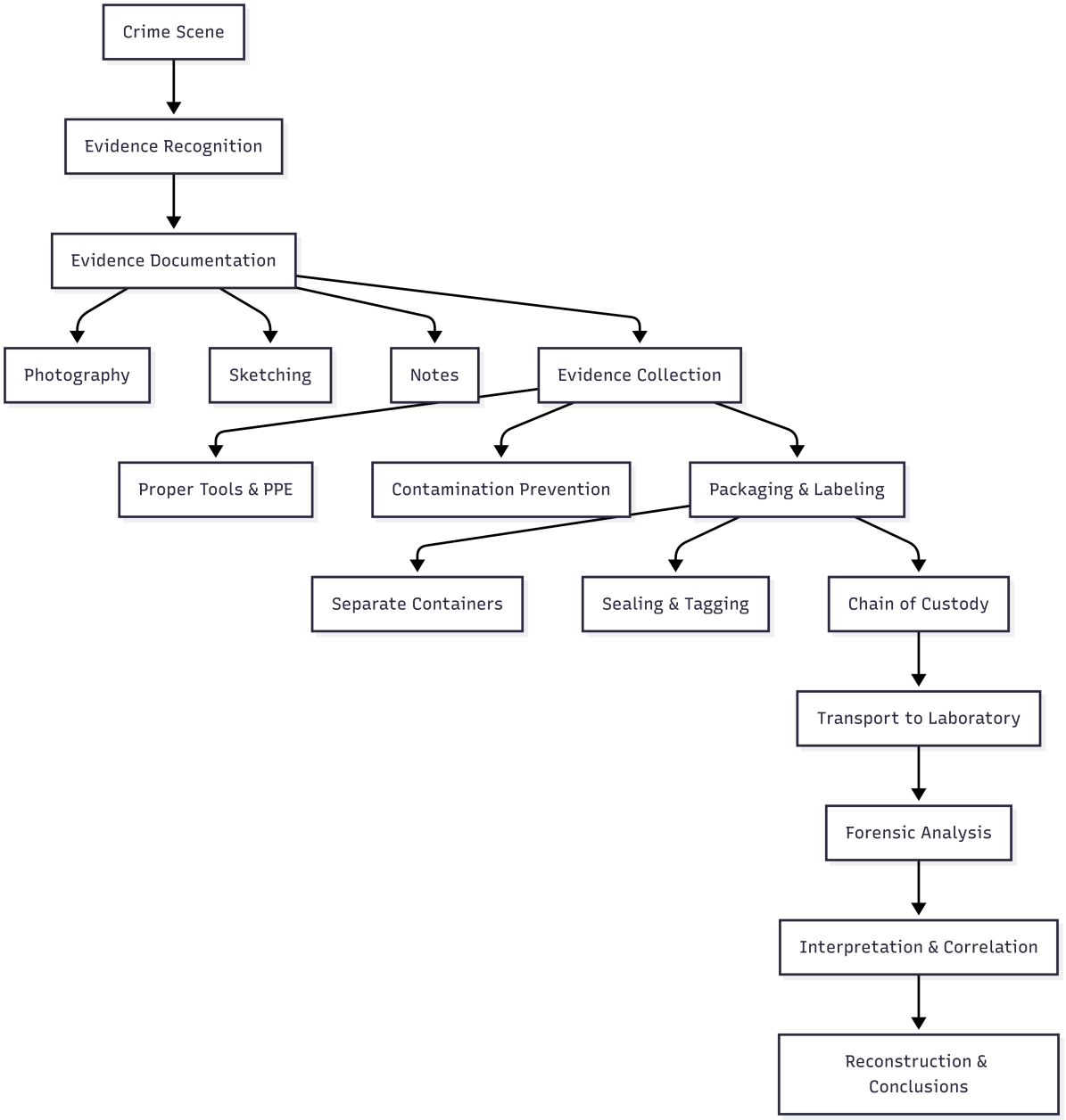


Figure 2.9: The crime scene investigation process: scene assessment, documentation, evidence collection, preservation, and analysis.

Types of Forensic Evidence

Physical evidence is the cornerstone of forensic investigation, providing objective information that can link suspects to crime scenes, corroborate or refute witness statements, and reconstruct criminal events[31]. Understanding evidence types is fundamental to effective crime scene investigation, as different categories require specific collection, preservation, and analysis procedures[32].

Forensic evidence can be classified according to multiple taxonomies. The most fundamental classification divides evidence into physical and testimonial categories, though crime scene investigators primarily focus on physical evidence. Physical evidence itself can be further categorized by its nature, origin, and the information it provides[33].

Classification by Physical Form

1. **Biological Evidence:** This category encompasses all materials of biological origin that can provide genetic information, species identification, or physiological data. Biological evidence includes:

    - *Blood and bloodstains:* Can determine blood type, DNA profile, blood spatter patterns indicating the nature and sequence of violent events, and potentially toxicological information.
    - *Bodily fluids:* Saliva, urine, and other fluids contain DNA and can establish presence, identity, or activities at the scene.
    - *Hair and fibers (biological):* Hair can provide DNA (if root is present), species identification, and information about origin on the body. Can be used for toxicological analysis.
    - *Tissue samples:* Can provide DNA identification and potentially information about cause of death or injury mechanisms.
    - *Plant material:* Seeds, leaves, pollen, and wood fragments can establish location, season, or linkages between scenes and suspects.

2. **Trace Evidence:** Small materials transferred during contact between persons, objects, or environments. Following Locard's Exchange Principle which states that every contact leaves a trace, trace evidence is often crucial for establishing connections[34]. Types include:

    - *Fibers (synthetic):* From clothing, carpets, or other textiles; can link suspects to victims or scenes through transfer.
    - *Glass fragments:* Can indicate direction of force, type of breaking event, and provide associative evidence through composition matching.
    - *Paint chips:* From vehicles, buildings, or objects; layering and composition can provide specific matches.
    - *Soil and dust:* Mineralogical and organic composition can link individuals to specific geographic locations.
    - *Gunshot residue (GSR):* Particles deposited on hands, clothing, or nearby surfaces when a firearm is discharged.

3. **Impression Evidence:** Patterns created when one object comes into contact with another, leaving a mark or indentation:

- *Fingerprints:* Ridge patterns on fingertips that are unique to each individual; can be patent (visible), latent (invisible, requiring development), or plastic (three-dimensional impressions).
- *Footwear impressions:* Can identify shoe brand, model, size, and individual wear patterns; help establish movement through the scene.
- *Tire tracks:* Indicate vehicle type, direction of travel, and potentially specific vehicle identification through wear patterns and damage.
- *Tool marks:* Impressions or striations left by tools; can match to specific implements and indicate method of entry or damage.
- *Bite marks:* Can potentially link to suspects through dental characteristics, though interpretation is controversial.

4. **Physical Objects and Materials:**
   - *Weapons:* Firearms, knives, blunt instruments; can be matched to injuries and may contain trace or biological evidence.
   - *Projectiles and cartridge cases:* Ballistic evidence linking firearms to shooting incidents through distinctive markings.
   - *Documents:* Handwriting, printed materials, forged documents; subject to questioned document examination.
   - *Digital devices:* Computers, smartphones, storage media containing potential digital evidence.
5. **Chemical Evidence:**
   - *Drugs and controlled substances:* Identification and quantification of illegal or prescription drugs.
   - *Accelerants:* Flammable liquids used in arson cases.
   - *Explosives:* Residues and components of explosive devices.
   - *Toxins and poisons:* Chemical substances that may have caused death or injury.

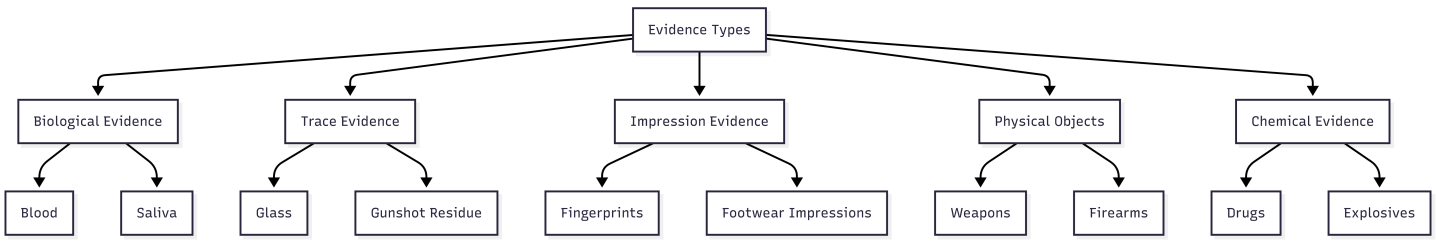


Figure 2.10: Classification of forensic evidence types with examples of each category.

Classification by Evidential Value

Beyond physical form, evidence is also classified according to the type of information it provides[35]:

- **Identifying Evidence:** Establishes identity of individuals, objects, or substances (e.g., DNA profiles, fingerprints, serial numbers).
- **Associative Evidence:** Links persons, places, or objects together (e.g., matching fibers, soil samples connecting suspect to scene).
- **Reconstructive Evidence:** Helps determine the sequence of events, positions of individuals, or mechanisms of injury (e.g., blood spatter patterns, bullet trajectories).
- **Exculpatory Evidence:** Tends to exclude or exonerate a suspect from involvement in the crime.

Evidence Handling Principles
Regardless of type, all physical evidence must be[32], [35]:

- *Recognized:* Investigators must identify potential evidence through systematic scene examination.
- *Documented:* Evidence location, appearance, and context must be thoroughly recorded through notes, photographs, sketches, and/or video.
- *Collected:* Using appropriate techniques and tools to prevent contamination, loss, or alteration.
- *Preserved:* Stored under conditions that maintain integrity (e.g., refrigeration for biological samples, sealed containers for trace evidence).
- *Tracked:* Chain of custody must be meticulously maintained to ensure admissibility in court.

Understanding these evidence categories and their specific requirements is essential for effective crime scene investigation and forms a core component of forensic education and training programs.

Evidence Tagging and Documentation Methods

Evidence tagging is a critical component of crime scene investigation that ensures proper identification, tracking, and chain of custody for all collected materials. Proper tagging methods are fundamental to maintaining evidence integrity and admissibility in legal proceedings[32], [36].

Theoretical Foundation

Evidence tagging serves multiple forensic purposes. First, it creates a unique identifier for each piece of evidence that links it to specific documentation (photographs, sketches, reports). Second, it establishes the chain of custody by recording who collected the evidence, when, and from where. Third, it prevents cross-contamination by maintaining clear separation between evidence items. Finally, it facilitates laboratory analysis by providing context about the evidence's origin and significance[31], [35], [36].

Standard Tagging Procedures

Professional crime scene investigation employs systematic tagging methods that typically include the following components:

1. **Evidence Markers (Placards):** Numbered or lettered markers are placed adjacent to evidence items before photography and collection. These markers serve as visual reference points in photographs and establish a systematic identification system. Common marker types include:

    - *Numbered tent markers:* Folded cards with sequential numbers, easily visible in photographs, illustrated in Figure 2.11



Figure 2.11: Numbered evidence tent markers used for crime scene documentation [37].

    - *Flag markers:* Stakes with numbered flags, useful for outdoor scenes or large areas. An example is shown in Figure 2.12



Figure 2.12: Evidence marking flags for outdoor crime scenes [38].

    - *Adhesive markers:* Flat numbered stickers for vertical surfaces or confined spaces. As seen in figure 2.13

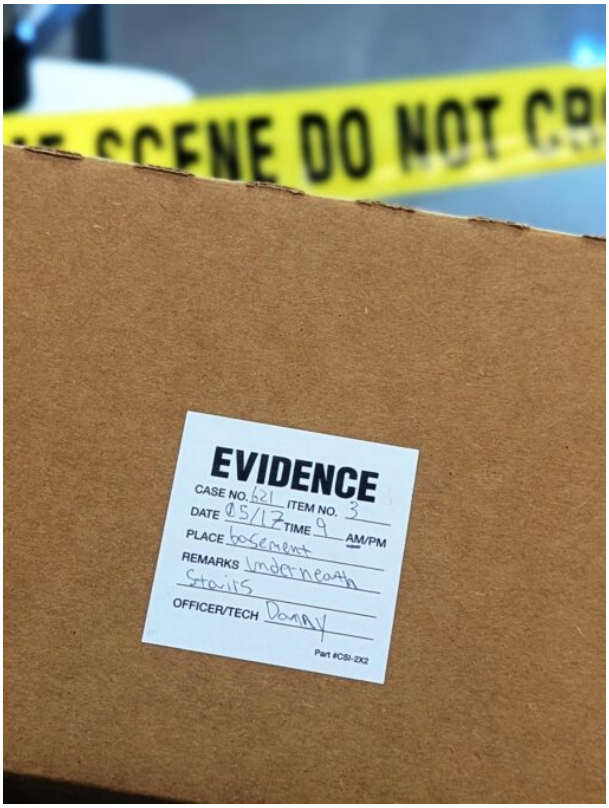


Figure 2.13: Adhesive evidence labels for tagging collected items [39].

- *Photo scales:* Rulers or calibrated scales placed alongside evidence to indicate size. As shown in Figure 2.14

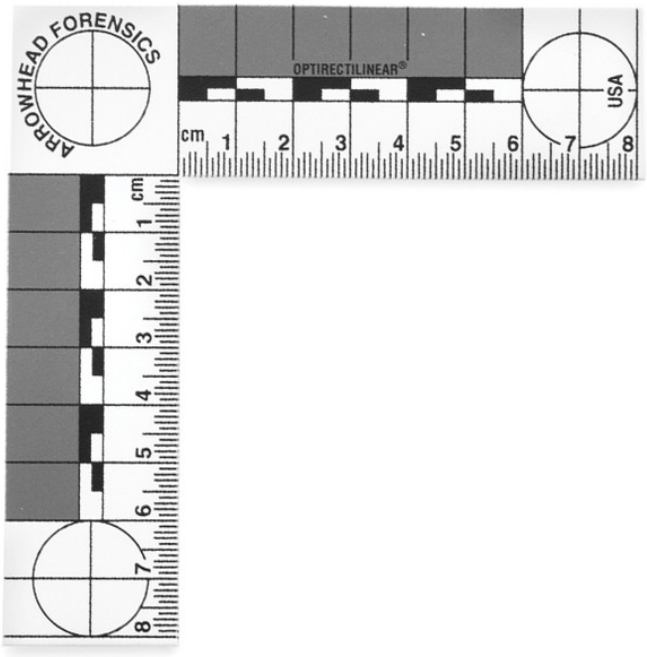


Figure 2.14: ABFO No. 2 photomacrographic scale used for accurate measurement in evidence photography [40].

2. **Evidence Tags and Labels:** After collection, physical tags are attached to evidence containers. These tags must include:
    - Unique evidence identifier (case number + item number)
    - Brief description of the item
    - Exact location where the evidence was found
    - Date and time of collection
    - Name and signature of the collecting officer
    - Handling and storage instructions (if applicable)
3. **Photographic Documentation:** Evidence must be photographed in three stages:
    - *Overview photographs:* Show the item's location within the overall scene context
    - *Mid-range photographs:* Show the item with an evidence marker in place
    - *Close-up photographs:* Detailed images of the item with a scale for size reference
4. **Sketches and Diagrams:** Crime scene sketches incorporate evidence marker numbers, showing the spatial relationships between evidence items. Measurements from fixed reference points are recorded to enable accurate reconstruction.
5. **Chain of Custody Documentation:** A continuous written record tracking every person who handled the evidence, the date/time of each transfer, and the purpose. This documentation is essential for legal admissibility.

Digital Evidence Tagging
Modern forensic practice increasingly incorporates digital tagging methods:

- **Digital photography with metadata:** Camera systems embed GPS coordinates, timestamps, and photographer identification directly in image files.
- **Barcode and RFID systems:** Evidence containers are tagged with scannable identifiers linked to digital databases, reducing manual documentation errors and enabling real-time tracking.
- **3D scanning with annotation:** Laser scanners and photogrammetry systems create 3D models where evidence locations are digitally tagged with associated metadata.
- **Evidence management software:** Centralized database systems link photographs, physical tags, laboratory results, and chain of custody records to each evidence item.

Common Tagging Errors and Prevention
Proper training must address common mistakes in evidence tagging:

- *Premature placement of markers:* Placing markers before initial photography can obscure critical details. Markers should only be positioned after overall scene documentation.

- *Inconsistent numbering:* Using multiple numbering schemes or skipping numbers creates confusion. A single, sequential system should be maintained throughout.
- *Obscured markers in photos:* Poor lighting or camera angles can make marker numbers unreadable. Multiple angles should be photographed to ensure clarity.
- *Incomplete tag information:* Missing critical fields (especially time, location, or collector signature) can compromise evidence integrity. Standardized forms should be used.
- *Cross-contamination during tagging:* Handling multiple evidence items without changing gloves or using contaminated markers can transfer trace evidence. Proper protective equipment and clean materials must be used.

Understanding and practicing these evidence tagging methods is fundamental to forensic training, as proper documentation directly impacts the success of criminal investigations and prosecutions.

Traditional crime scene training: methods & limitations

Traditional crime scene investigation training has historically relied on classroom instruction, printed manuals, lectures, and hands-on demonstrations using mock crime scenes. These approaches emphasize foundational knowledge of evidence types, procedural standards, and legal considerations. Trainees learn core competencies such as scene assessment, documentation methods, pattern recognition, and proper evidence handling through structured exercises and supervised practice. [41]

However, traditional training methods have several limitations:

1. **Limited realism:** Mock scenes often lack the complexity, unpredictability, and environmental variability of real crime scenes. This can restrict exposure to irregular or challenging conditions.
2. **Resource constraints:** Setting up high-fidelity training environments requires time, materials, and physical space. Institutions with limited budgets may be unable to provide repeated, diverse training scenarios.
3. **Inconsistent replication:** Because scenarios are created manually, achieving consistent conditions across training groups or sessions is difficult, leading to variability in trainee experience and assessment.
4. **Restricted scalability:** Hands-on, instructor-led training is inherently limited by the availability of qualified personnel and the number of trainees who can be supervised effectively.

2.1.6 Virtual Reality in Forensic Science

Virtual Reality (VR) represents an immersive technological shift in forensic practice, moving beyond traditional two-dimensional (2D) documentation to provide three-dimensional (3D) environments for examination and presentation [42]. While initially introduced for straightforward visualization and plotting of distances at crime scene, its applications have rapidly expanded across forensic education, research, and complex criminal investigation. [42]

Applications

The application of VR in forensic science primarily focuses on high-resolution visualization and accurate scene reconstruction. One major application is in **immersive crime scene reconstruction**, where VR systems, paired with multi-angle and omnidirectional spatial data collection devices (such as laser scanners and structured light scanners), capture the entire scene [43]. This approach allows the recreation of events, in order to understand the narrative [43].

Benefits

The integration of VR offers several crucial benefits over conventional forensic documentation methods. The most significant advtange is the ability to overcome the fundamental limitation of visualing 3D data on a 2D computer screen, which often results in the loss of vital details. VR, as an immersive technology, resolves this by allowing the user to step into the scene, supporting visualization and deeper spatial comprehension [42]. For courtroom purposes, the shift to immersive VR presentation enhances the clarity of evidence, providing a more insightful idea to judges and juries [42], [43].

Limitations and challenges

The implementation of VR technology in forensic contexts faces both technical and ethical challenges. The primary goal of VR in forensics is to address the inefficiency of traditional methods for effectively recontrsucting crime scenes, however that task itself is inherently complex, requiring specialized systems and expertise to be able to do it effectively. Furthermore, a challenge associated with highly realistic visualization tools, is the potential to put the viewer in a "seeing-is-beliving" attitude, which can pose ethical concerns regarding the interpretation of visual evidence in court [42].

2.1.7 Relevant Articles

Using Virtual Reality as Support to the Learning Process of Forensic Scenarios

Khalilia et al. investigate the use of Virtual Reality (VR) as an educational support tool in forensic science training, with a specific focus on crime scene investigation courses [44]. The study was conducted at Al-Istiqlal University during the Spring 2021 semester and involved students enrolled in a crime scene investigation module. Data were collected through an online questionnaire designed to assess students' perceptions of different learning strategies, including collaboration, discussion-based learning, field-based activities, and problem-solving, when combined with VR technology. The findings indicate that integrating VR into forensic education enhances student engagement and supports the effective application of modern learning strategies. The authors conclude that VR contributes positively to the learning process by improving students' understanding of crime scene scenarios and increasing their involvement in practical forensic training. Overall, the study highlights VR as a valuable instructional tool for enriching forensic science education and improving learning outcomes in investigative training environments [44].

Virtual Reality Forensics: Forensic Analysis of Meta Quest 2

Raymer, MacDermott, and Akinbi [45] present a forensic-focused study on the Meta Quest 2 VR headset, a widely used consumer virtual reality device developed by Meta Platforms Inc. The headset enables users to engage in fully immersive virtual environments, many of which support online interaction and shared experiences. While these environments offer new forms of social engagement, they also introduce security and safety concerns, as users may encounter unlawful activities such as online harassment, exploitation, or cyberbullying. To address these challenges, the authors investigate the feasibility of performing sound digital forensic analysis on VR hardware. Their study demonstrates a complete forensic acquisition and examination of a Meta Quest 2 device, revealing recoverable artifacts related to user behavior, system configuration, and stored data. The findings show that valuable evidential data can be extracted without compromising forensic integrity. The key contributions of this work include documenting the forensic acquisition methodology, identifying relevant internal storage locations, and analyzing digital artifacts that may assist investigators in VR-related crime investigations [45].

Using Virtual Reality for Forensic Examinations of Injuries

Koller et al. explore the application of Virtual Reality (VR) in combination with three-dimensional photogrammetry for improving injury measurement and documentation in forensic investigations [46]. The study highlights the limitations of traditional documentation techniques that rely on two-dimensional photographs with reference scales, which often suffer from loss of spatial information and reduced measurement accuracy due to issues such as perspective distortion, depth limitations, and image clarity. To evaluate the effectiveness of VR-based analysis, the researchers conducted experiments using a mannequin containing wounds of predefined dimensions. Injury measurements obtained through conventional photography were compared with measurements derived from 3D photogrammetry and VR-based visualization. The results indicate that VR-based measurements were significantly more accurate than those obtained from standard photographs, while photogrammetric measurements performed on a desktop system showed the highest overall precision. The authors conclude that both VR and photogrammetry provide reliable and effective approaches for injury documentation, with 3D models offering improved spatial understanding and analytical accuracy, thereby supporting enhanced forensic documentation and training practices [46].

A Toolbox for the Rapid Prototyping of Crime Scene Reconstructions in Virtual Reality

Sieberth et al. present a Virtual Reality (VR)-based toolbox designed to improve the efficiency of crime scene reconstruction in forensic investigations [47]. The study addresses the limitations of conventional 3D reconstruction workflows, where interaction with three-dimensional data is typically performed on two-dimensional displays using standard input devices. This approach often results in time-consuming and cumbersome model positioning and adjustment despite the availability of rich 3D data. The proposed system enables forensic investigators to visualize, explore, and manipulate incident scenes directly within an immersive three-dimensional environment. It supports the integration of heterogeneous data sources, including laser-scanned crime scenes, surface scans of physical objects, and medical imaging data such as CT and MRI. By allowing intuitive

interaction with complex spatial data, the toolbox facilitates rapid prototyping of crime scene reconstructions and supports more efficient examination of forensic scenarios. The authors demonstrate that the use of VR enhances spatial understanding, improves scene exploration, and enables more interactive and realistic reconstructions when compared to traditional screen-based methods, highlighting the potential of VR to support forensic analysis and investigative decision-making [47].

Virtual Reality for Teaching and Learning in Crime Scene Investigation

Mayne and Green discuss the use of virtual reality (VR) as a teaching tool to improve the learning of crime scene investigation skills [48]. This article focuses on a custom VR application designed to offer immersive, hands-on learning experiences for students. The study shows that VR provides cost-effective training by allowing students to interact with virtual crime scenes, improving their investigation skills without the limitations of traditional methods. The app received positive feedback from both students and staff, with minimal adverse effects reported. The article concludes that VR has significant potential to expand forensic education, though challenges such as development costs and scalability need to be addressed. The goal is to use VR to provide students with practical, immersive experiences that are both engaging and accessible.

Designing and Evaluation of a Mixed Reality System for Crime Scene Investigation Training: A Hybrid Approach

Albeedan, Kolivand, and Hammady present the design and evaluation of a Mixed Reality (MR) system for training crime scene investigators using 3D scanning and photography techniques [49]. The system was developed using Microsoft HoloLens 2.0 and FARO 3D scanner, aiming to provide interactive and cost-effective training for investigators in a virtual environment that simulates real-life crime scenes. The results showed that the system helps improve investigative skills and enhances the interaction and learning experience for users, with students and staff reporting high levels of satisfaction. The article also aims to expand the system's use in the future to become a key tool in training investigators by improving accessibility and interaction in various educational environments.

2.2 Existing Applications

2.2.1 StaplesVR - Forensic Training Simulator

StaplesVR [50] is a photorealistic VR training environment developed in collaboration with New Zealand Police and ESR, using LiDAR and photogrammetry to recreate real crime scenes.

Features

- Photorealistic scenes built from real crime houses
- Complete crime scene workflow with evidence collection and documentation
- Protocol-driven training aligned with NZ Police standards
- Contamination prevention mechanics with glove-changing simulation

Limitations

- No chain-of-custody management
- No real-time procedural feedback
- Single-user only with no collaborative support



Figure 2.15: Screenshot from StaplesVR Forensic Training Simulator [50].

2.2.2 Vizitech USA - VR Forensics

VR Forensics by Vizitech USA [51] offers protocol-driven training with 36 different crime scene scenarios, emphasizing blood spatter analysis and contamination control.

Features

- Extensive library of 36 crime scenarios with high replay value
- Protocol-oriented training with protective equipment checks
- Virtual evidence marker placement tools
- Supports collaborative classroom environments

Limitations

- No chain-of-custody tracking
- No evidence tagging with forensic metadata
- No real-time procedural feedback or adaptive guidance



Figure 2.16: Screenshot from Vizitech USA VR Forensics [51].

2.2.3 VictoryXR - CSI & Forensics (VXRLabs)

VictoryXR's CSI experience [52] provides hyper-realistic 3D environments with synchronous multi-user support for collaborative forensic training.

Features

- Hyper-realistic 3D crime scenes, police stations, and laboratories
- Multi-user support for collaborative team-based investigations
- Progressive learning with five distinct cases and lab modules
- Contamination prevention with proper gloving procedures

Limitations

- No chain-of-custody tracking system
- No evidence tagging with forensic metadata
- No real-time procedural feedback or adaptive guidance



Figure 2.17: Screenshot from VictoryXR CSI & Forensics [52].

2.2.4 Home Detective

Home Detective [53] is a mixed-reality game for consumer VR devices that overlays virtual crime scene elements onto the user's real living space.

Features

- Mixed reality support using passthrough technology
- Hand tracking for natural interactions
- Performance evaluation with letter-grade scoring
- Game-style achievements and progressive cases

Limitations

- Entertainment-focused rather than professional training
- No protocol training or chain-of-custody management
- Lower visual realism compared to professional tools
- Single-player only



Figure 2.18: Screenshot from Home Detective [53].

2.2.5 Virtual Crime Scene Simulation in Palestine

The virtual crime scene simulation conducted in Palestine [54] provided forensic biology students with a shooting-incident scenario meant to strengthen practical investigation skills and apply theoretical learning in a local context.

Features

- Localized crime-scene content tailored to regional context
- Helps bridge theoretical forensic education with practical application
- Provides exposure to realistic investigative scenarios for students

Limitations

- Technical realism or simulation fidelity not documented
- Appears to be a one-time training initiative rather than a full application

2.2.6 Comparison of Existing Applications

Feature	StaplesVR	Vizitech USA	VictoryXR	Home Detective	Our Project
Complete Crime Scene Workflow	✓	✓	✓	✗	✓
Evidence Collection	✓	✓	✓	✓	✓
Procedural / Protocol Training	✓	✓	✓	✗	✓
Chain of Custody Management	✗	✗	✗	✗	✓
Evidence Tagging & Documentation	✓	✗	✗	✗	✓
Real-Time Procedural Feedback	✗	✗	✗	✗	✓
NPC-Based Adaptive Hint System	✗	✗	✗	✗	✓
Performance Evaluation & Reports	✗	✗	✗	✓	✓
High Visual Realism	✓	✓	✓	✗	✓
Mixed Reality Support	✗	✗	✗	✓	✗
Narrative / Engagement Focus	✓	✓	✓	✓	✓
Gamification	✗	✗	✗	✓	✓

Table 2.1: Feature-based comparison of existing forensic VR training applications.

2.2.7 Analysis of Existing Applications

After reviewing the current landscape of forensic VR applications, several critical gaps and limitations emerge. While existing solutions provide valuable training experiences, they exhibit deficiencies across multiple dimensions that limit their educational effectiveness and accessibility.

2.2.8 Identified Limitations

Through systematic analysis of existing applications, the following key limitations have been identified:

1. **Limited Visual Fidelity:** Applications like StaplesVR and Home Detective utilize low-polygon models and simplified textures, reducing immersion and failing to replicate the visual complexity of actual crime scenes. This diminished realism can negatively impact learning transfer to real-world scenarios.

2. **Restricted Procedural Depth:** Many applications focus on isolated forensic tasks without providing comprehensive end-to-end workflow training. For example, Home Detective emphasizes basic evidence collection but lacks proper chain of custody documentation and evidence analysis procedures.

3. **Lack of Adaptive Learning Support:** Most applications follow rigid, linear progression without intelligent guidance systems. This creates barriers for novice learners who may require contextual assistance and scaffolding to understand complex forensic concepts and procedures.

4. **Limited Engagement Mechanisms:** While VictoryXR incorporates task-based progression, most applications lack sustained engagement features such as meaningful progression systems, performance feedback, or motivational elements that encourage continued learning and skill development.

5. **Accessibility and Usability Barriers:** Applications like Vizitech require guided facilitation, limiting independent learning opportunities. Additionally, many systems lack intuitive interfaces, creating steep learning curves for VR interaction itself, which detracts from forensic skill acquisition.

2.2.9 Planned Improvements in Our Project

Our project directly addresses these limitations through a comprehensive set of targeted improvements designed to enhance educational effectiveness, accessibility, and cultural relevance.

Enhanced Realism and Immersion

We will implement photorealistic 3D assets and high-resolution textures based on authentic crime scene photographs and forensic documentation. Environmental details such as dynamic lighting, ambient sounds, and realistic material properties will create a convincing sense of presence. This heightened realism serves multiple pedagogical purposes: it reduces the cognitive gap between training and practice, supports better visual pattern recognition skills, and increases learner engagement through emotional investment in realistic scenarios.

Comprehensive Procedural Training

Our application will guide users through complete forensic workflows, from initial scene assessment and documentation through evidence recognition, collection, and preservation. Each procedure will be broken down into discrete steps with clear success criteria, ensuring learners understand not only what to do but why each action matters in the broader investigative context. Critical concepts such as contamination prevention, chain of custody maintenance, and evidence integrity will be reinforced throughout the experience.

NPC-Based AI-Powered Adaptive Assistance

Applying the theoretical foundations of AI-powered learning support discussed earlier, we will implement an interactive NPC (non-player character) that serves as an on-demand forensic advisor. When prompted by the user, this virtual assistant will answer forensics-related questions and provide contextual hints using a large language model (LLM) that receives real-time information about the current scene state, the user's progress, and the specific task at hand.

Students can interact with the NPC using either voice input or text input through a virtual keyboard, making the system accessible and flexible. This approach gives learners control over when to request assistance, encouraging independent problem-solving while ensuring help is available when needed. The LLM-powered system will be able to provide nuanced, situation-specific guidance that adapts to the particular evidence present, the

investigative stage, and the user's demonstrated understanding, making the application accessible to students with varying levels of prior forensic knowledge.

Gamification and Sustained Engagement

The application will incorporate gamification elements designed to motivate continued practice and skill mastery. Features will include performance scoring based on accuracy and efficiency, progressive difficulty levels, achievement badges for procedural milestones, and optional challenges for advanced learners. These elements will be carefully balanced to enhance motivation without compromising the serious educational purpose of the simulation.

Intuitive Interface Design

Recognizing that VR interaction itself can be a barrier to learning, we will prioritize intuitive, natural interaction methods. Clear visual feedback, haptic responses, and audio cues will ensure users always understand their available actions and the results of their interactions.

Figure 2.19 illustrates the high-level architecture of the proposed system and the interaction between users, system components, and generated outputs.

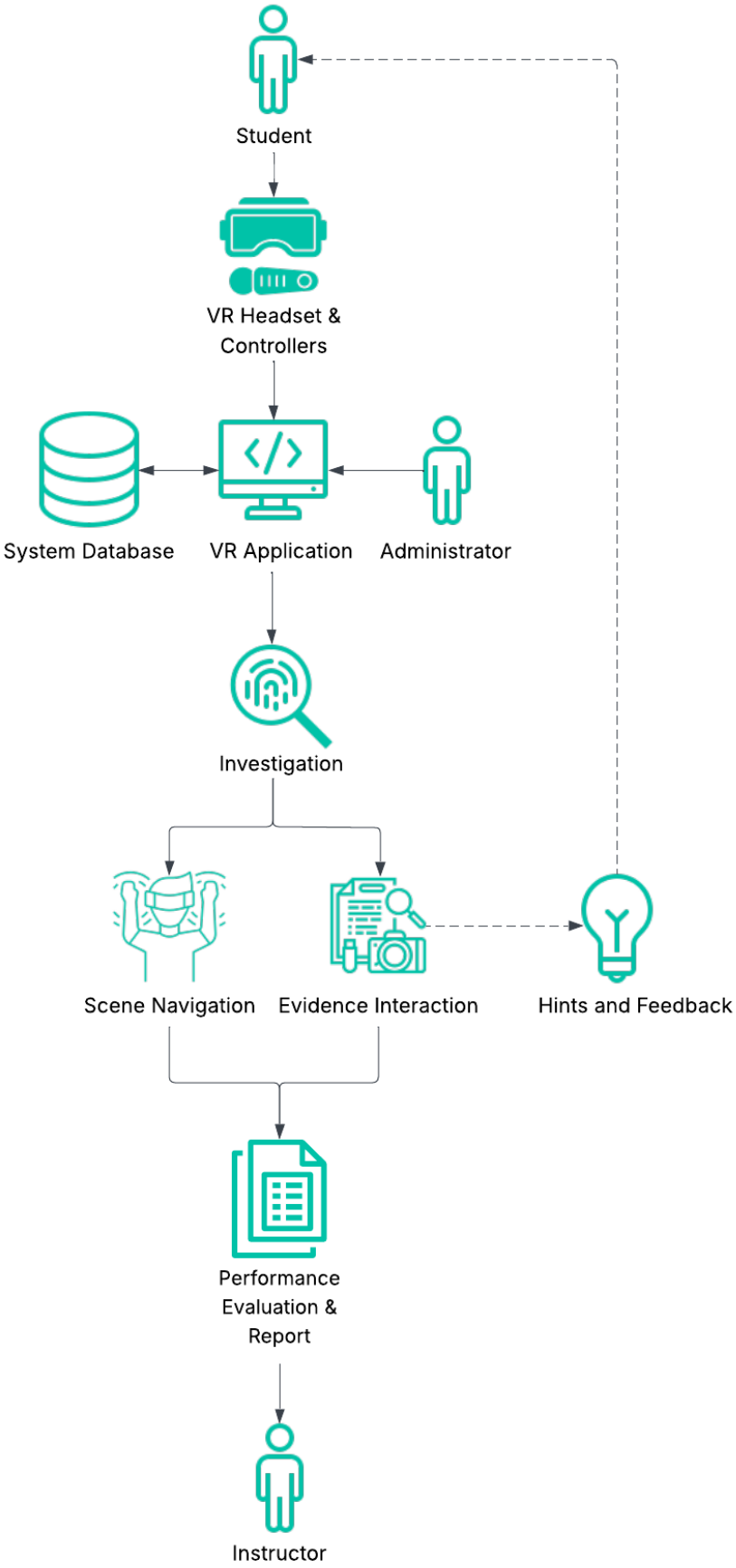


Figure 2.19: System Overview Figure.

2.3 Tools and Technology

The development of the VR-based system relies on a combination of software tools and development frameworks chosen to support realism, interactivity, and cross-platform VR compatibility.

Unity 3D Game Engine will serve as the primary development environment for constructing the virtual crime scenes and implementing all core interactions. Unity offers a flexible editor, physics-based environmental behaviour, and high-fidelity rendering, all of which are essential for accurately representing forensic evidence and scene layouts. Its widespread adoption in VR research and training simulations further supports reproducibility and extensibility.

OpenXR Framework is employed as the underlying interoperability standard to ensure consistent behaviour across various VR devices. By targeting OpenXR instead of proprietary SDKs, the system remains hardware-agnostic, enabling deployment to multiple headsets (Meta, HTC, Windows MR, etc.) without major architectural changes. This is crucial for educational institutions with diverse equipment.

XR Interaction Toolkit provides the foundation for implementing VR-specific interactions such as grabbing objects, teleportation, and controller-based input. The toolkit includes pre-built components that simplify development and enable realistic manipulation of forensic evidence and scene elements.

C# Scripting will be employed to implement all system logic, including collision detection, evidence-handling rules, scoring mechanisms, spatial tracking, and scene progression. The object-oriented structure of C# enables clear encapsulation of forensic actions and facilitates expandability, such as adding new crime scene modules or assessment features.

For 3D modelling and assets both **Blender** and **Unity Asset Store** will be utilized for obtaining, creating, and refining 3D assets, including interior environments, forensic evidence items, and props. Realistic models will contribute directly to the immersion and accuracy of forensic training scenarios. Blender's wide range of features and the Unity Asset Store will provide a variety of assets to be optimized and utilized keeping these key points in mind.

Unity Version Control will be incorportaed to manage project files, track changes, and support collaboration between project members. UVC ensures easy and stable development, testing, and refinement between every team member seamlessly.

Firebase will be employed as the backend service for data storage, authentication, and persistence. Firebase is a cloud-based platform developed by Google that provides a suite of backend services accessible directly from client applications without requiring a dedicated server. Three Firebase services are utilised in this system:

- Firebase Authentication: manages user identity and access control, supporting email and password authentication. It handles session tokens, credential validation, and role-based access enforcement for student, instructors, and administrators without requiring custom authentication infrastructure.

- Cloud Firestore: serves as the primary database, storing scene configurations, evidence definitions, investigation session records, performance reports, and user profiles. Firestore is a NoSQL document-oriented database that organises data into collections of documents, where each document is a flexible JSON-like structure that can contain nested fields and sub-collections. This model suits the system's

data naturally: an investigation session, for example, contains a variable number of evidence interactions, tool usages, and procedural steps that differ between scenes and students, making a fixed relational schema unnecessarily rigid.

- Firebase Storage: is used to store binary assets generated at runtime, specifically the photographs captured by students during the three-stage evidence documentation procedure. These images are stored per session and linked to their corresponding evidence records in Firestore.

A relational database (SQL) was considered but ultimately not chosen. Traditional SQL databases such as PostgreSQL or MySQL require a dedicated backend server to sit between the client application and the database, meaning the team would need to design, implement, and host a custom API layer in addition to the database itself. Firebase eliminates this entirely — its Unity SDK allows the VR client to communicate with Firestore directly, removing the need for custom middleware and significantly reducing development overhead. Furthermore, Firebase Authentication is natively integrated with Firestore, meaning role-based access control for students, instructors, and administrators is handled within a single platform rather than requiring a separate authentication system wired to a custom backend. Firebase also provides real-time data synchronisation, which allows the instructor dashboard to reflect student session data as it is written, without requiring manual refresh or polling logic. Finally, at the scale of this project — a small number of concurrent users in a university setting — Firebase's free tier is sufficient, avoiding the cost and complexity of provisioning and maintaining a dedicated server entirely.

Chapter 3

System Analysis and Design

The proposed system is a Virtual Reality-based simulation designed for forensic science students. The system aims to provide students with an immersive and interactive learning experience by offering a training environment that simulates realistic crime scenes. Within the crime scene students can use the VR headset and controllers to look around, freely navigate the scene and identify potential evidence.

The system supports progressive learning through multiple interaction modes. The base observational mode provides students with an environment where they explore crime scenes through movement and 360-degree viewing without interactive evidence handling. This mode supports both fully modeled 3D environments and 360-degree photographic crime scenes, allowing students to develop scene assessment and spatial awareness skills. The advanced interactive mode unlocks comprehensive forensic interaction, enabling students to perform all investigative procedures including evidence identification, collection, tagging, and documentation.

The system allows users to document and tag evidence items following professional forensic standards, including placement of evidence markers, comprehensive labeling with chain of custody tracking, and three-stage photographic documentation. This implementation simulates real-world crime scene investigation workflows and ensures students learn proper evidence handling procedures.

To enhance replayability and prevent memorization-based learning, the system implements dynamic evidence placement. Each time a student initiates an investigation session for a particular crime scene, evidence items are randomly positioned at predefined spawn points throughout the environment. This randomization ensures that students cannot rely on memorizing evidence locations from previous sessions, promoting genuine investigative skills development including systematic scene searching, critical observation, and evidence recognition based on forensic indicators rather than spatial memory.

The system is intended mainly as a training tool, which it renforces by implementing immediate procedural feedback using visual cues and haptic feedback. However, assessment is also supported providing users with post-session feedback including scoring and detailed evaluation reports that measure both investigative accuracy and proper evidence documentation procedures.

A system administration component enables authorized administrators to manage the crime scene repository, including selecting from pre-existing interactive 3D scenes embedded in the system and uploading new 360-degree recorded scenes, configuring mode settings, and controlling scene availability for students. This administrative foundation supports future expansion of content management and customization capabilities.

Unlike traditional learning methods that rely on videos or static demonstrations, the proposed system emphasizes hands-on learning through interaction and immediate feedback. By combining immersive exploration with performance evaluation and professional documentation standards, the system aims to enhance practical forensic training and better prepare students for real-world crime scene investigations.

3.1 System Model and Architecture

3.1.1 Product Description

System Objectives

- Provide realistic, navigable virtual crime scene environments for user exploration.
- Support multiple interaction modes, including observational-only mode and full interactive forensic investigation mode.
- Enable integration of 360-degree recorded crime scenes for base-level observational training.
- Enable users to identify, collect, tag and record virtual evidence naturally within the virtual environment following professional forensic documentation standards.
- Implement dynamic evidence placement with randomized positioning across multiple spawn points to ensure unique scene configurations for each investigation session.
- Generate performance measures including personal score and evaluation reports.
- Deliver real-time procedural feedback to guide user actions during the investigation process.
- Offer optional guidance mechanisms to assist users during investigations without disrupting immersion.
- Provide system administrators with tools to select from available 3D crime scenes, upload 360-degree recorded scenes, and manage crime scene availability and configuration.

System Main Features

The proposed system's features are all built using VR technology to ensure realistic crime-scene interaction, authentic evidence handling, and hands-on skill development.

- **Multiple Interaction Modes:** The system supports progressive learning through distinct interactive modes. The base mode provides an observational learning experience where students can navigate and observe the scene using natural movement and 360-degree viewing but cannot interact with evidence. This mode is designed for initial familiarization with crime scene assessment and spatial awareness. The advanced mode unlocks all interactive forensic capabilities including evidence collection, tagging, documentation, and analysis, providing comprehensive investigative training.

- **360-Degree Recorded Crime Scenes:** In addition to fully interactive 3D environments, the system supports integration of 360-degree photographic or video-based crime scenes. These recorded scenes can be added by system administrators and are accessible in the base observational mode, allowing students to explore real or staged photographed crime scenes through immersive VR viewing. Students can look around in all directions and move between predefined viewpoints to observe scene details, but without the ability to manipulate objects or collect evidence. This feature provides cost-effective access to diverse crime scenes that would be difficult or impossible to recreate as full 3D environments, supporting observational training and scene assessment skills.

- **Scene Navigation:** Users naturally navigate the scenes using teleportation locomotion, natural head movements and a comfortable 360° inspection of environments and objects using a VR headset.

- **Evidence Observation and Inspection:** Students can grab evidence items, rotate them and closely inspect them using VR controllers. This ensures detailed examination of texture, material and placement.

- **Placement of Evidence Markers:** Students select numbered markers and position them next to relevant items. VR enables realistic placement by allowing the user to kneel, adjust angles, and visually confirm that each marker is correctly positioned.

- **Evidence Photography:** The system allows students to document evidence by photographing them using a virtual camera tool, VR supports realistic camera handling, framing, zooming as needed in a professional forensic setting.

- **Fingerprint Dusting and Lifting:** VR tools simulate brushing surfaces with powder, revealing fingerprints on the surface, and lifting them using virtual adhesive tape. Haptic feedback and visual cues help students learn the delicacy required in fingerprint recovery.

- **UV Light Examination:** Students use a virtual UV flashlight to detect otherwise invisible biological traces, fibers, or surface residues. VR supports natural hand movement and angle adjustment, allowing learners to sweep the light across surfaces, observe fluorescence, and identify hidden evidence that would not be visible under normal lighting conditions.

- **Evidence Collection and Tagging:** The system implements professional evidence tagging and documentation procedures. Users collect evidence using VR grabbing mechanics and place items into appropriate virtual containers. The system provides comprehensive evidence tagging functionality including:

  - Placement of numbered evidence markers (tent cards, flags, or adhesive markers) adjacent to evidence before collection
  - Digital evidence labels containing unique identifiers, item descriptions, exact location coordinates, and collection timestamp
  - Chain of custody tracking that automatically records the student's identification and time of collection

  - Guided three-stage photographic documentation (overview, mid-range with marker, and close-up with scale)
  - Visual and procedural feedback when tagging steps are performed incorrectly or out of sequence

  This comprehensive tagging system mirrors real-world forensic procedures and ensures students learn proper evidence documentation that maintains admissibility standards.

- **Dynamic Evidence Placement:** The system implements randomized evidence placement to enhance educational value and prevent memorization-based learning. Each crime scene is configured with multiple predefined spawn points for each evidence item, representing forensically plausible locations where that type of evidence could realistically appear. When a student initiates an investigation session, the system randomly selects one spawn point for each evidence item, ensuring a unique crime scene configuration for each investigation attempt. This randomization promotes genuine investigative skills by requiring students to conduct systematic scene searches, apply forensic principles to identify evidence, and develop recognition abilities based on visual indicators, material properties, and contextual clues rather than spatial memory from previous sessions. The dynamic placement system maintains forensic realism while significantly increasing replay value and ensuring that repeated practice sessions continue to challenge and develop student capabilities.

- **Procedural Feedback System:** Real-time feedback is delivered through VR controller haptics, helping students recognise errors and reinforcing correct investigative procedures following standard forensic protocols.

- **NPC-Based Hint System:**

  An interactive virtual forensic advisor (NPC) that students can approach to request assistance. Students can ask forensics-related questions or request hints using either voice input or text input through a virtual keyboard. The NPC responds with contextual guidance, offering personalized assistance without breaking immersion.

  AI Hint System Subsystem:

  The AI-based hint system is designed to provide real-time assistance to the user during the investigation process.

  One of the main challenges when integrating systems based on Large Language Models (LLMs) into real-time VR environments is avoiding the issue of system freezing (AI agent blocking), where the system becomes unresponsive while waiting for the model's response.

  To address this issue, the hint system is implemented using an asynchronous communication model. When the user requests a hint, the request is sent to the AI model in the background without interrupting the main system's operation.

  This ensures that the virtual reality environment remains fully interactive, allowing the user to continue navigating and interacting with evidence without interruption.

  While waiting for the response, the NPC performs a "thinking" animation (such as head movements or subtle gestures) to indicate that the system is processing the request, which helps maintain a sense of immersion.

Once the response is received, the NPC presents the hint to the user either as text or voice in a smooth and natural manner.

This approach follows non-blocking design principles commonly used in real-time interactive systems such as games and VR applications.

- Prevents system freezing
- Improves responsiveness
- Enhances the overall user experience

- **Scoring and Evaluation:** Upon completing a scene, the system compiles a summary of collected evidence, decisions made, errors encountered, and procedural accuracy. Students can view their performance report in a VR panel, supporting reflection and instructor assessment.
- **Administrative Scene Management:** System administrators can manage the crime scene repository through an administrative interface. This includes selecting from pre-existing 3D crime scene environments already embedded in the system, uploading and integrating new 360-degree recorded scenes, configuring scene mode settings, and controlling scene availability for student access. This management system provides the foundation for future capabilities such as customized evidence configurations and learning objective alignment.

Operating Environments

Based on manufacturer specifications, Unity/OpenXR documentation, and Firebase requirements:

Hardware Requirements:

- **VR Headsets:** Meta Quest 2/3/Pro, HTC Vive, Valve Index, or OpenXR-compatible devices
- **Controllers:** VR motion controllers with 6DOF tracking and haptic feedback
- **PC (for PC VR) [55]:** Intel i5-4590/AMD Ryzen 5 1500X, 8GB RAM, GTX 1060/RX 580, USB 3.0
- **Network:** 5+ Mbps internet for Firebase synchronization

Software Platform:

- **VR Runtime:** OpenXR 1.0+
- **Framework:** Unity 6000.4 with XR Interaction Toolkit
- **Backend:** Firebase Authentication and Cloud Firestore
- **OS:** Windows 10/11 64-bit

Constraints

Derived from VR industry standards, platform documentation, and usability research:

Technical Constraints:

- **Performance:** Must maintain 90 FPS, limiting 3D model complexity and physics calculations
- **Hardware:** Standalone headset processing power restricts simultaneous high-polygon objects
- **OpenXR:** Cross-platform compatibility may limit platform-specific features

Development Constraints:

- **Unity:** Constrained by engine capabilities and VR SDK compatibility
- **Firebase:** Free tier limits on operations, storage, and connections
- **Assets:** 3D scene creation requires specialized skills and time
- **360° Media:** Requires professional equipment; large file sizes

Usability Constraints:

- **Motion Sickness:** Interaction design avoids rapid movements and acceleration
- **Session Duration:** VR fatigue limits sessions to 20-30 minutes
- **Learning Curve:** First-time users need tutorials and intuitive design
- **Physical Space:** Requires adequate room for safe VR interaction

Deployment Constraints:

- **Connectivity:** Requires internet for authentication and synchronization
- **Hardware:** Institutional deployment requires VR headset

User and System Requirements

The following requirements define what the system must enable users to do (user requirements, UR) and the corresponding technical implementations that satisfy them (system requirements, SR). Each system requirement is listed as a subset of the user requirement it directly fulfills.

- **UR-1:** Users must be able to select between different interaction modes before starting an investigation.
  - **SR-1:** The system shall implement multiple interaction modes, including base observational mode and advanced interactive mode, with appropriate feature restrictions for each level.
- **UR-2:** Users must be able to navigate virtual crime scenes using teleportation-based movement and natural head motion.

  - **SR-3:** The system shall provide teleportation-based locomotion and 360° head tracking compatible with VR headsets.

- **UR-3:** Users must be able to explore 360-degree recorded crime scenes with full viewing freedom in base observational mode.
  - **SR-2:** The system shall support integration and display of 360-degree photographic or video-based crime scenes for observational learning in base observational mode.
  - **SR-17:** The system shall support both fully interactive 3D crime scene environments and 360-degree recorded crime scenes within the same framework.

- **UR-4:** Users must be able to inspect the crime scene environment and objects from multiple angles in a 360° view.
  - **SR-3:** The system shall provide teleportation-based locomotion and 360° head tracking compatible with VR headsets. (The 360° head tracking component of SR-3 directly enables multi-angle scene inspection.)

- **UR-5:** Users must be able to pick up, rotate, and closely examine evidence items (in advanced interactive mode).
  - **SR-4:** The system shall support real-time interaction with virtual objects using VR controllers, allowing evidence to be grabbed, rotated, and inspected (in advanced interactive mode).

- **UR-6:** Users must be able to place numbered evidence markers accurately next to relevant items in the scene.
  - **SR-5:** The system shall provide selectable numbered evidence markers (tent cards, flags, adhesive markers) and enable precise placement through VR hand tracking.

- **UR-7:** Users must be able to photograph evidence using a virtual camera with appropriate framing and zoom, following three-stage documentation procedures.
  - **SR-6:** The system shall include a virtual camera tool that supports three-stage photographic documentation: overview, mid-range with marker, and close-up with scale.

- **UR-8:** Users must be able to dust for fingerprints and lift them using correct forensic procedures.
  - **SR-7:** The system shall simulate fingerprint dusting and lifting using virtual forensic tools with visual and haptic feedback.

- **UR-9:** Users must be able to examine surfaces using a virtual UV light to detect hidden biological traces or residues.
  - **SR-8:** The system shall include a virtual UV light tool that reveals fluorescent evidence when the light is applied to compatible surfaces at appropriate distances.

- **UR-10:** Users must be able to collect evidence, store it in appropriate containers, and attach comprehensive digital evidence tags including all required documentation fields.
  - **SR-9:** The system shall allow users to place collected evidence into appropriate virtual containers.
  - **SR-10:** The system shall implement comprehensive digital evidence tagging including unique identifiers, item descriptions, location coordinates, timestamps, and chain of custody information.
  - **SR-11:** The system shall validate evidence tagging procedures and provide feedback when steps are performed incorrectly or out of sequence.
  - **SR-15:** The system shall support randomized placement and selection of evidence elements within predefined crime scene configurations to ensure scenario variability each time a session is initiated.
- **UR-11:** Users must receive immediate feedback on their investigative actions and evidence tagging procedures.
  - **SR-12:** The system shall deliver real-time procedural feedback using visual indicators and VR controller haptics.
- **UR-12:** Users must be able to interact with a virtual forensic advisor NPC to ask forensics-related questions and request guidance using voice input or text input during the investigation.
  - **SR-13:** The system shall implement a virtual forensic advisor NPC that accepts user questions via voice input or text input and generates contextual responses to forensics-related questions and hints using a large language model (LLM) provided with scene state, user progress, and current task information.
- **UR-13:** Users must be able to view a performance summary and evaluation after completing a crime scene investigation.
  - **SR-14:** The system shall generate a post-investigation performance report summarising collected evidence, tagging accuracy, procedural errors, and overall score, displayed within a VR interface.
- **UR-14:** System administrators must be able to select from pre-existing 3D interactive crime scenes and upload new 360-degree recorded scenes to the system.
  - **SR-16:** The system shall provide an administrative interface that allows authorized administrators to select from pre-existing 3D crime scenes, upload new 360-degree recorded scenes, configure mode settings, and manage scene availability.
  - **SR-17:** The system shall support both fully interactive 3D crime scene environments and 360-degree recorded crime scenes within the same framework. (SR-17 is also required by UR-3.)
- **UR-15:** System administrators must be able to configure scene mode settings and manage scene availability for students.

- **SR-16:** The system shall provide an administrative interface that allows authorized administrators to select from pre-existing 3D crime scenes, upload new 360-degree recorded scenes, configure mode settings, and manage scene availability. (SR-16 also satisfies UR-14.)

Non-Functional Requirements

1. The system shall be portable across major consumer VR headsets by supporting the OpenXR standard.
2. The system shall enforce user authorization and role-based access to prevent unauthorized access to features and user data.
3. The system shall support usability by allowing users to start an investigation in three interactions or less from the main menu.
4. The system shall support comfortable VR usage by employing interaction and movement techniques that minimize user discomfort during investigation sessions.
5. The system shall provide maintainability by allowing adding new scenes without requiering modification to system components and applicaiton logic.
6. The system shall demonstrate reliability during investigation sessions, minimizing unexpected interruptions or crashes under normal usage conditions.
7. The system shall be compatible with commonly used VR hardware components and input devices supported by the target platforms.
8. The system shall support scalability by allowing additional crime scene scenarios to be added without degrading system performance.
9. The system shall maintain acceptable performance by loading available scenes and initializing investigation sessions within a few seconds under normal operating conditions.
10. The system shall target a stable frame rate of 90 frames per second to ensure smooth and comfortable VR interaction during investigation sessions.

3.2 Functional Decomposition

3.2.1 Actors

The system includes the following actors:

Student

The primary user of the system. Students interact with the system by navigating the crime scene environment, identifying and documenting evidence, receiving real-time feedback, and reviewing score/ performance evaluations after completing an investigation.

Instructor

A secondary system user. Instructors review and assess generated student preformance evaluation reports. However, instructors do not directly interact with the virtual crime scene during investigations.

System Administrator

Manages the crime scene database and controls the availability of crime scenes for student access. The system administrator can select from pre-existing 3D interactive crime scenes already embedded in the system and configure their availability for students. Additionally, administrators can upload and integrate new 360-degree recorded crime scenes for observational learning. Administrators configure scene interaction modes, manage scene availability for students, and maintain the scene repository. This role provides the foundation for future expansion of administrative capabilities such as customizing evidence placement, setting learning objectives per scene, and analyzing aggregate student performance across multiple scenes.

LLM Service An external AI service actor representing the Large Language Model (LLM) backend integrated into the NPC-based hint system. The LLM receives structured contextual data — including current scene state, student progress, and the specific investigative task — and generates forensics-specific guidance in response to student queries. It does not initiate interactions; it responds exclusively when invoked by the student through UC-11.

3.2.2 Use Cases

The following use cases describe the primary interactions between system actors and the VR forensic training system. Use cases are organized by actor and include relationship specifications («include», «extend», and generalization) to illustrate dependencies and extensions. Detailed use case specifications including preconditions, postconditions, and alternative flows are provided in Appendix A.

Common Use Cases

1. **UC-0: Login and Authenticate**

   User (Student, Instructor, or Administrator) enters credentials to authenticate and access the system. The system validates credentials and grants access based on user role and permissions.

   Related Requirements: (Implied security requirement)

   Actors: Student, Instructor, Administrator

   Postcondition: User is authenticated; role is determined; main menu is displayed

Student Use Cases

2. **UC-1: Select Mode and Scene**

   Student selects the appropriate mode (base observational or advanced interactive) and crime scene before starting an investigation.

   Related Requirements: UR-1, SR-1

   Precondition: Student is logged in (UC-0)

   Postcondition: Investigation session is initialized; scene begins loading

3. **UC-2: Navigate Crime Scene**

Student explores the crime scene using teleportation-based movement and natural head motion for 360° viewing.

Related Requirements: UR-2, SR-3

Precondition: Crime scene has been loaded (UC-1)

Note: This use case is «included» in all investigation activities

Concurrency Notes: The session timer runs continuously during navigation. The FeedbackSystem monitors user movement in the background. UC-11 (Request Hints) can be triggered at any time without interrupting navigation.

4. **UC-3: Explore 360-Degree Recorded Scene**

   Student navigates and observes a 360-degree photographic crime scene in base observation mode, moving between predefined viewpoints and looking in all directions without object interaction.

   Related Requirements: UR-3, SR-2

   Precondition: Base observational mode selected; 360-degree recorded scene loaded (UC-1)

   Note: Generalization of UC-2 for recorded scenes; restricted to base observational mode

5. **UC-4: Observe Evidence**

   Student visually observes and identifies potential evidence items without physical interaction. This initial observation enables students to recognize evidence that requires documentation and collection, following proper forensic protocol of not disturbing evidence before it is properly documented.

   Related Requirements: UR-4, SR-4

   Precondition: Advanced interactive mode selected; student is navigating the scene (UC-2)

   Concurrency Notes: The session timer runs continuously during observation. The student may request guidance via UC-11 at any point without interrupting the observation activity. Observation actions are logged concurrently by the system for performance tracking.

6. **UC-5: Place Evidence Markers**

   Student selects and places numbered evidence markers (tent cards, flags, or adhesive markers) adjacent to relevant evidence items following proper forensic sequencing procedures.

   Related Requirements: UR-6, SR-5

   Precondition: Evidence has been observed (UC-4)

   Note: This use case is «included» in UC-6 (Photograph Evidence)

   Concurrency Notes: The session timer continues running while placing markers. The student may invoke UC-11 (Request Hints) during marker placement without interrupting the task, Additionally, since UC-5 is included within UC-6, marker placement may occur while the photography session is already active.

7. **UC-6: Photograph Evidence**

   Student uses a virtual camera to document evidence following the three-stage photographic procedure: overview shot, mid-range shot with evidence marker visible, and close-up shot with measurement scale.

   Related Requirements: UR-7, SR-6

   Precondition: Evidence marker has been placed (UC-5)

   Note: This use case «includes» UC-5 (Place Evidence Markers)

   Concurrency Notes: The session timer runs concurrently across all photography stages. The student may request a hint via UC-11 during photography without aborting the current stage. Additional investigative actions such as UC-7 (UV Light) and UC-8 (Fingerprint Detection) may be initiated while the photography process remains active.

8. **UC-7: Examine Evidence Using UV Light**

   Student uses a virtual UV light to reveal hidden biological traces, fibers, or fluorescent residues on surfaces and objects.

   Related Requirements: UR-9, SR-8

   Precondition: Evidence has been observed (UC-4); evidence marker placed (UC-5)

   Note: «extends» UC-6 (Photograph Evidence) for optional specialized examination before physical contact with evidence

9. **UC-8: Recover Fingerprints**

   Student performs fingerprint recovery procedures using virtual dusting powder and lifting tape, with haptic feedback to simulate proper technique.

   Related Requirements: UR-8, SR-7

   Precondition: Evidence has been observed (UC-4); evidence marker placed (UC-5)

   Note: «extends» UC-6 (Photograph Evidence) for optional specialized processing before physical contact with evidence

10. **UC-4a: Inspect Evidence Physically**

    Student picks up, rotates, and closely examines evidence items using VR controllers. This physical inspection occurs after proper documentation has been completed.

    Related Requirements: UR-5, SR-4

    Precondition: Evidence has been photographed (UC-6); optional specialized examinations (UC-7, UC-8) completed if applicable

    Note: Precedes physical collection; must follow UC-6 and any optional UC-7 or UC-8

    Concurrency Notes: The session timer continues during inspection. The EvaluationMetrics system logs interaction time and actions concurrently.

11. **UC-9: Collect Evidence**

    Student physically collects evidence items from the crime scene, storing them in appropriate virtual containers following proper documentation.

Related Requirements: UR-10, SR-9

Precondition: Evidence has been documented and photographed (UC-6); physical inspection (UC-4a) may be performed

Note: «includes» UC-2 (Navigate Crime Scene)

12. **UC-10: Tag Evidence**

    Student attaches comprehensive digital evidence tags to collected items immediately upon collection, including unique identifiers, descriptions, location coordinates, timestamps, and chain of custody information.

    Related Requirements: UR-10, SR-10, SR-11

    Precondition: Evidence is being collected (UC-9); marker placement (UC-5) and photography (UC-6) already completed

    Note: Tagging occurs immediately during or after collection to maintain chain of custody

13. **UC-11: Request Hints and Guidance**

    Student approaches and interacts with the virtual forensic advisor NPC to ask forensics-related questions or request context-aware guidance. The student can input questions using either voice input or text input via a virtual keyboard. The NPC responds with contextual answers and hints generated by an LLM that is provided with the current scene state, user progress, and task details.

    Related Requirements: UR-12, SR-13

    Precondition: Student is in an active investigation session (UC-1 initiated)

    Note: «extends» all investigative use cases as optional assistance, «includes» UC-17 (Generate Contextual Forensic Response) — every hint request triggers the AI Advisor to generate a response

    Concurrency Notes: This use case executes concurrently with any active investigative use case (UC-4 through UC-10). The current task is not interrupted. The LLM service processes requests asynchronously while the student continues interaction.

14. **UC-12: Complete Investigation**

    Student explicitly completes and submits the investigation session, triggering the generation of performance evaluation and ending the active investigation.

    Related Requirements: SR-14

    Precondition: Investigation has been started (UC-1)

    Postcondition: Performance report is generated (UC-13)

15. **UC-13: View Performance Evaluation**

    Student views a post-investigation performance summary displayed in VR, including score, evidence identification accuracy, tagging correctness, procedural errors, and detailed evaluation metrics.

    Related Requirements: UR-13, SR-14

    Precondition: Investigation has been completed (UC-12)

    Postcondition: Performance report is available for instructor review (UC-14)

Instructor Use Cases

15. **UC-14: Review Student Performance Reports**

    Instructor accesses and reviews student performance evaluation reports from completed investigations, including investigation summaries, scores, procedural assessments, and detailed metrics for individual students or entire classes.

    Related Requirements: UR-13, SR-14

    Precondition: Instructor is logged in (UC-0); at least one student has completed an investigation

    Actors: Instructor

    Note: Similar to UC-13 but with extended access to all student reports and aggregate statistics

Administrator Use Cases

16. **UC-15: Manage Crime Scene Repository**

    System administrator selects from available pre-existing 3D interactive crime scenes embedded in the system and uploads new 360-degree recorded photographic scenes to expand the training repository.

    Related Requirements: UR-14, SR-16, SR-17

    Precondition: Administrator is logged in (UC-0)

    Actors: Administrator

    Note: Precedes scene configuration (UC-16)

17. **UC-16: Configure Scene Settings**

    System administrator configures mode settings for crime scenes and manages scene availability for students.

    Related Requirements: UR-15, SR-16

    Precondition: Administrator is logged in (UC-0); crime scenes exist in repository (UC-15)

    Actors: Administrator

    Note: Follows repository management (UC-15); configuration takes effect for new sessions

LLM Service Use Cases

18. **UC-17: Generate Contextual Forensic Response**

    The LLM receives a structured prompt containing the student's question, current scene state, evidence found so far, and the active investigative step. It generates a contextually appropriate forensic hint or explanation and returns it to the NPC system for display to the student.

    Related Requirements: SR-13

    Precondition: Student has submitted a query via UC-11

Postcondition: A contextual response is returned and displayed through the NPC interface

Actors: LLM Service

Note: «include» relationship with UC-11 — UC-11 includes UC-17, since generating a response is a necessary part of every hint request

Use Case Relationships

The system's use cases have several key relationships that define dependencies and extensions:

«include» Relationships:

- UC-6 (Photograph Evidence) «includes» UC-5 (Place Evidence Markers) - Evidence markers must be placed before photographic documentation
- UC-9 (Collect Evidence) «includes» UC-2 (Navigate Crime Scene) - Evidence collection requires scene navigation
- UC-11 (Request Hint and Guidance) «includes» UC-17 (Generate Contextual Forensic Response) - every hint request triggers the LLM service to generate a response

«extend» Relationships:

- UC-7 (Examine Evidence Using UV Light) «extends» UC-6 (Photograph Evidence) - UV examination is an optional specialized technique performed during the documentation phase before physical contact
- UC-8 (Recover Fingerprints) «extends» UC-6 (Photograph Evidence) - Fingerprint recovery is an optional specialized technique performed during the documentation phase before physical contact
- UC-11 (Request Hints and Guidance) «extends» all investigation use cases - Hints provide optional assistance when needed

Generalization Relationships:

- UC-3 (Explore 360-Degree Recorded Scene) generalizes UC-2 (Navigate Crime Scene) - Specialized navigation for photographic scenes
- UC-13 (View Performance Evaluation - Student) and UC-14 (Review Student Performance Reports - Instructor) share similar functionality but with different access levels and scope

Sequential Dependencies:

- UC-1 (Select Interaction Mode and Scene) → UC-2 through UC-12 (Investigation activities) → UC-12 (Complete Investigation) → UC-13 (View Performance Evaluation)
- UC-4 (Observe Evidence) → UC-5 (Place Evidence Markers) → UC-6 (Photograph Evidence) → UC-4a (Inspect Evidence Physically) → UC-9 (Collect Evidence) + UC-10 (Tag Evidence) - Evidence handling follows a mandatory forensic procedural sequence
- Optional: UC-7 (UV Light) or UC-8 (Fingerprints) may extend UC-6 before proceeding to UC-4a

3.2.3 Use Cases Diagram

The use case diagram for the VR crime scene investigation system presents a high-level view of the system's functionality from the perspective of its users. It identifies the primary actors and the main tasks they perform while interacting with the system. This diagram defines the system boundary and clarifies how each actor engages with the VR application, focusing on what the system provides rather than how it is implemented. As part of requirements analysis, the use case diagram ensures that the system's functionality aligns with educational and training objectives and serves as a foundation for subsequent behavioral and interaction diagrams.

Figure 3.1 shows the system's comprihensive use cases diagram illustrating the interactions between system actors and functionalities.

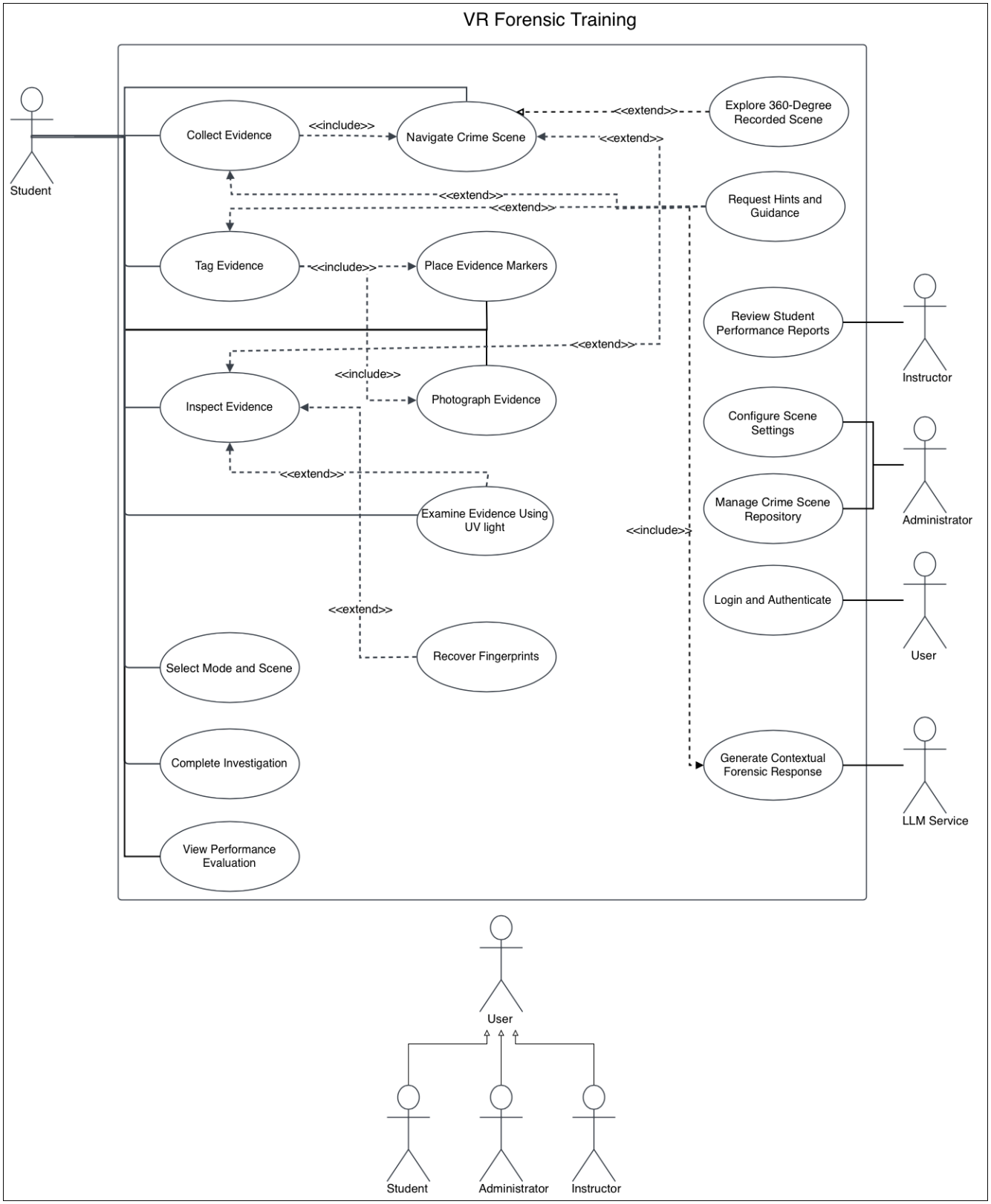


Figure 3.1: Use Case Diagram showing interactions between actors and system functionality.

3.3 System Models

3.3.1 Class Diagram

The class diagram presents the structural design of the VR forensic training system, illustrating the relationships between users, crime scenes, evidence, forensic tools, and evaluation components. The diagram employs inheritance hierarchies for extensibility, composition for complex relationships, and follows object-oriented design principles to support the system's core functionality.

The complete class diagram is organized into three complementary views that work together to present the entire system architecture. Figure 3.2 provides the **System Overview**, showing the high-level architecture and how major subsystems interconnect. This top-level view establishes the relationships between user roles, core domain entities, and system infrastructure components.

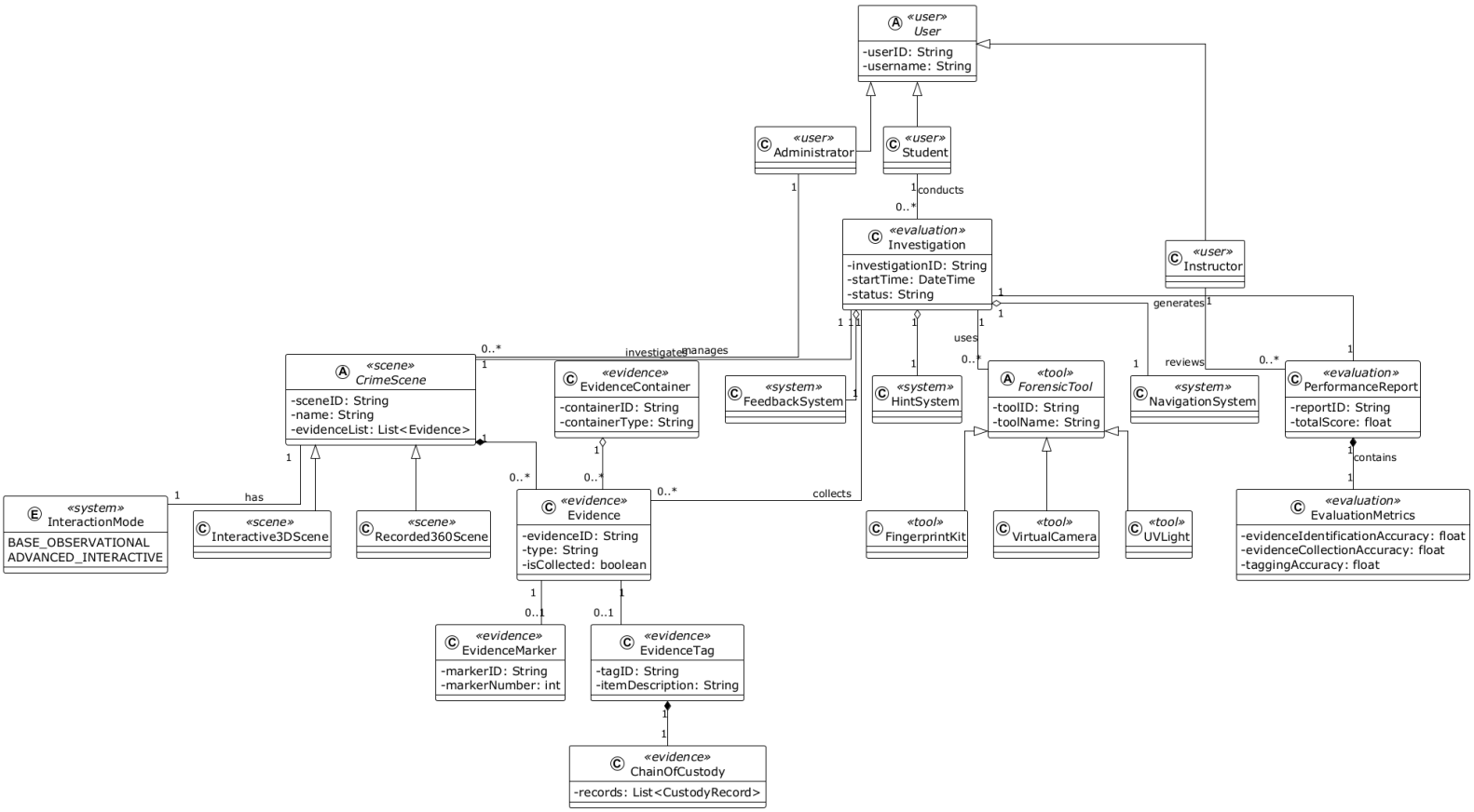


Figure 3.2: System Overview - Class Diagram

Figure 3.3 presents the **Domain Model**, focusing on the forensic investigation domain entities including crime scenes, evidence, forensic tools, and their detailed relationships. This diagram captures the essential forensic concepts and workflows that the system must support.

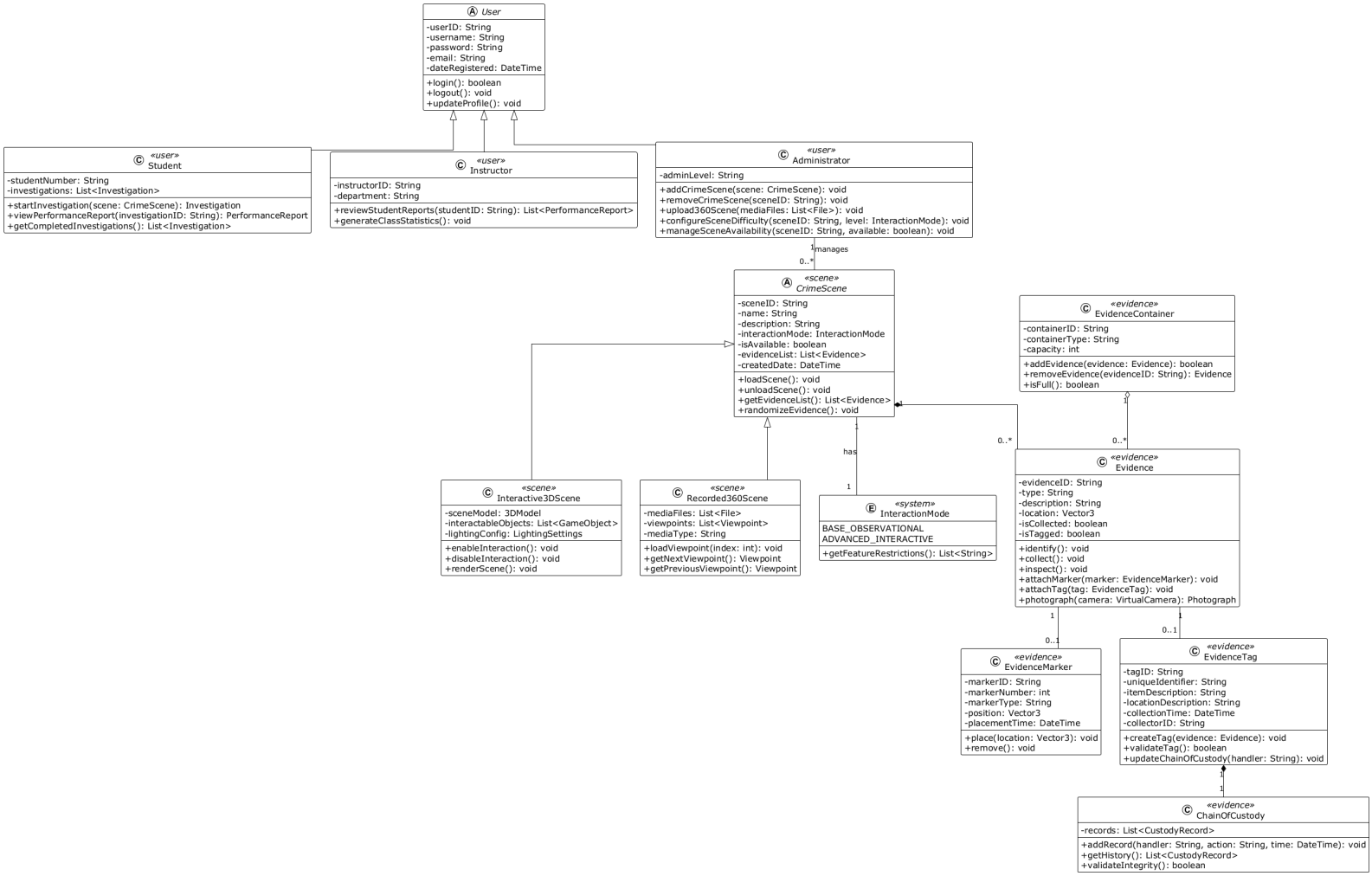


Figure 3.3: Domain Model - Class Diagram

Figure 3.4 illustrates the **Investigation System**, detailing how investigations are conducted, tracked, and evaluated, including the performance assessment and reporting mechanisms.

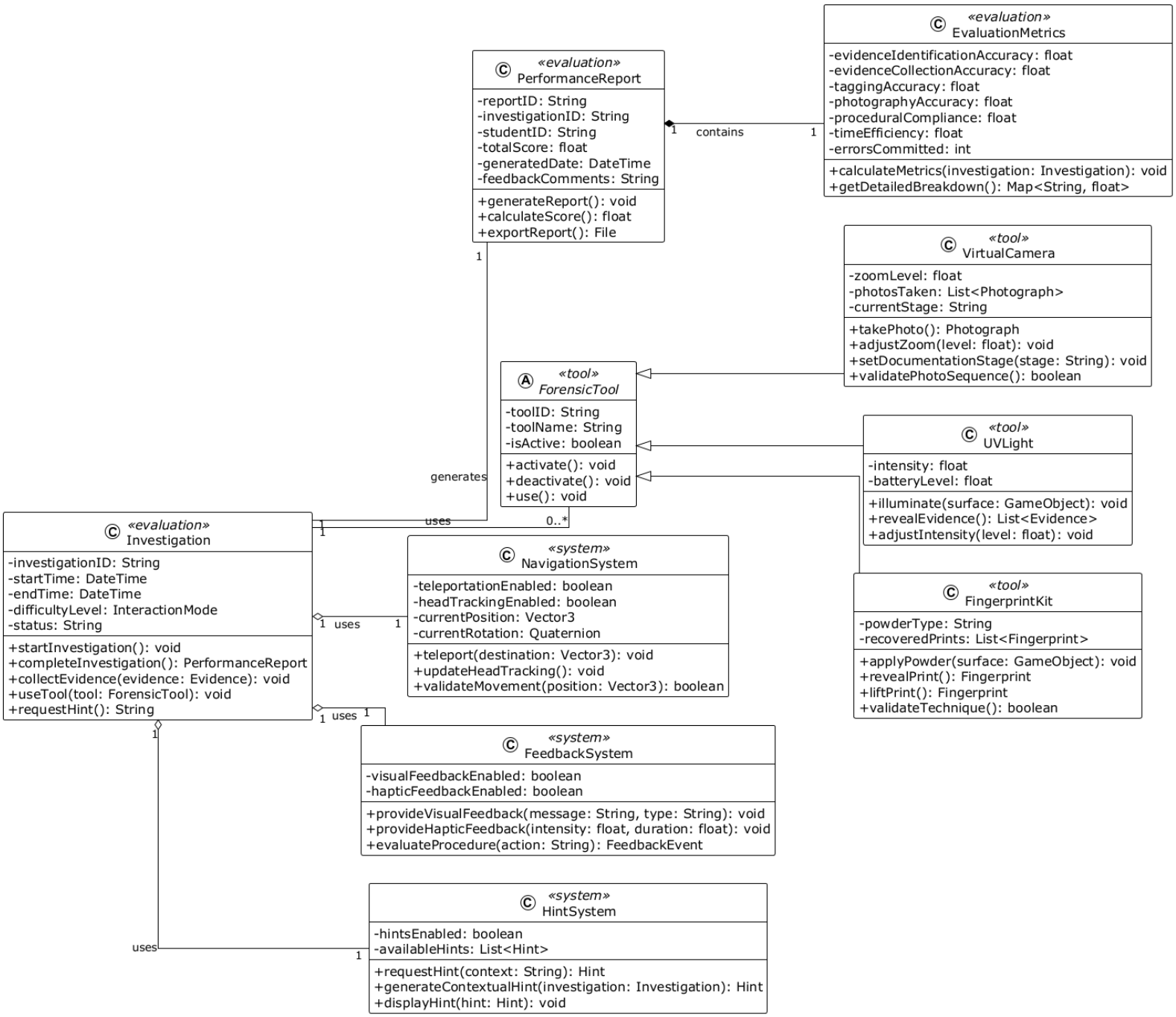


Figure 3.4: Investigation System - Class Diagram

Together, these three diagrams provide a complete structural specification of the system. The System Overview establishes the architectural foundation and user interactions, the Domain Model defines the core forensic entities and their behaviors, and the Investigation System specifies how these domain entities are orchestrated during actual training sessions. This decomposition allows each diagram to focus on specific aspects of the system while maintaining clear relationships between components across all three views.

Core Domain Classes

User Hierarchy:

- `User` (abstract): Base class for all system users with common attributes (userID, username, password).
- `Student`: Inherits from User. Represents the primary system user who conducts investigations. Contains methods for investigation activities and maintains a list of completed investigations.
- `Instructor`: Inherits from User. Reviews student performance reports and evaluations.
- `Administrator`: Inherits from User. Manages crime scene repository and system configuration.

Crime Scene Hierarchy:

- `CrimeScene` (abstract): Base class for all crime scene types. Contains common attributes (sceneID, name, description, interaction modes) and methods for scene management.
- `Interactive3DScene`: Fully modeled 3D environment supporting complete forensic interaction including evidence collection and manipulation.
- `Recorded360Scene`: 360-degree photographic or video-based scenes for observational learning, containing viewpoints and media files.

Evidence Management:

- `Evidence`: Represents physical evidence items with attributes including type, location, description, and collection status. Supports identification, collection, and inspection operations.
- `EvidenceMarker`: Numbered markers placed adjacent to evidence items following forensic documentation standards.
- `EvidenceTag`: Comprehensive digital documentation attached to collected evidence, including unique identifier, description, location, timestamp, and chain of custody information.
- `EvidenceContainer`: Virtual storage containers for collected evidence items, categorized by type (e.g., paper bags, plastic bags, sealed boxes).
- `ChainOfCustody`: Tracks complete handling history of evidence items, maintaining forensic integrity.

Investigation and Evaluation:

- `Investigation`: Represents a single investigation session conducted by a student, tracking start time, end time, associated crime scene, and all collected evidence.
- `PerformanceReport`: Generated after investigation completion, containing comprehensive evaluation metrics, scores, and procedural accuracy assessments.

- `EvaluationMetrics`: Detailed metrics measuring investigation quality including evidence identification accuracy, tagging correctness, procedural compliance, and time efficiency.

Forensic Tools Hierarchy

- `ForensicTool` (abstract): Base class for all virtual forensic tools with common attributes (toolID, toolName, isActive) and interaction methods.
- `VirtualCamera`: Implements three-stage photographic documentation (overview, mid-range, close-up) with zoom and framing capabilities.
- `UVLight`: Simulates ultraviolet light examination for detecting biological traces and fluorescent evidence.
- `FingerprintKit`: Provides fingerprint dusting and lifting functionality with powder application and tape lifting simulation.

System Support Classes

- `Interaction Mode`: Enumeration defining investigation modes (Base Observational, Advanced Interactive) with associated feature restrictions.
- `NavigationSystem`: Manages user movement through teleportation and head tracking within virtual environments.
- `FeedbackSystem`: Delivers real-time procedural feedback through visual cues and haptic responses.
- `HintSystem`: Provides context-aware guidance and optional assistance during investigations.

Key Relationships

- Student conducts multiple Investigations (1 to many).
- Investigation is associated with one CrimeScene and generates one PerformanceReport.
- CrimeScene contains multiple Evidence items (composition).
- Evidence can be associated with EvidenceMarker, EvidenceTag, and stored in EvidenceContainer.
- Investigation uses multiple ForensicTools during evidence processing.
- FeedbackSystem and HintSystem support Investigation activities.

3.3.2 Sequence Diagram

The sequence diagram describes how the different components of the VR forensic system interact during a single investigation scenario, focusing on the order in which actions and system responses occur. It shows how a student initiates an investigation, how the system authenticates the user, loads the selected crime scene, and enables interaction with evidence objects. The diagram also illustrates communication between the VR application, internal logic modules (such as evidence handling and scoring), and backend services responsible for retrieving scene data and storing investigation results. By modeling these interactions in a time-ordered manner, the sequence diagram clarifies component responsibilities and ensures that system behavior follows the intended investigation flow. Figure 3.5 shows the sequence of interactions between the student and system components during a VR investigation.

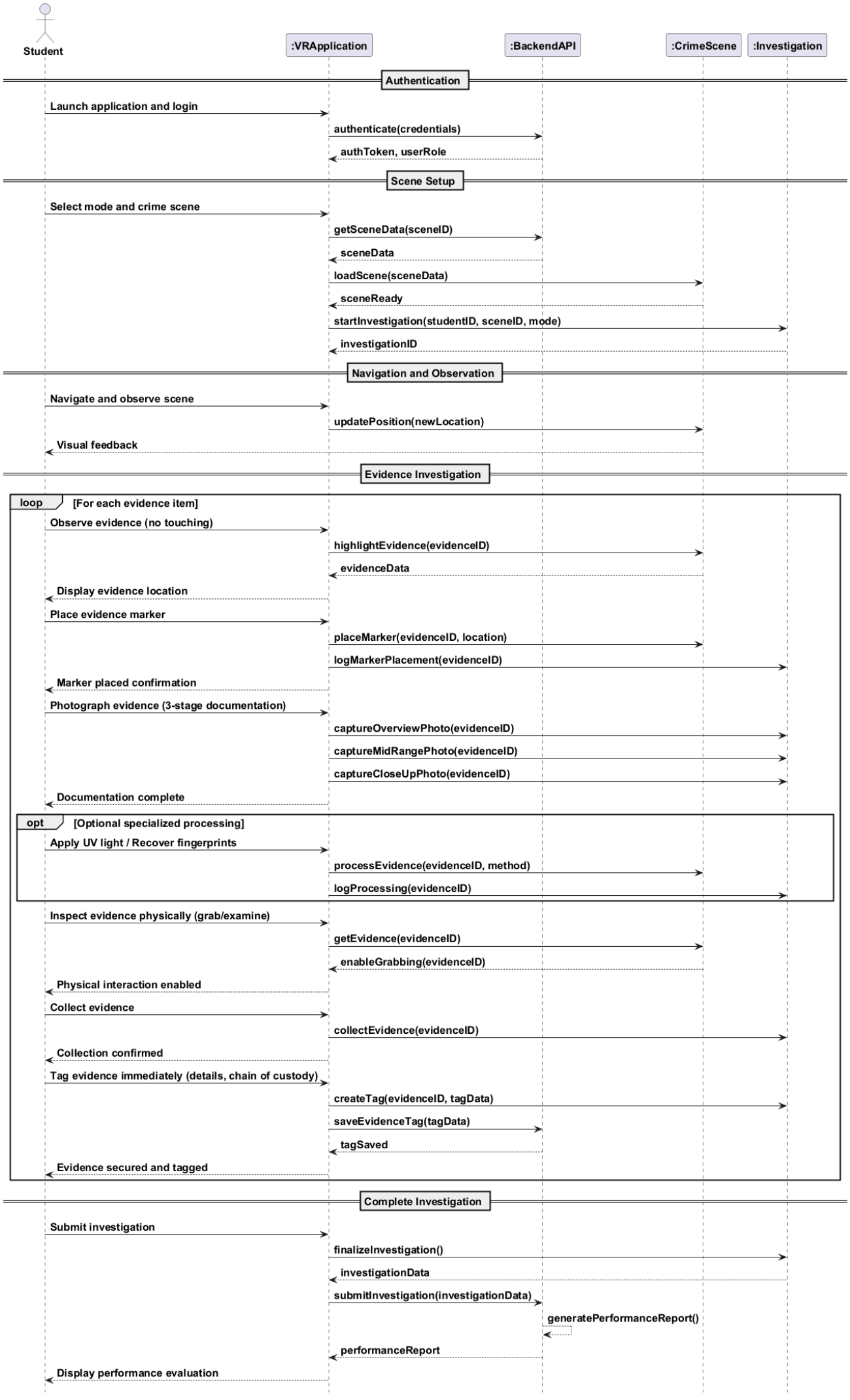


Figure 3.5: Sequence diagram of the investigation flow.

3.3.3 Activity Diagram

The activity diagram describes the flow of actions involved in performing a virtual crime scene investigation in the VR system. It outlines the steps a student follows, starting from launching the application and selecting a crime scene, then navigating the environment, inspecting and tagging evidence, and finally completing the investigation. Decision points are used to represent alternative paths, such as switching between interaction modes or revisiting evidence. This diagram clarifies how investigation tasks progress within the system and how different actions are logically connected. Figure 3.6 shows the workflow within the VR forensic system.

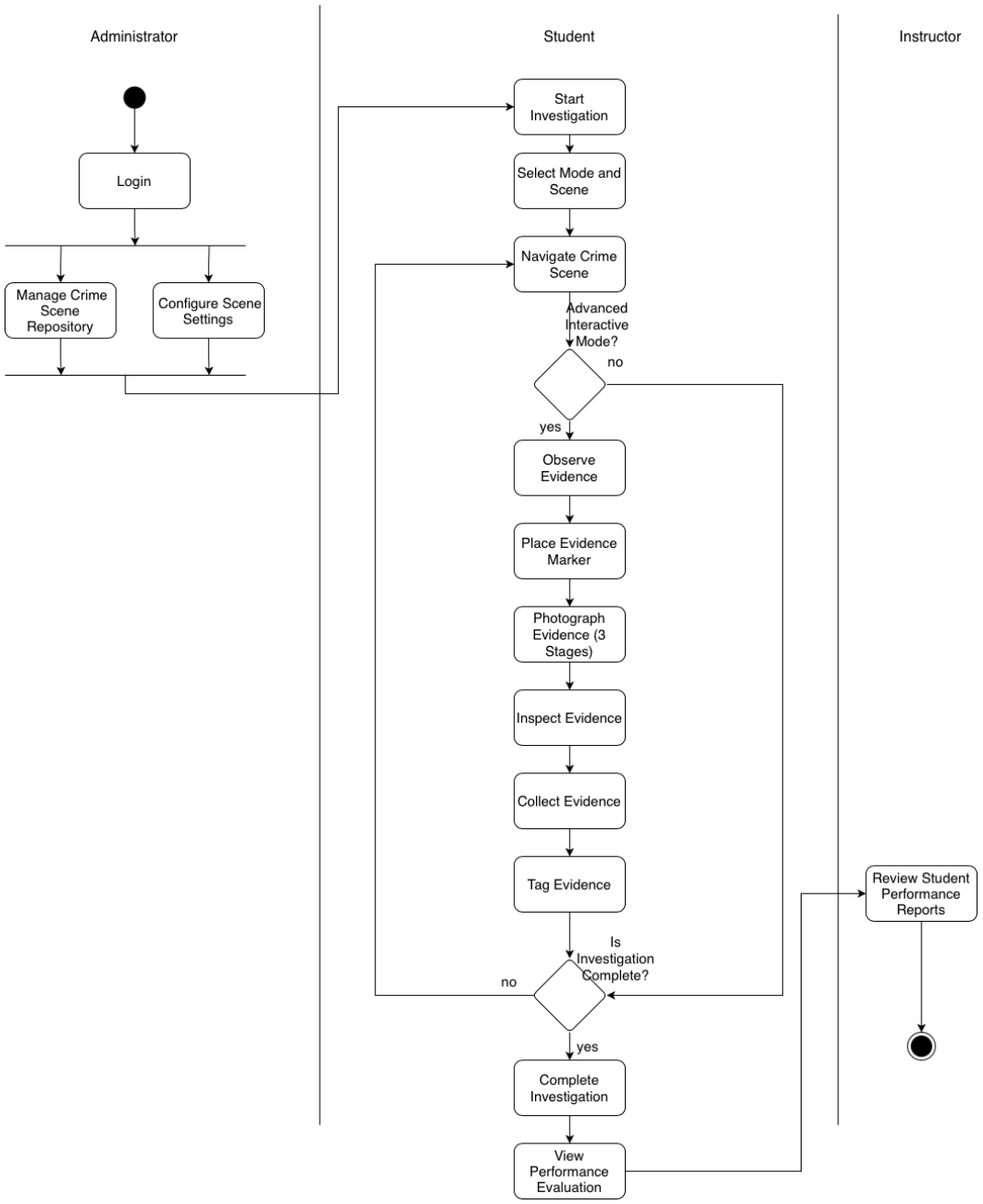


Figure 3.6: Activity diagram of the system workflow.

3.3.4 State Chart Diagram

State chart diagrams are used to model system behavior by defining the states a system or object can be in and the transitions between those states. They help ensure that actions are performed only when the system is in a valid state. In this project, state charts are used to control investigation flow and enforce correct system behavior during user interaction.

Figure 3.7 illustrates the overall system statechart of the application. The diagram models the system behavior from application launch and user authentication to role-based navigation for students, instructors, and administrators. It shows how students initiate and complete investigation sessions, how instructors access and review reports, and how administrators manage and configure crime scenes. This statechart provides a high-level view of application flow and clarifies how different user roles interact with the system.

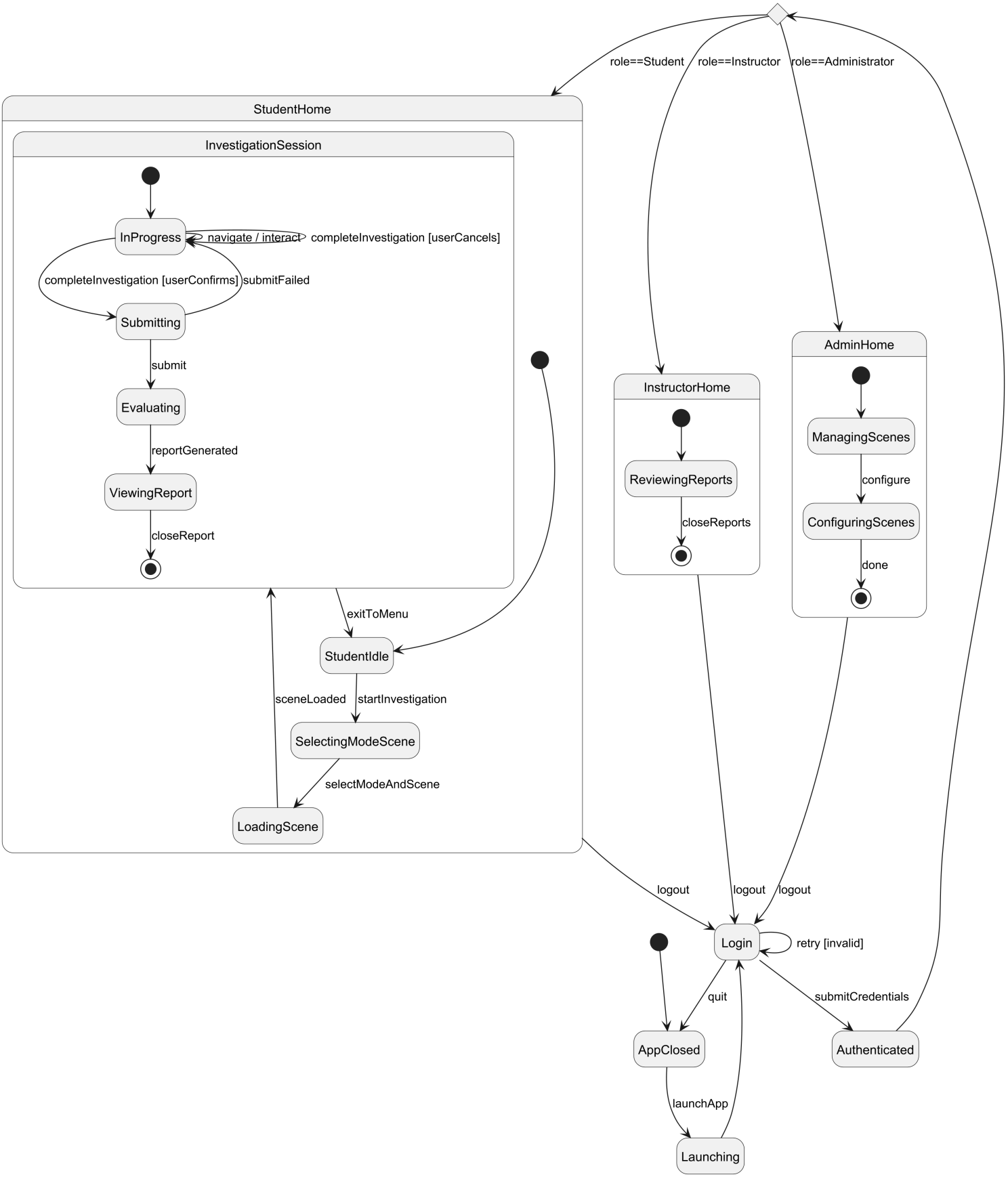


Figure 3.7: Overall system statechart of the application.

Figure 3.8 presents the high-level investigation statechart, which models the behavior of a single investigation session. The diagram shows the main phases of an investigation from initialization through completion, including the navigation and evidence handling states. The Active state contains the primary investigation workflow, with a reference to the detailed evidence handling flow shown separately in Figure 3.9. This simplified view focuses on the overall investigation progression, while the detailed evidence handling procedures are documented in the separate evidence handling state diagram.

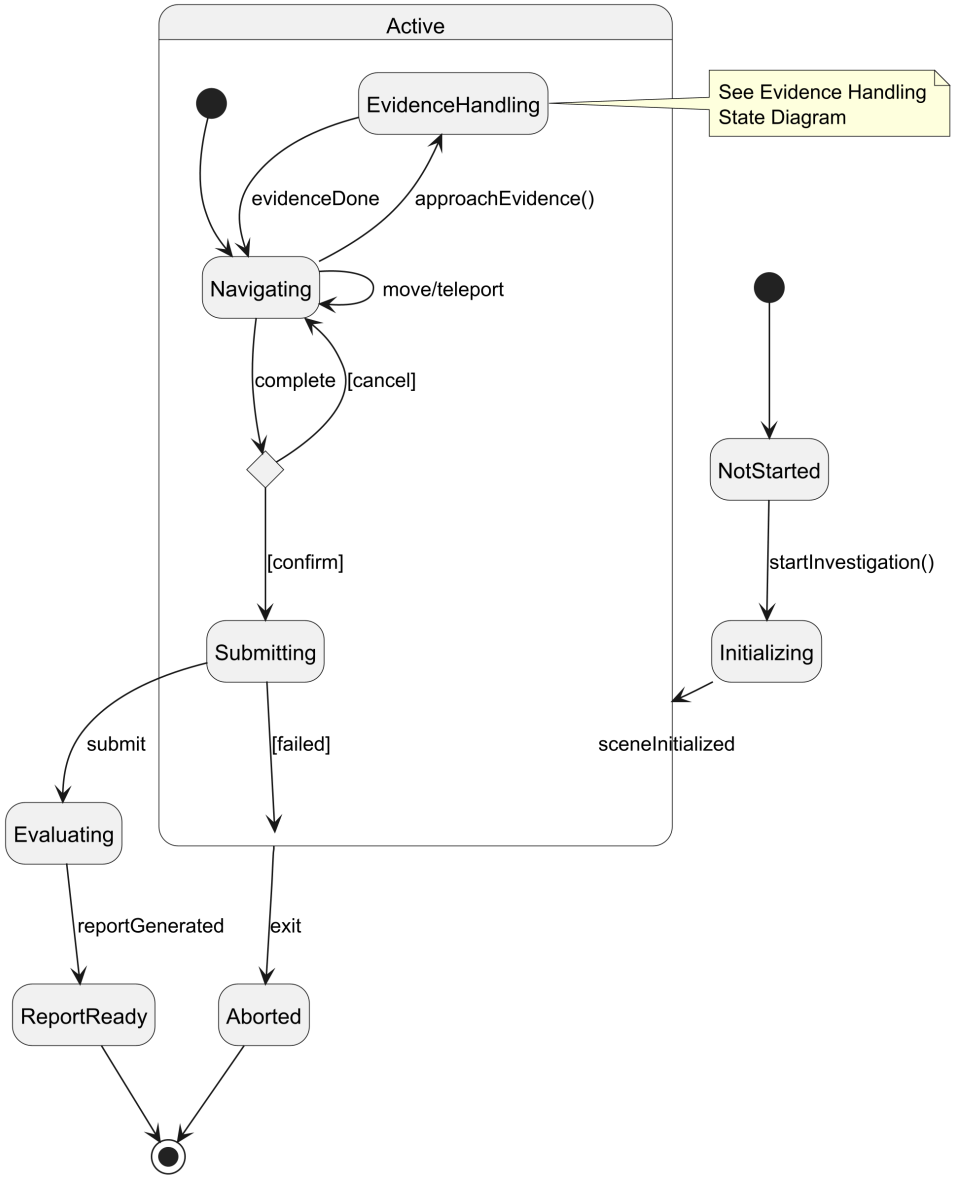


Figure 3.8: High-level investigation statechart showing main investigation phases.

Figure 3.9 presents the detailed evidence handling statechart, which models the complete workflow for handling individual evidence items during an investigation. The diagram shows the progression through inspection, optional specialized examination techniques (UV light and fingerprint recovery), collection, storage, marker placement, three-stage photographic documentation, and comprehensive evidence tagging. It includes validation states to ensure proper tagging procedures are followed before evidence handling is considered complete. This statechart provides detailed guidance for the evidence handling subsystem and ensures students follow correct forensic procedures.

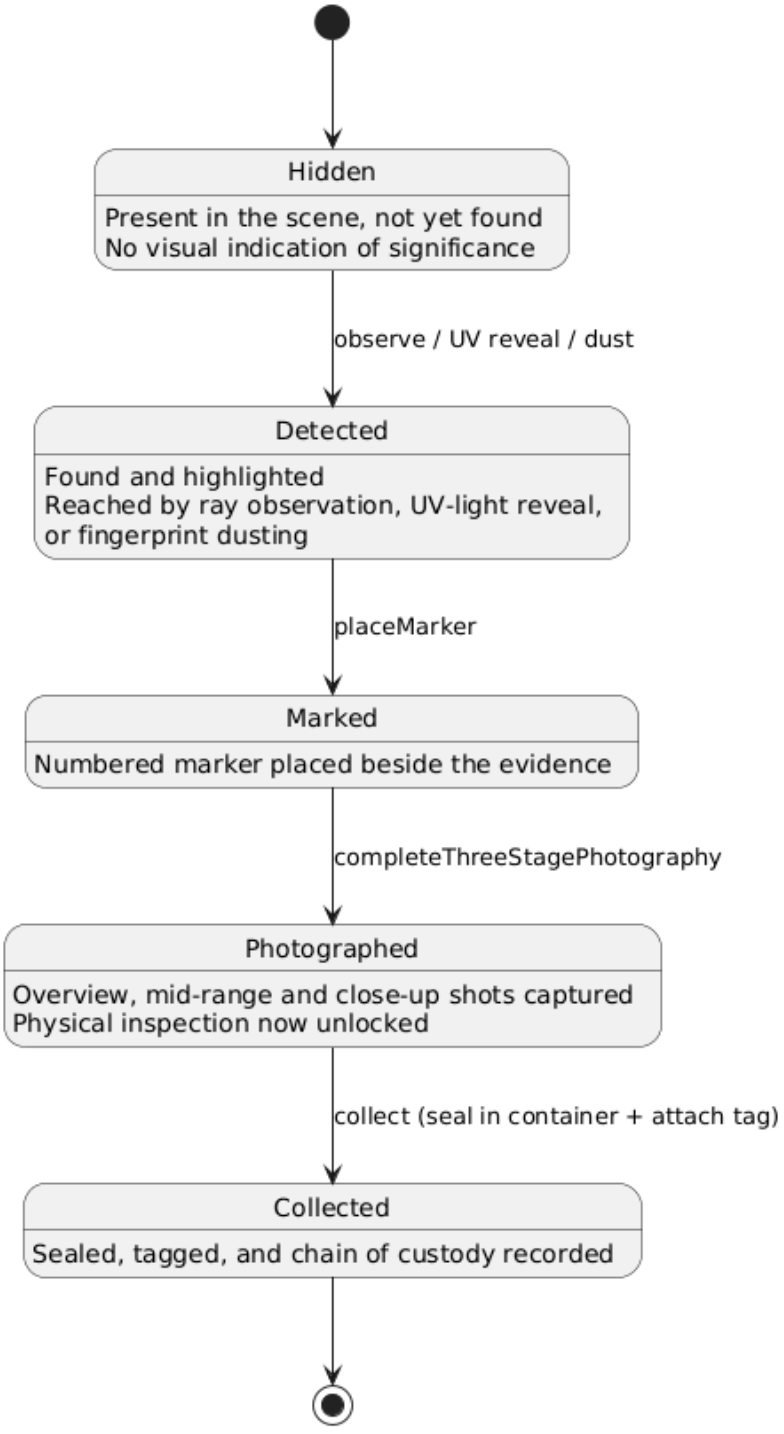


Figure 3.9: Evidence handling statechart showing detailed evidence handling workflow.

3.4 System Architecture

3.4.1 Sub-System

The system is composed of 8 sub-systems, each responsible for a specific function.

VR Interaction Subsystem

Handles all VR hardware input/output and user interaction within the virtual environment.

Components:

- `InputHandler`: Processes controller inputs including triggers, grips, thumbsticks, and buttons.
- `TeleportationManager`: Manages locomotion via teleportation, and handles smooth transitions to prevent motion sickness.
- `HandController`: Tracks hand positions in 3D space, enables grabbing and releasing of objects, and manages realistic hand poses for natural interaction.
- `HapticFeedbackController`: Provides tactile feedback for user interactions to enhance immersion.
- `GazeController`: Tracks head orientation for look-based interactions and UI targeting.

Figure 3.10 illustrates the VR Interaction Subsystem component diagram.

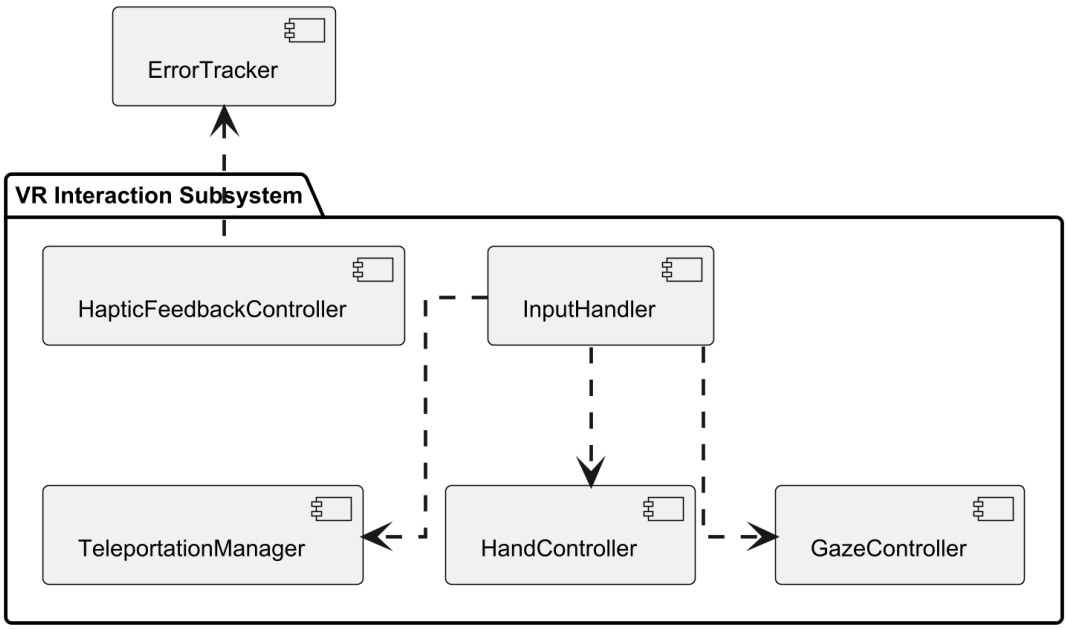


Figure 3.10: VR Interaction Subsystem Component Diagram

Scene Management Subsystem

Loads, manages, and transitions between crime scenes and environments, coordinating mode configuration and environmental settings.

Components:

- `SceneLoader`: Asynchronously loads scene assets and configurations while managing loading progress and resource allocation.
- `ModeController`: Manages Base Observational versus Advanced Interactive modes, enabling or disabling features according to selected mode.
- `SceneConfiguration`: Stores scene metadata including evidence locations, investigation objectives, and difficulty settings.
- `SpawnPointManager`: Manages student spawn locations and initial orientations for consistent investigation starts.

Figure 3.11 illustrates the Scene Management Subsystem component diagram.

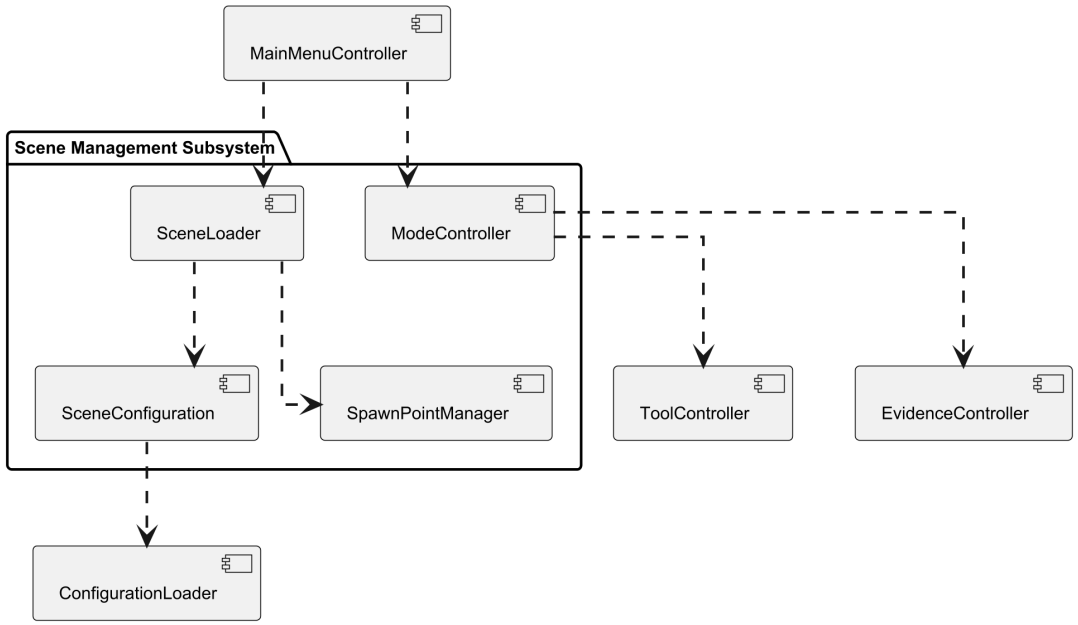


Figure 3.11: Scene Management Subsystem Component Diagram

Evidence Management Subsystem

Core domain subsystem managing all evidence-related operations including discovery, inspection, collection, and state tracking.

Components:

- `EvidenceRegistry`: Maintains a catalog of all evidence items in the scene and tracks their discovery status.
- `EvidenceController`: Manages the evidence state machine (Hidden, Detected, Marked, Photographed, Collected) and enforces state transitions.

- `InspectionHandler`: Manages grab, rotate, and examine interactions with zoom functionality for detailed observation.
- `CollectionManager`: Validates and processes evidence collection requests, enforcing procedural requirements such as PPE usage.
- `ContainerManager`: Manages evidence containers and validates container-evidence compatibility.

Figure 3.12 illustrates the Evidence Management Subsystem component diagram.

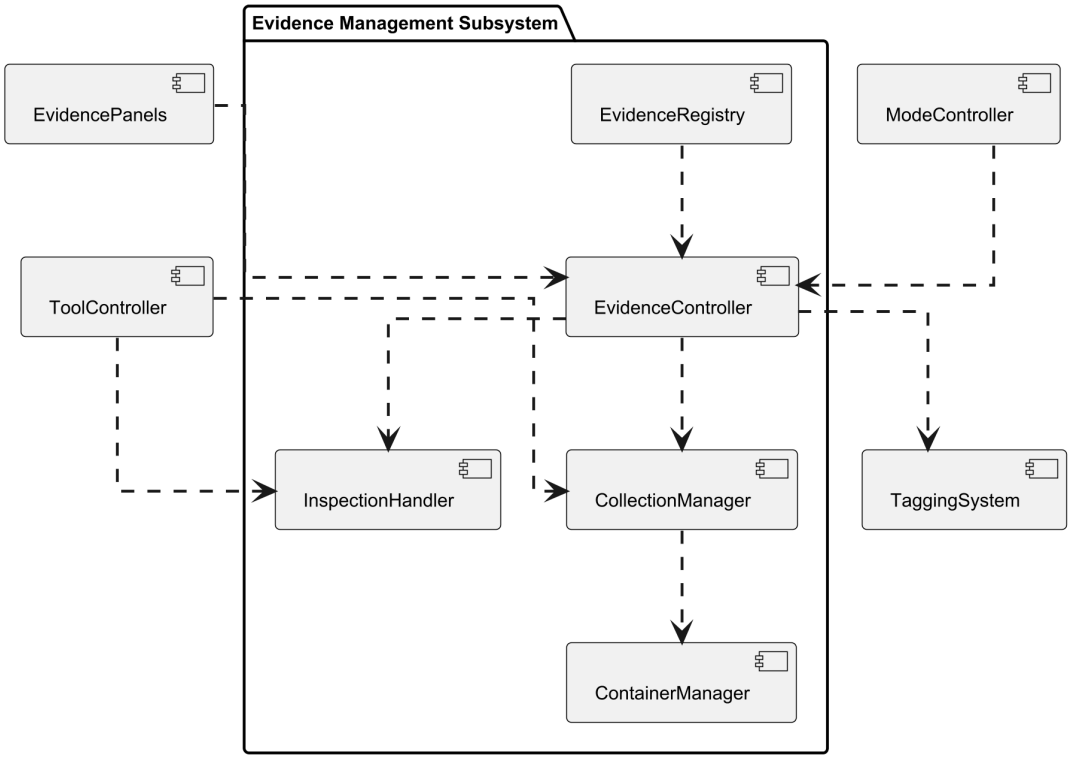


Figure 3.12: Evidence Management Subsystem Component Diagram

Documentation Subsystem

Handles all forensic documentation procedures including photography, evidence marking, tagging, and chain of custody tracking.

Components:

- `VirtualCamera`: Captures photographs with viewfinder rendering, zoom control, and flash functionality.
- `PhotographyValidator`: Validates three-stage photography completion (overview, mid-range, close-up) and checks framing requirements.
- `EvidenceMarkerSystem`: Places and manages numbered markers, tracking marker-evidence associations.
- `MeasurementScaleTool`: Provides scale references for close-up photography with multiple scale sizes.

- `TaggingSystem`: Creates and attaches digital evidence tags with comprehensive metadata entry.
- `ChainOfCustodyLogger`: Tracks evidence handling timeline and records all custody transfers.

Figure 3.13 illustrates the Documentation Subsystem component diagram.

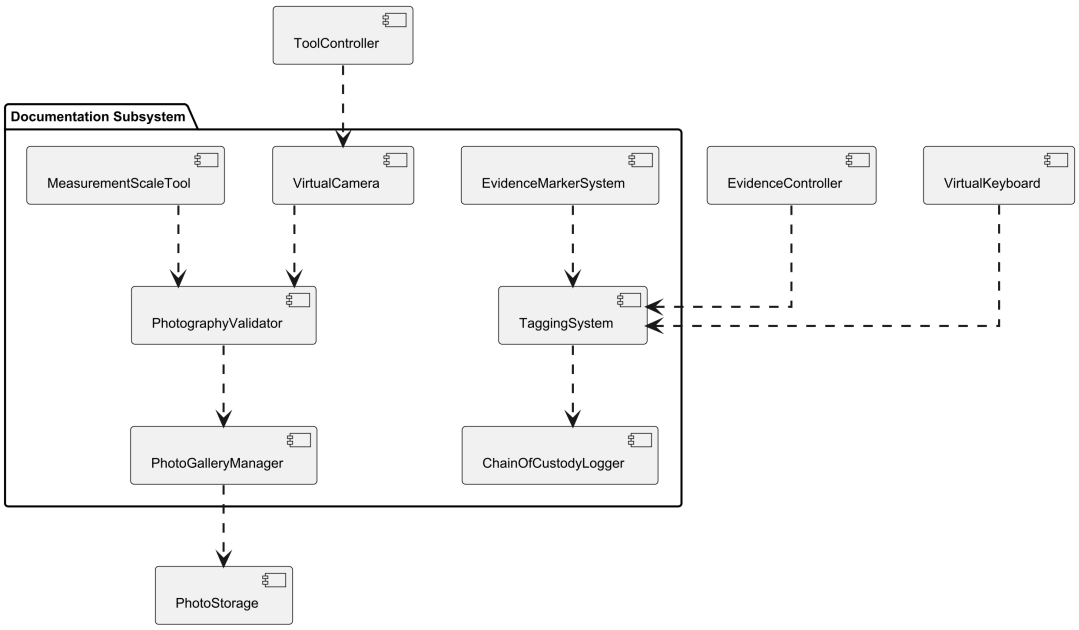


Figure 3.13: Documentation Subsystem Component Diagram

Tool Management Subsystem

Manages the inventory, selection, and usage of virtual forensic tools required for investigation procedures.

Components:

- `InventoryManager`: Maintains the student's tool inventory and tracks available/equipped tools.
- `ToolSelector`: Handles tool selection from radial menus or inventory interfaces.
- `ToolController`: Manages tool behavior, activation, and interaction with evidence.

Supported Tools: VirtualCamera, UVLight, FingerprintKit, EvidenceMarker, MeasurementScale, EvidenceContainers, PPE (gloves, booties)

Figure 3.14 illustrates the Tool Management Subsystem component diagram.

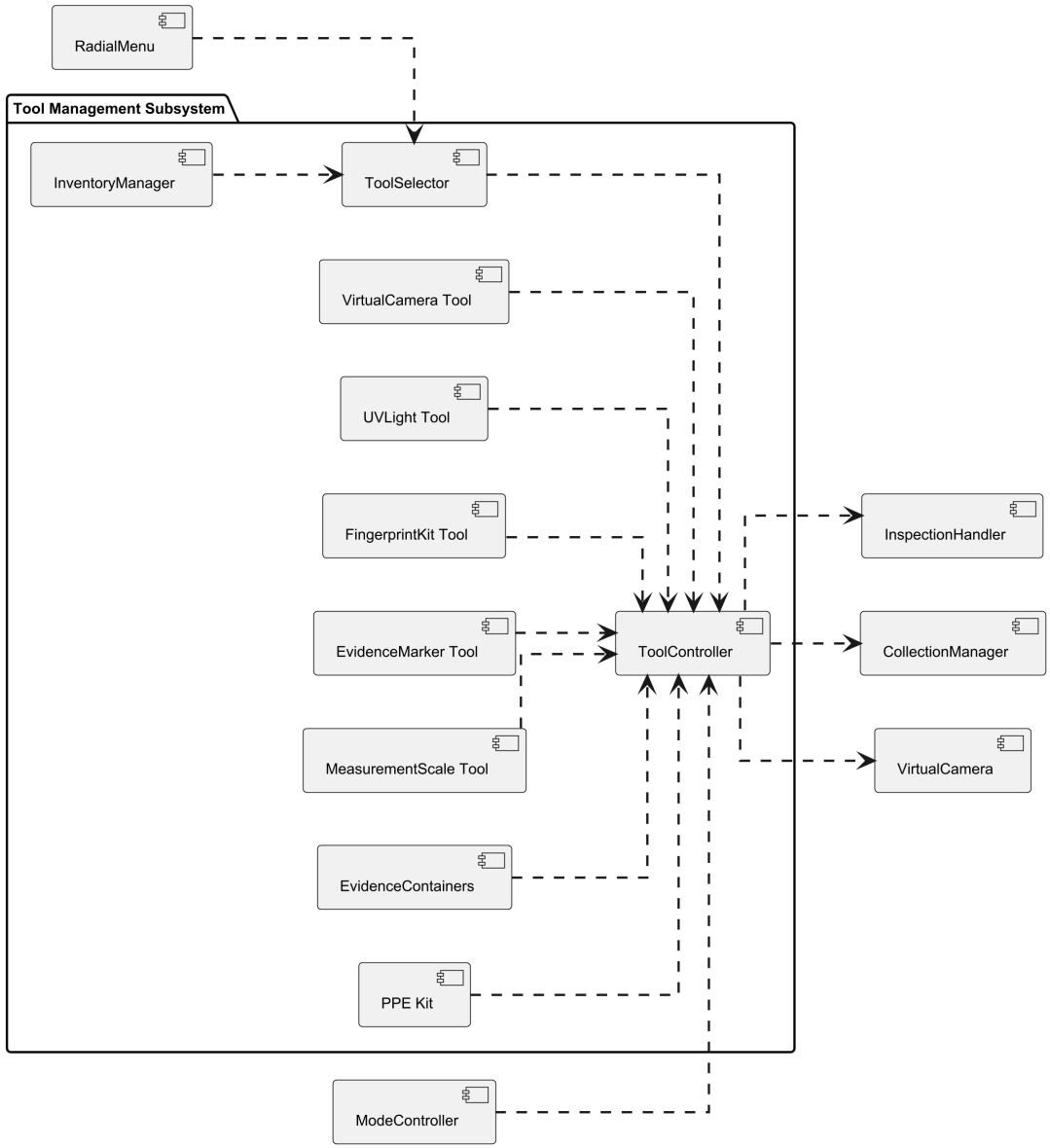


Figure 3.14: Tool Management Subsystem Component Diagram

User Interface Subsystem

Provides all user interface elements including menus, forms, HUD displays, and feedback presentation in the VR environment.

Components:

- `MainMenuController`: Handles scene selection, mode selection, and investigation initiation.
- `HeadsUpDisplay (HUD)`: Shows investigation status, objectives, timer, and collected evidence count.
- `EvidencePanels`: Displays detailed evidence information during inspection.
- `VirtualKeyboard`: Provides text input capability within the VR environment.
- `RadialMenu`: Implements radial selection menus for tool selection and quick actions.

- `NotificationSystem`: Displays procedural feedback, warnings, and success messages.
- `TutorialOverlay`: Provides contextual guidance and tutorial information when enabled.

Design Principles: Minimal HUD clutter, spatial UI placement, clear visual hierarchy, accessibility considerations, and audio confirmation feedback

Figure 3.15 illustrates the User Interface Subsystem component diagram.

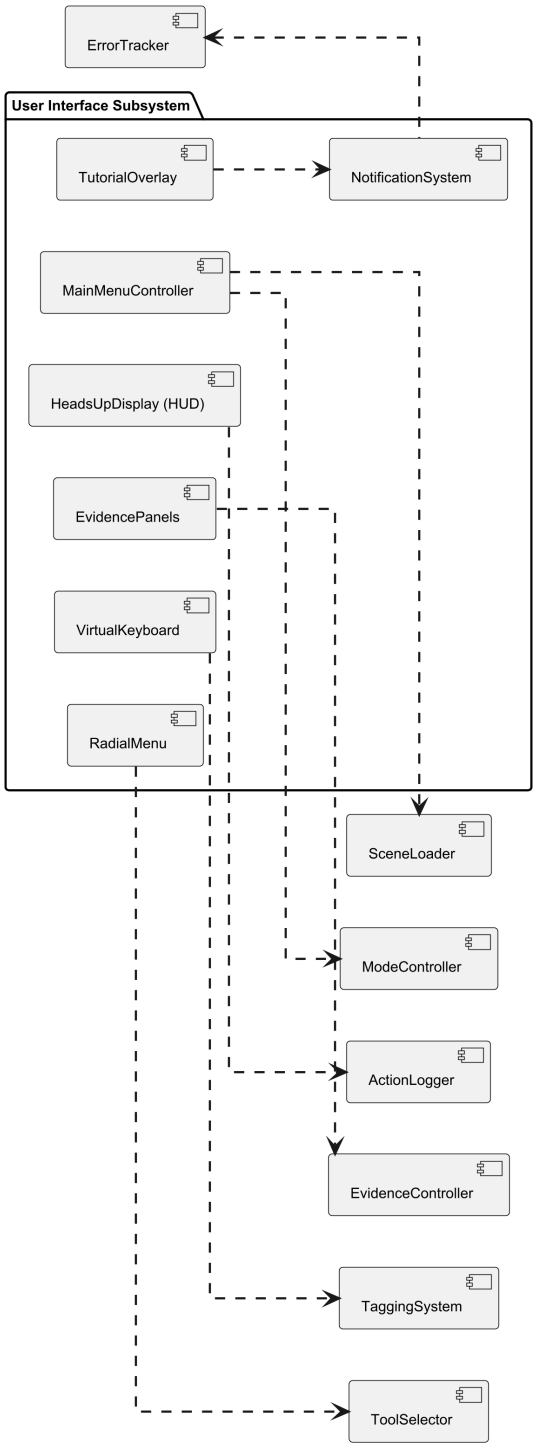


Figure 3.15: User Interface Subsystem Component Diagram

Evaluation Subsystem

Tracks student actions, calculates performance scores, identifies errors, and generates comprehensive evaluation reports.

Components:

- `ActionLogger`: Records all student actions with timestamps for complete investigation replay capability.
- `ErrorTracker`: Identifies and classifies procedural errors (critical, moderate, minor, warnings).
- `ScoreCalculator`: Computes scores across multiple dimensions.
- `ReportGenerator`: Creates performance reports with visualizations and detailed feedback.

Evaluation Metrics: Evidence identification accuracy, collection procedural compliance, documentation completeness, contamination incidents, time efficiency, and adherence to forensic best practices

Figure 3.16 illustrates the Evaluation Subsystem component diagram.

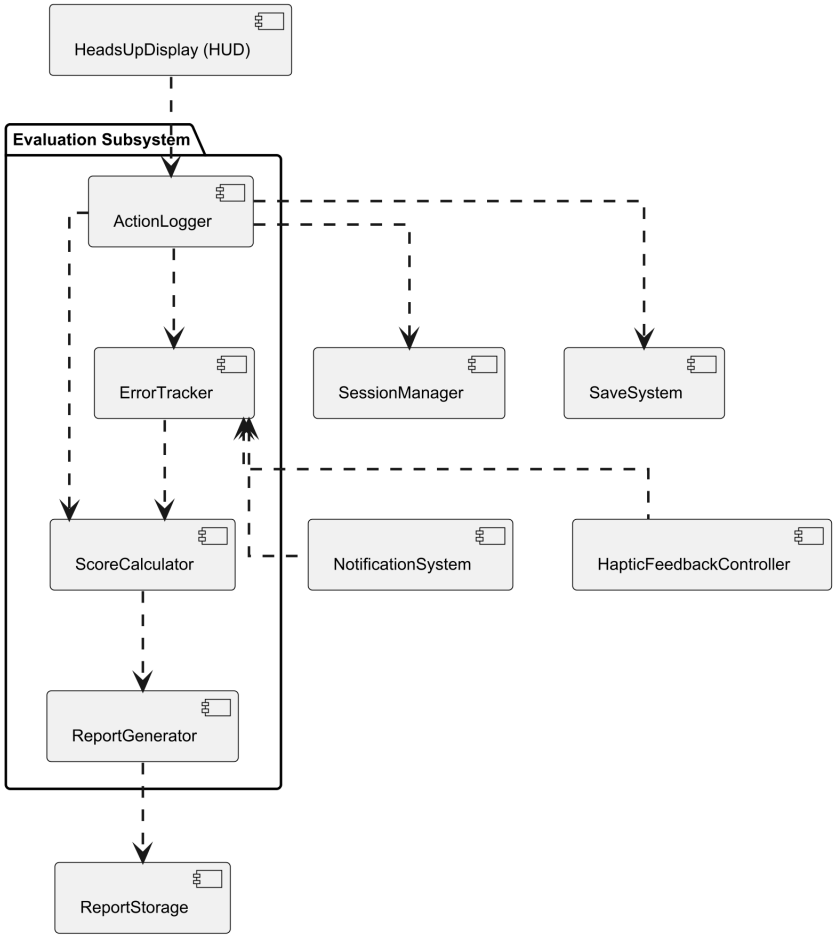


Figure 3.16: Evaluation Subsystem Component Diagram

Data Persistence Subsystem

Provides persistent storage and retrieval services for all system data including scenes, user profiles, investigation sessions, and evaluation reports.

Components:

- `SessionManager`: Manages investigation session data persistence and retrieval.
- `SaveSystem`: Handles save game functionality including session state and progress.
- `ReportStorage`: Stores and retrieves performance evaluation reports.
- `ConfigurationLoader`: Loads scene configurations and system settings.

Figure 3.17 illustrates the Data Persistence Subsystem component diagram.

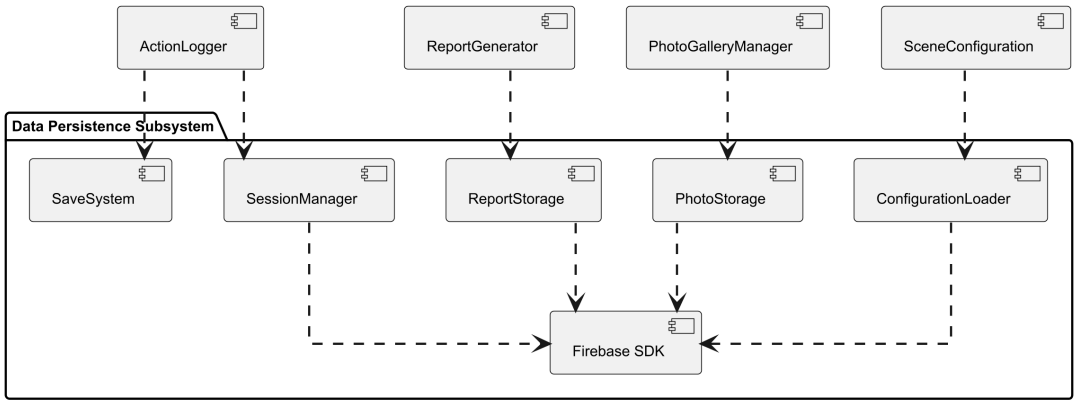


Figure 3.17: Data Persistence Subsystem Component Diagram

3.4.2 Software Architecture

The system follows a layered architecture, which separates the system into four layers: the Interface Layer, the Application Layer, the Domain Layer, and the Data Layer. This separation is component-based to improve maintainability by allowing each layer to be developed, tested, and updated independently, while keeping responsibilities clear and minimizing coupling between VR interaction code and core forensic logic.

Interface Layer

This layer contains all user-facing interaction components and presentation logic. In the context of the VR forensic training system, it is responsible for:

- **VR user interface (Student):** menus for selecting mode and scene, in-scene HUD panels, evidence details panels, tagging forms, and confirmation dialogs.
- **VR interaction handling:** controller input, teleportation/navigation, object highlighting, ray-casting/selection, grabbing/placing objects, and interaction prompts.
- **Feedback presentation:** visual cues (e.g., correctness indicators, warnings for procedural mistakes), haptic feedback triggers, and audio cues where applicable.
- **Administrator & Instructor interface:** scenes for managing crime scenes (enable/disable availability, configure modes, upload or register 360-degree scenes), and reviewing performance reports.

Application Layer
This layer coordinates system between the system's interface and the core doamin logic. It implements the application services that manage sessions and investigation flow, such as:

- **Investigation session management:** start/end investigation sessions, load the selected scene and mode, and maintain session state.
- **Use case execution:** navigation logging, evidence identification workflow, evidence collection requests, tagging steps, and documentation steps (e.g., photography stages).
- **Scoring and assessment generation:** triggering scoring calculations, collecting metrics (time, missed evidence, incorrect procedure steps), and generating post-session summaries.
- **Authorization management:** ensuring only admins can access management functions students can access training sessions, and Instructors can access reports.

The Application Layer calls the Domain Layer to enforce rules and uses the Data Layer for persistence, but it remains independent of how data is physically stored.

Domain Layer
This layer contains the core business concepts and rules of the forensic training system. It models the forensic investigation process and maintains procedural integrity regardless of UI or storage details. Typical domain elements include:

- **Entities and domain/business models:** CrimeScene, Evidence, Investigation, User.
- **Forensic rules and validations:** procedural constraints (e.g., documentation steps order).
- **Evaluation logic:** definitions of what constitutes correct/incorrect actions, scoring rules, and mapping actions to rubric items.

Data Layer
This layer provides persistence and data access services, hiding storage details from the rest of the system. In this application, it is responsible for:

- **Scene repository access:** registering and retrieving interactive 3D scenes and 360-degree recorded scenes, including metadata (scene name, difficulty, available mode).
- **User storage:** user accounts/roles.
- **Session logging and reports:** storing investigation action logs, generated evaluation metric results, and exported reports.

This separation enables changing or future-expanding the underlying storage mechanism without modifying domain rules or interface behavior, supporting scalability and future expansion.

Figure 3.18 presents the layered architecture of the system, showing the separation between interface, application, domain, and data layers.

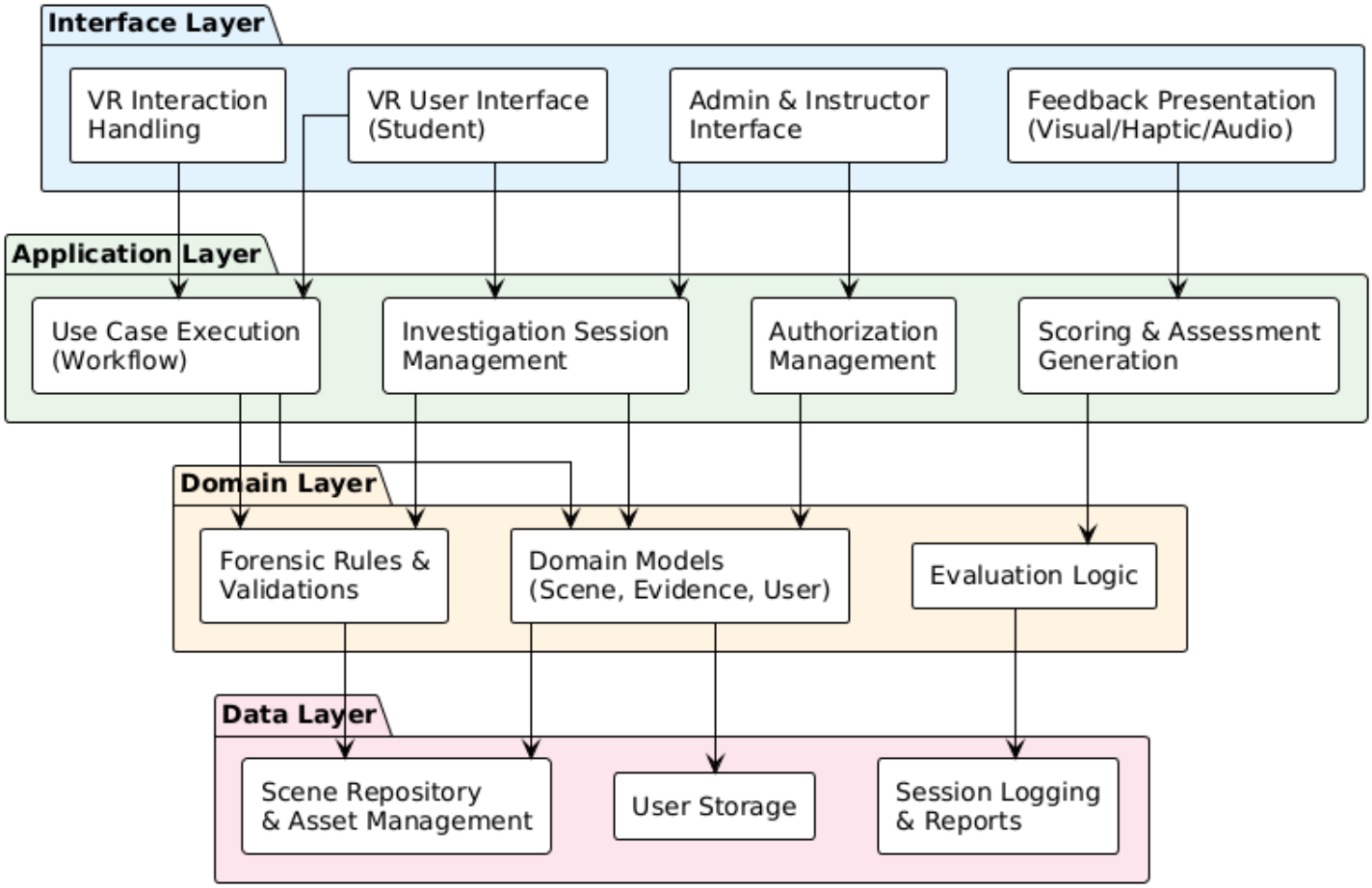


Figure 3.18: Architechture Diagram Showing Layered Architecture.

3.4.3 Deployment Diagram

The deployment diagram represents the physical architecture of the VR forensic training system by illustrating the distribution of software components across hardware nodes and the communication links between them. The system follows a client-cloud architecture pattern, leveraging Firebase as the backend infrastructure for authentication and data storage.

Figure 3.19 presents the deployment architecture showing the VR client application and Firebase cloud services.

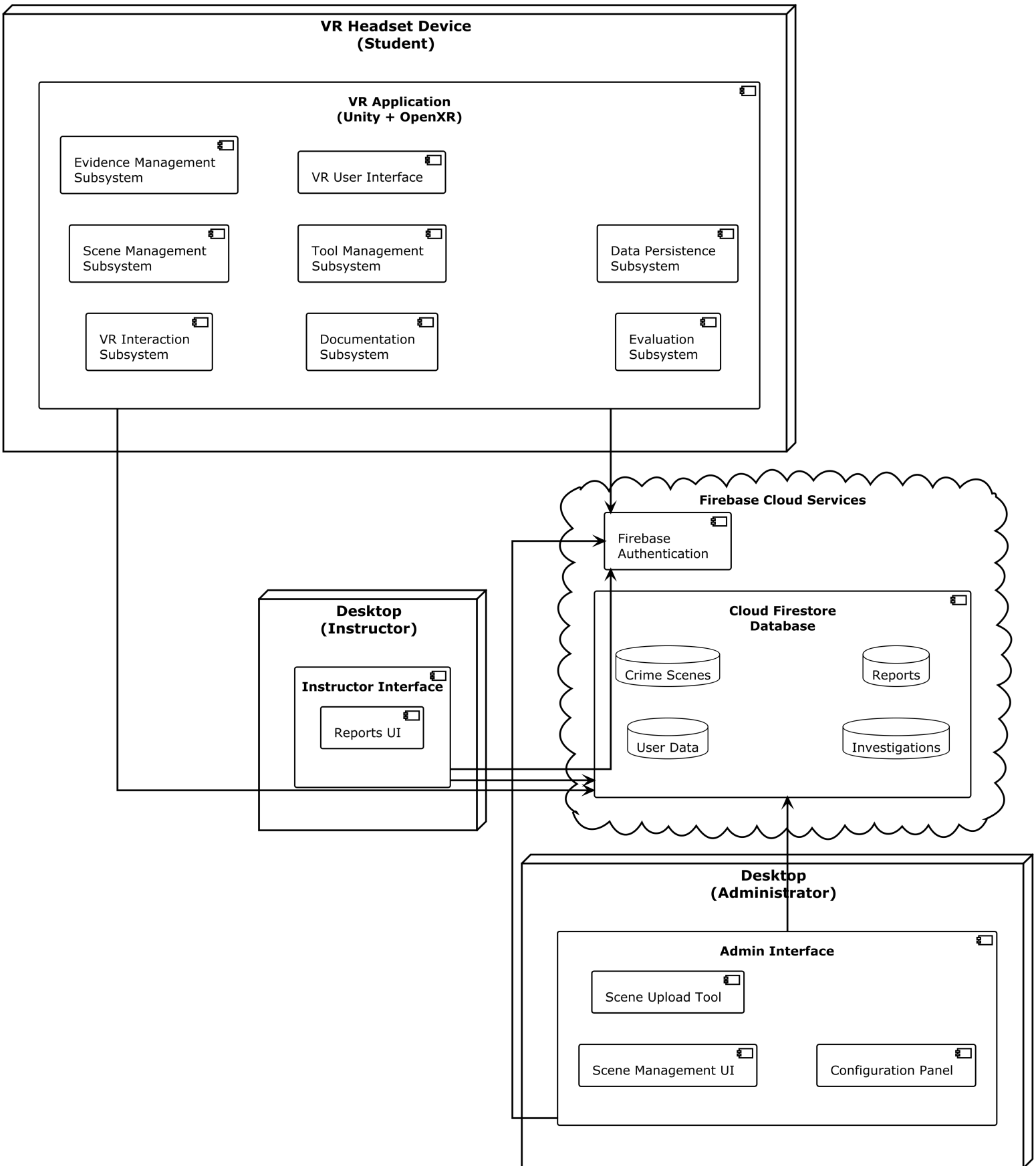


Figure 3.19: Deployment diagram showing system architecture.

Hardware Nodes

Host PC / Workstation:

The machine that executes the VR application. It runs the Unity-based VR engine, handles real-time rendering at 90 FPS, processes the input received from the VR controllers, and communicates with Firebase cloud services. The PC also hosts the separate forensic-advisor backend service used by the NPC advisor. A VR-capable PC is used so that the application can meet the rendering and performance demands of the simulation.

VR Headset and Controllers:

The display and input device through which the user experiences the simulation, connected to the host PC through an OpenXR-compatible VR runtime. Supported headsets include Meta Quest 2/3/Pro (over a PC link), HTC Vive, Valve Index, and other OpenXR-compatible devices. The headset presents the rendered stereoscopic view and tracks head movement, while the controllers provide tracked input. The system is used by all actors — students, instructors, and administrators — with the application presenting role-based interfaces according to the authenticated user's role.

Software Components

VR Application:

The main client application built using Unity with OpenXR support for cross-platform VR compatibility. The application serves all user roles through role-based interfaces. The application contains:

- *Unity VR Engine:* Handles 3D rendering, physics simulation, and VR interaction mechanics
- *OpenXR Runtime:* Provides standardized VR hardware abstraction layer
- *Firebase SDK:* Enables direct communication with Firebase services from the VR application
- *Core Modules:* Scene management, evidence handling, navigation, and feedback systems
- *Forensic Tools:* Virtual camera, UV light, and fingerprint kit implementations
- *User Interface:* VR menu system with role-specific interfaces including student investigation interface, instructor report viewing interface, and administrator scene management interface

Forensic Advisor Backend Service:

A lightweight local service, separate from the Firebase backend, that serves the NPC forensic advisor. It exposes endpoints for chat, speech-to-text, and text-to-speech, hosting the language model (Gemini Flash), the speech-recognition model (Whisper), and the speech-synthesis model (Piper) behind a single interface, while keeping the models' API credentials off the client.

Firebase Cloud Services

Firebase Authentication:

Provides user authentication services supporting email/password authentication with role-based access control. Validates user credentials and manages user sessions for the VR application. Enforces role-based permissions ensuring students, instructors, and administrators have appropriate access levels to different application features and data.

Cloud Firestore:

NoSQL document database storing all system data. The database implements security rules based on Firebase Authentication tokens to enforce role-based access control at the data layer.

Communication Protocols

All communication between the VR application and Firebase services uses HTTPS protocols with TLS encryption ensuring data security during transmission. The VR application communicates with Firebase Authentication via REST APIs for user login and token management, and with Cloud Firestore for data queries, reads, and writes.

The Firebase SDK embedded in the VR application handles authentication token management and the underlying network communication with Firebase services.

Scalability and Performance

Firebase's cloud infrastructure provides automatic scaling to accommodate varying numbers of concurrent users without manual server provisioning. Cloud Firestore scales horizontally to handle increased database read and write operations as the number of students, investigations, and stored reports grows over time.

3.5 Data Management

The system manages structured data related to users, crime scenes, investigations, evidence items, and reports. The data model is designed to ensure data integrity and an organized full documentation of investigations. Although the system uses Cloud Firestore rather than a relational database, the ERD is presented here as a logical data model to formally define system entities, their attributes, and relationships at the conceptual level. The physical Firestore implementation will adapt this model to suit document-based storage patterns.

User Entity:

User data is stored in a User entity, to manage authentication, role-based access, and activity tracking. For students investigation activity data, evaluation metrics, and performance reports are all stored to provide a personalized experience and to allow instructors to review student performance reports.

Crime Scene Entity:

Crime scene data is stored in a CrimeScene entity, to define the virtual environments allowed within the system. This includes scene configurations, such as the scene name, type, and allowed modes. This enables administrators to manage scene repository and student access to each scene.

Evidence and Spawn Point Entities:
Evidence are stored in an Evidence entity which represents an evidence item associated with a scene. Their meta data includes their type - biological, trace, physical, chemical - and description. Each evidence has a list of spawn points associated with it, which are candidate locations in the scene that the evidence can appear in. They are stored in the EvidenceSpawnPoint entity which stores location coordinates. This separation supports the randomization of evidence placement in a scene.

Investigation Entity:
Investigation sessions are stored in the Investigation entity storing the start, end time, and interaction mode, which provides data for evaluation. Each investigation is associated with the user that initiated it and the selected scene. This design allows scene replayability and proper documentation of investigations.

Spawned Evidence and Hint Request Entities:
When an investigation is initiated, each evidence's location is stored in the SpawnedEvidence entity. This entity is used to form an association between an investigation session, ensuring that evidence interactions are stored per user and per session. Requests for assistance related to spawned evidence are recorded in the HintRequest entity, enabling tracking of user dependency on hints.

Investigation Action Entity:
The InvestigationAction entity records each investigative interaction the user performs, this includes actions such as inspecting evidence, placing markers, collecting evidence, tagging, and requesting hints or asking questions (via voice or text) by interacting with the NPC advisor. Each action is timestamped and evaluated for correctness, enabling detailed analysis of user behavior and procedural accuracy.

Evidence Processing Entities:
Marker placement actions generate records in the EvidenceMarkerPlacement entity, which stores the location of the marker, the correctness of it, and the spawned evidence associated with it. Evidence collection actions generate entries in the CollectedEvidence entity, capturing collection details and container information. Collected evidence may be further classified using the EvidenceTag entity, which stores unique identifiers and descriptive metadata.

Performance Evaluation Entities:
Performance outcomes are summarized through the PerformanceReport entity, which is generated for each completed investigation. This entity stores overall scores and accuracy measures related to evidence handling, tagging, and time efficiency. Detailed evaluation data is maintained in the EvaluationMetrics entity, allowing detailed analysis of user performance, error patterns, and procedural mistakes.

The Entity Relationship Diagram (ERD) shown in Figure 3.20 defines how these entities relate to one another, ensuring that all investigation data is consistently linked to users, crime scenes, and evidence items. Proper data modeling supports accurate report generation and future analytical extensions.

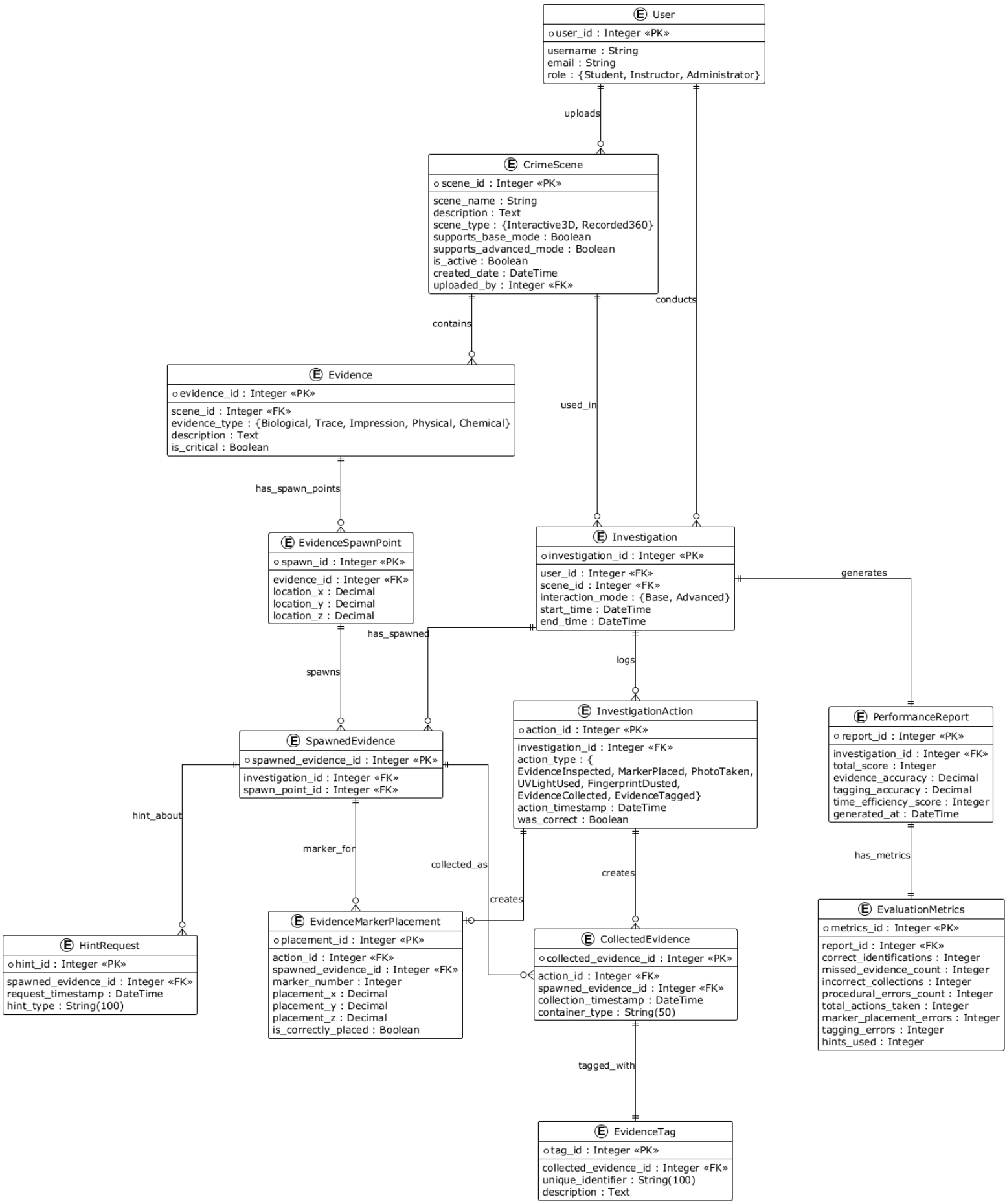


Figure 3.20: Entity-Relation Diagram showing database entities and relationships.

Figure 3.21 presents the Firestore data model for the system. The diagram reflects Firestore's document-oriented structure rather than a relational schema. Solid lines indicate subcollection paths, representing hierarchical parent-child document relationships where subcollections are always queried relative to their parent document. Dashed arrows indicate cross-collection reference fields, where a document stores another document's ID as a string field to express a logical relationship. The model defines three root collections: users, crime_scenes, and investigations. Evidence items and their candidate spawn points are stored as a subcollection under crime_scenes. All investigation-related data — including spawned evidence, actions, marker placements, collected evidence, and hint requests — are stored as direct subcollections of the investigation document, ensuring that all data for a given session can be queried efficiently without traversing unrelated documents.

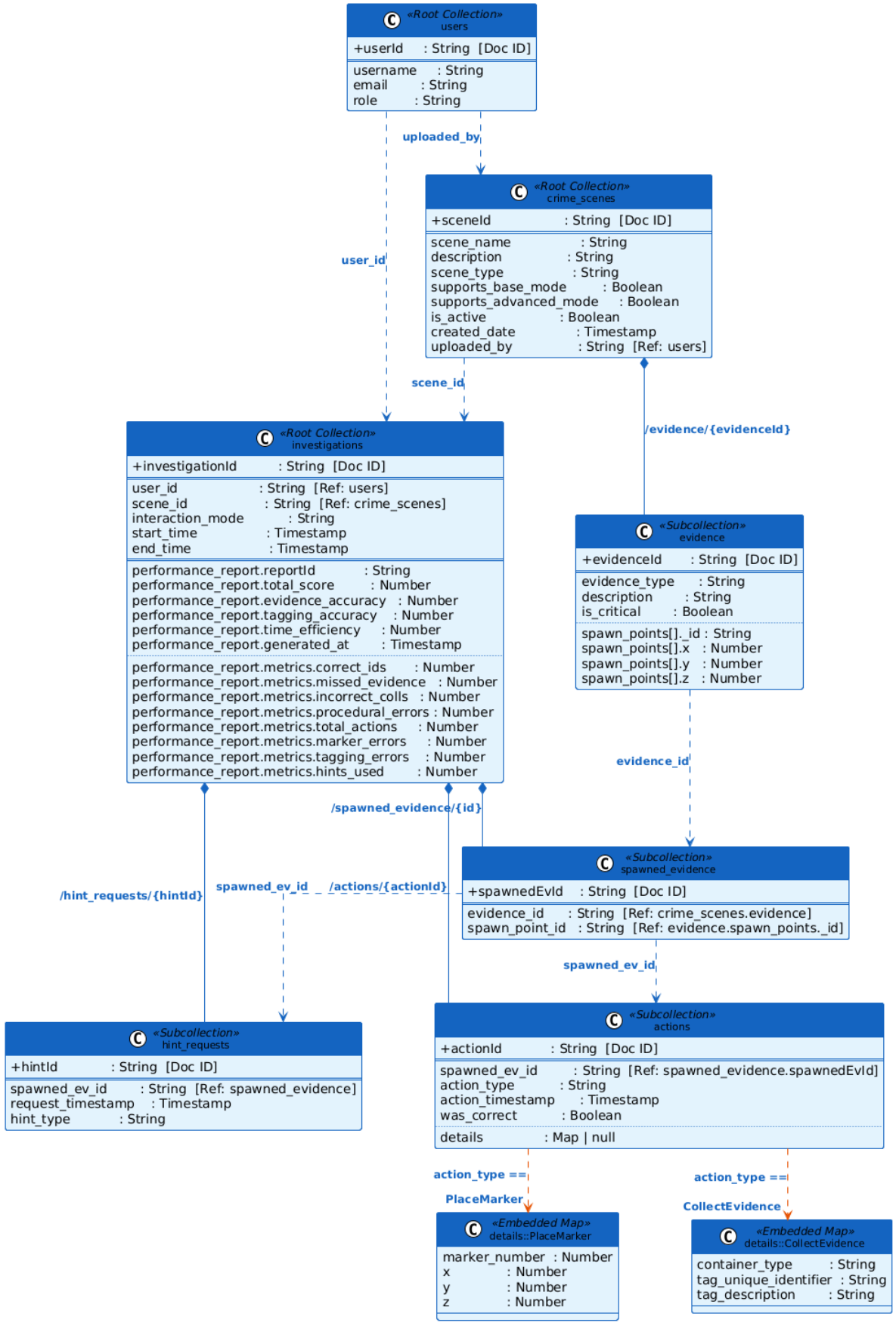


Figure 3.21: Firebase Data Model

Chapter 4

System Implementation

4.1 Overview

This chapter describes the implementation of the Crime Scene Virtual Reality Simulation, detailing the technical decisions, tools, and processes involved in translating the system design, presented previously, into a functional application. The implementation followed the Agile iterative methodology, progressing through a series of defined development stages from environment construction through to AI integration and testing. The system was built using Unity 3D as the primary development environment, targeting the OpenXR framework for cross-platform VR compatibility, with Firebase providing backend authentication and data persistence. Development proceeded in clearly scoped sprints: the core VR environment and navigation were established first, followed by the evidence interaction and forensic tool systems, then the scoring, feedback, and report generation components, and finally the NPC-based AI hint system. This ordering ensured that each layer of functionality was validated before dependent systems were built on top of it.

4.2 Development Environment and Project Setup

4.2.1 Unity and Rendering Pipeline

The application was developed using Unity 6000.4, selected for its updated and efficient XR tooling. Unity's Universal Render Pipeline (URP) was chosen over the Built-in Render Pipeline for its support for shader graph authoring required by the UV light and fingerprint reveal systems, and its compatibility with the XR Interaction Toolkit.

4.2.2 OpenXR Configuration

Cross-platform VR compatibility was achieved by targeting the OpenXR standard rather than any proprietary SDK. OpenXR interaction profiles were configured for the Meta Quest Touch controllers and the HTC Vive wand controllers, mapping trigger, grip, thumbstick, and button inputs to the system's input actions through Unity's Input Action Asset. This hardware-agnostic approach directly satisfies Non-Functional Requirement 1, ensuring the application can be deployed across the range of OpenXR-compatible headsets available at the institution without architectural changes. The XR Interaction

Toolkit (version 3.4) was used as the foundation for all VR-specific interactions, providing pre-built components for ray interactors, direct interactors, teleportation, and grab mechanics. This reduced low-level input handling complexity and allowed development effort to focus on forensic-specific interaction logic rather than controller abstraction.

4.2.3 Firebase SDK Integration

Firebase was integrated into the Unity project using the Firebase Unity SDK, which provides native C# bindings for Firebase Authentication and Cloud Firestore. The google-services.json configuration file exported from the Firebase console was placed in the Assets directory, from which the SDK reads project credentials at runtime. Firebase was initialised asynchronously at application startup using FirebaseApp.CheckAndFixDependenciesAsync(), ensuring that all dependent systems wait for a confirmed valid Firebase connection before attempting any authentication or database operations.

4.3 VR Environment and Scene

4.3.1 Asset Gathering and Creation

Blender

Blender is a free, open-source 3D creation suite that supports the full asset development pipeline, including modelling, UV unwrapping, texturing, rigging, and animation. It was employed throughout this project for several distinct purposes. Assets downloaded from external sources such as the Unity Asset Store and free repositories were imported into Blender for modification and re-unwrapping UVs where texture seams were unsuitable for close-up forensic inspection. Select evidence items and environment props that could not be sourced externally were modelled from scratch to match the specific requirements of the crime scene. Additionally, the NPC forensic advisor character was rigged with an armature and animated within Blender, producing the idle and thinking animation clips that were subsequently exported as FBX files and imported into Unity for use with the Animator Controller. All assets were exported using the FBX format with embedded textures, ensuring material data was preserved on import into Unity's Universal Render Pipeline.

Unity Asset Store and External Repositories

The Unity Asset Store and Sketchfab were used to source assets that would have been prohibitively time-consuming to produce from scratch within the project timeline. Furniture, interior props, and room architecture — elements that establish the environmental context of the crime scene but are not themselves subjects of forensic interaction — were sourced externally, allowing development effort to be concentrated on the forensic evidence items and interaction systems. Forensic evidence assets, including weapons and other scene-specific items, were also sourced externally where suitable models were available, and subsequently refined in Blender as described above. All externally sourced assets were selected based on PBR material compatibility with Unity's Universal Render Pipeline and licensing terms permitting use in an academic project.

4.3.2 Scene Construction and Lighting

The crime scene environment was constructed within Unity by assembling the processed assets into a coherent interior layout reflecting a realistic investigative setting. Room geometry, furniture placement, and prop positioning were arranged to create a plausible spatial narrative, with evidence spawn points distributed across forensically meaningful locations. Colliders were applied to all static geometry to support both physics interactions and teleportation boundary validation, ensuring students cannot navigate outside the intended scene area.

A mixed lighting approach was adopted to balance visual fidelity with performance. Static scene geometry — walls and floors — was lit using Unity's Lightmapper to produce baked lightmaps, which store precomputed lighting and ambient occlusion information directly in textures and impose no runtime cost. Real-time lights were reserved for dynamic elements where baked lighting is insufficient. This hybrid strategy allows high-quality indirect lighting across the majority of the scene while preserving the GPU headroom needed for dynamic tool interactions.

Support for 360-degree recorded crime scenes, used in the Base Observational mode, was implemented using Unity's skybox system. Equirectangular images are assigned to a skybox material applied to the scene's background, surrounding the student with the photographed environment. Navigation within these scenes is not provided since they are for observational purposes only. Additionally, since no 3D geometry underlies these scenes, all evidence interaction features are automatically disabled by the Mode Controller when a recorded scene is loaded. Figure 4.1 displays the skybox environment as it appears to the student in Base Observational mode.

Figures/360viewer.png

Figure 4.1: 360-degree recorded scene

4.3.3 Crime Scene Scenario

The main crime scene, *The Landlord's Apartment*, is the scene used to demonstrate the Advanced Interactive mode. It depicts an indoor residential homicide: a small single-floor apartment, comprising a living room and an adjoining kitchen, in which a victim is found on the living-room floor. The scene was authored as an original training scenario rather than a real case, with each evidence item placed so that it plausibly results from the perpetrator's contact with the scene, giving the student a coherent set of traces to reconstruct.

Eight evidence items are distributed across the apartment, spanning the forensic categories the system supports — biological, trace, chemical, and physical — so that a single scene exercises every tool and procedure. The items are listed in Table 4.1. Some are visible on observation, while others are latent and appear only once the appropriate tool

is used: the latent fingerprints are recovered with the fingerprint kit, and the hair fiber and cleaning-fluid residue are revealed with the UV light, rewarding students who search the scene methodically rather than relying on what is immediately visible.

ID	Evidence item	Type	How it is found
E-01	Bloodstain pool	Biological	Visible on observation
E-02	Blood transfer smear	Biological	Visible on observation
E-03	Latent fingerprint (glass)	Trace	Latent; recovered with the fingerprint kit
E-04	Latent fingerprint (handle)	Trace	Latent; recovered with the fingerprint kit
E-05	Shoe impression	Trace	Visible on observation
E-06	Hair fiber	Trace	Hidden; revealed with the UV light
E-07	Blunt instrument (wrench)	Physical	Visible object; the scene's primary weapon
E-08	Cleaning fluid residue	Chemical	Hidden; revealed with the UV light

Table 4.1: Evidence items in Scene 01, The Landlord's Apartment.

Because evidence positions are randomised through the spawn-group system described later in this chapter, each of the eight items is authored with several candidate locations and one is selected at the start of every session, so the same scenario yields a different spatial configuration on each playthrough. Collectively the eight items require the full investigative workflow — observation, marker placement, three-stage photography, fingerprint and UV processing, collection into the correct container, and tagging — making this scene a complete exercise of the mechanics described throughout this chapter.

4.3.4 User Interface

All user interface elements in the system are implemented as world-space canvases — Unity Canvas components configured in World Space render mode — positioned and scaled within the 3D scene rather than overlaid on the screen. This approach is standard practice in VR interface design, as screen-space overlays cause visual discomfort by appearing to float at a fixed depth inconsistent with the surrounding stereoscopic environment. Interaction with all panels is performed using the XR Interaction Toolkit's ray interactor, which casts a visible ray from the controller and triggers UI pointer events on contact with canvas elements.

Menus

The main menu is presented to the student upon successful login and provides scene and mode selection before an investigation is initiated. It is implemented as a world-space panel displaying buttons for selection between Base Observational and Advanced Interactive modes. When more scenes are added this menu will also display a list of available scenes as retrieved from Firestore. Scene availability is enforced at this stage by

filtering the retrieved scene list against the selected interaction mode and the availability field set by the administrator, ensuring students only see scenes that have been explicitly enabled. Figure 4.2 shows a screenshot of the Main Menu the student is shown after logging in.

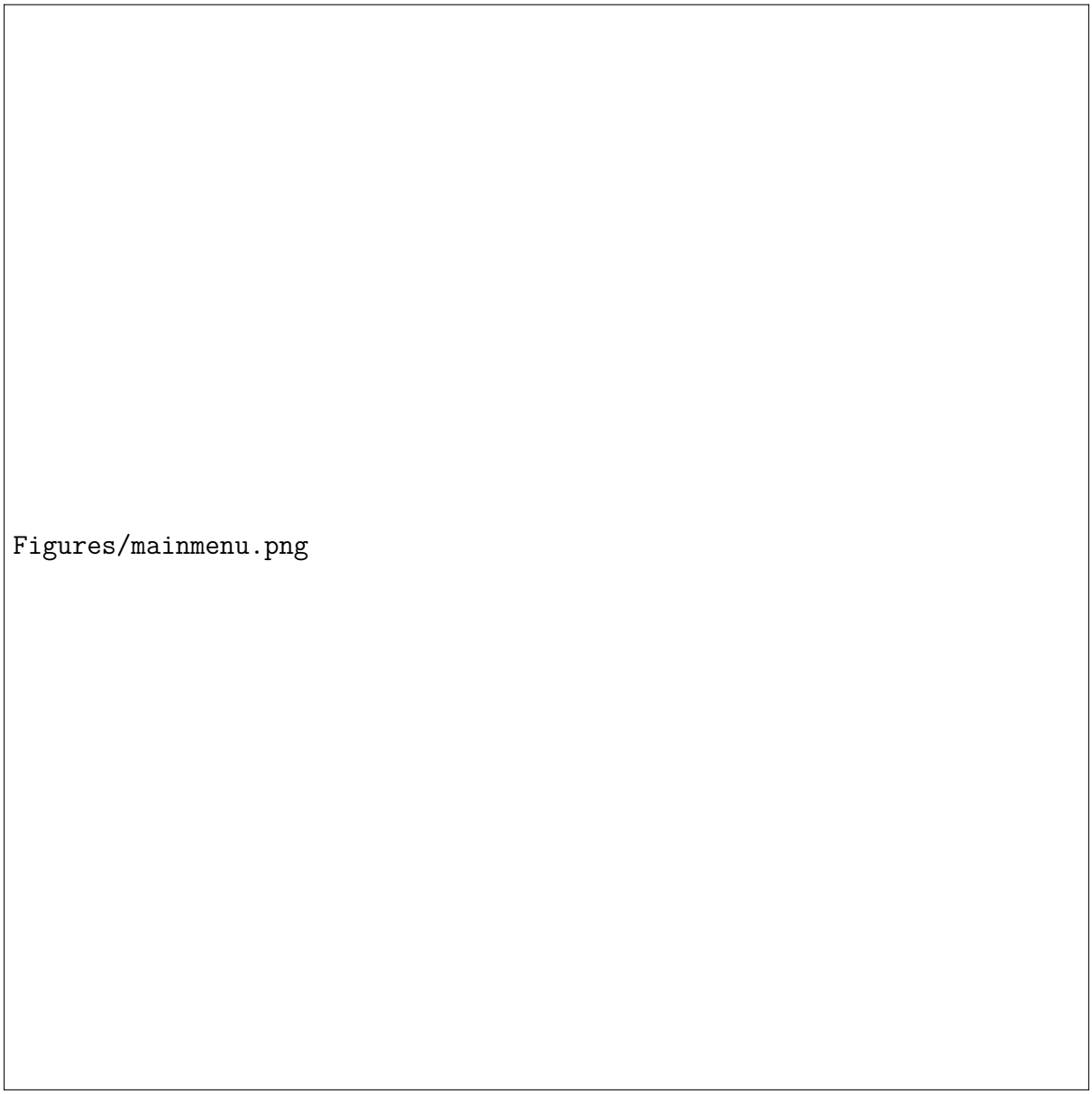


Figure 4.2: Student Main Menu

A separate administrative panel, accessible only to users authenticated with the admin role, provides scene management controls including enabling and disabling scene availability and configuring mode settings. Figure 4.3 provides a screenshot of this dashboard



Figure 4.3: Admin Dashboard

Additionally, an instructor dashboard was implemented to allow users with instructor role to view student performance reports and investigation sessions. This dashboard is displayed in Figure 4.4



Figure 4.4: Instructor Dashboard

HUD

The heads-up display provides persistent contextual information throughout an active investigation session without obstructing the student's view of the crime scene. It is implemented as a world-space canvas anchored at a fixed position relative to the player's head transform, so that it remains consistently visible regardless of where the student looks. The HUD displays the current investigation timer, the current selected tool, the count of evidence items collected relative to the total present in the scene, and a live score indicator to support the gamification objectives of the system. Information density on the HUD was deliberately kept low to avoid cognitive overload and maintain focus on the forensic environment, consistent with the interface design principles outlined previously.

Figure 4.5 shows an example of how the HUD looks, to the player, while the investigation is in progress.

Figures/hud.png

Figure 4.5: Heads Up Display

4.4 Player Input and Locomotion

4.4.1 Controller Input Handling

All controller input is handled through Unity's Input System, with the physical controls expressed as named input actions in the Input Action Asset rather than read as hardware-specific buttons, which lets the same control logic run unchanged across the different OpenXR controllers configured earlier. The XR Interaction Toolkit consumes these actions to drive the core VR interactions — controller tracking, the ray and direct interactors used to select and grab objects, and locomotion — while the project's own systems bind the actions specific to the forensic task: the right-hand trigger activates whichever tool is equipped, with the ToolEquipManager forwarding a single shared ac-

tivate action to the held tool so the same trigger places a marker, takes a photograph, toggles the UV beam, or dusts and lifts a print; the grip grabs and collects evidence; the thumbsticks drive locomotion, rotate a held item during close inspection, and select a slot in the tool radial menu; a face button toggles the close-inspection view; holding the thumbstick click opens the radial menu; and a push-to-talk button records a question for the forensic advisor.

4.4.2 Teleportation System

Locomotion is implemented using both movement-based and teleportation-based movement types. The teleportation-based locomotion type was selected for its low cognitive overhead and its established effectiveness in reducing motion sickness during extended VR sessions, both of which are important for a training tool intended to be used in sessions of up to 20–30 minutes. The XR Interaction Toolkit's TeleportationProvider component, combined with teleportation area component applied to the scene's floor geometry, forms the basis of this system. When the student tilts the thumbstick forward, a curved ray is projected from the controller and visualized as an arc terminating in a reticle that indicates the prospective destination.

Boundary validation is performed continuously while the arc is active. The destination point is checked against the navigable floor area of the scene; if the student aims the arc at an invalid location, such as outside the scene bounds or through a wall, the reticle changes colour to provide immediate visual feedback that the location cannot be selected, and teleportation is prevented until a valid target is found. On activation, the player's position is updated instantly to the selected point, with the screen briefly fading to black and back to reduce the disorientation that can result from an abrupt change in viewpoint. This arc-feedback is demonstrated in Figure 4.6, showing different feedback based on the targeted location.

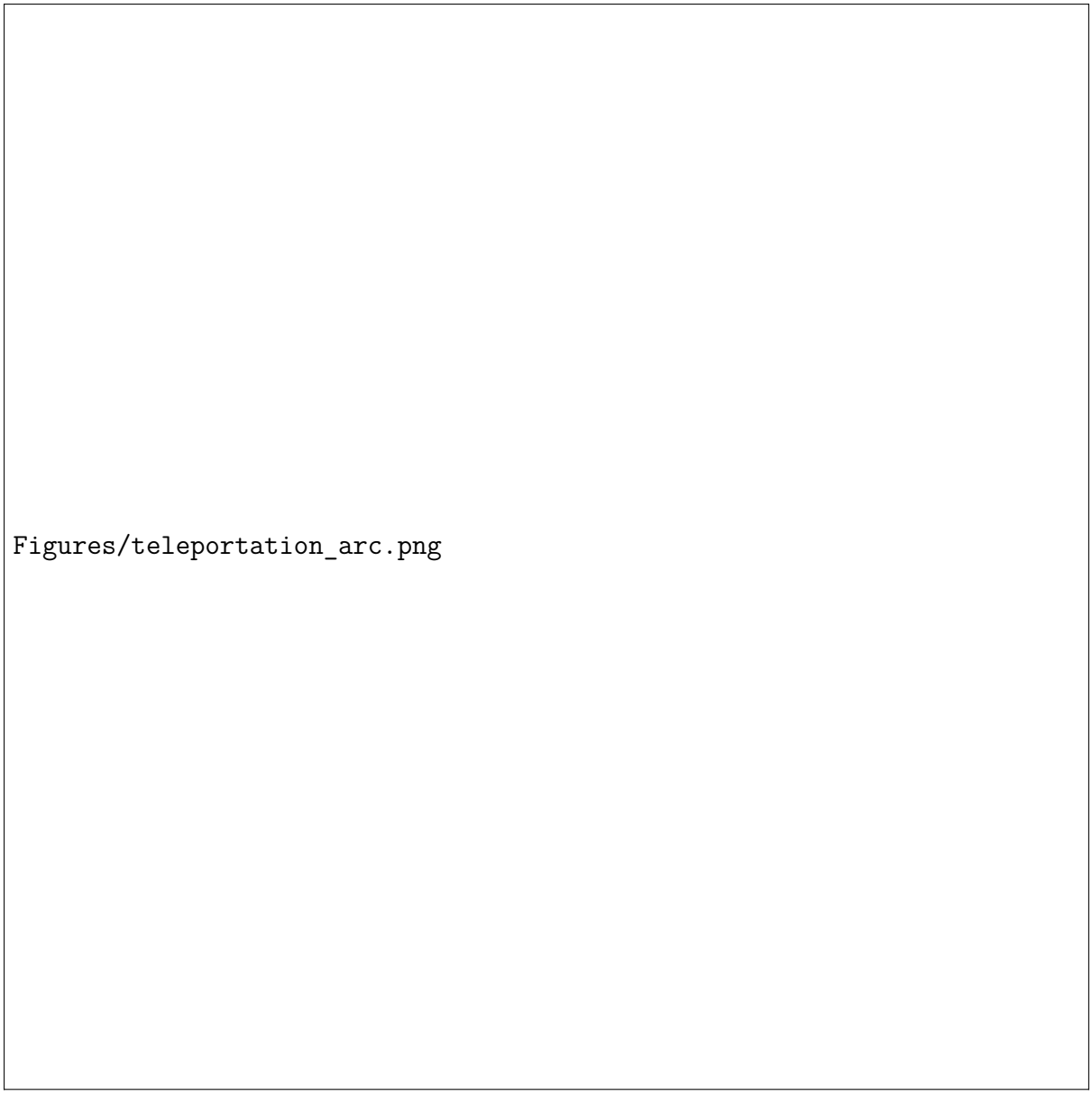


Figure 4.6: Teleportation arc

(the teleportation arc with valid (green) and invalid (red) target indicators shown side by side)

In addition to teleportation, natural movement is supported continuously—providing a movement option that some users may find more intuitive and convenient—allowing students to move and look around freely in all directions at any time, and students may physically crouch to lower their viewpoint when inspecting evidence at floor level. Together, these mechanisms allow students to explore the crime scene comfortably while supporting the close, deliberate positioning required for marker placement, photography, and other documentation tasks.

4.5 Evidence Interaction

4.5.1 Evidence Highlight and Observation

Before any evidence item can be physically interacted with, students must first observe it, in keeping with the forensic principle of not disturbing a scene before it has been properly documented. To support this initial observation stage, every evidence object in the scene is fitted with a hover highlight effect implemented using a custom monobehaviour script. When the student's controller ray enters the interaction range of an evidence item, the EvidenceHighlight component detects the hover and enables an emissive highlight on the object's materials, making the evidence glow in a distinct colour. Flat surface evidence such as blood pools, smears, and impressions has no emissive mesh to light up, so for these items the same component instead draws a coloured outline around the evidence's footprint, producing an equivalent cue. Evidence that can only be found under ultraviolet light is deliberately exempt from this hover highlight until the UV beam has revealed it, so that a sweeping ray cannot betray its location prematurely. In each case the visual cue signals to the student that the item is a recognised piece of evidence without requiring any prior physical contact, allowing observation to take place purely through movement and viewing angle.

While an item is highlighted, the student can approach it, view it from multiple angles using natural head movement, and visually assess characteristics such as size, colour, texture, and position relative to the surrounding scene. This observation drives the item into its Detected state, marking it as identified for later use in the evaluation metrics. Figure 4.7 shows how an evidence item is highlighted when the student hovers over it to detect/observe it.

Figures/highlighted_evidence.png

Figure 4.7: highlighted evidence

(screenshot of a highlighted evidence item)

4.5.2 Physical Inspection

Students are permitted to pick it up and examine it closely using the controllers—once an evidence item has been documented—this physical inspection stage is implemented using the XR Interaction Toolkit's XRGrabInteractable component, attached to each evidence item . Track rotation is enabled so that rotating the controller correspondingly rotates the held item, allowing the student to inspect all sides of the object. Throw physics is disabled on all evidence items, since allowing evidence to be thrown or to retain momentum after release would be inconsistent with the careful handling expected in a forensic context; instead, releasing the grip simply returns the item smoothly to its original position and orientation in the scene.

Haptic feedback is provided at several points during this interaction. A short pulse is issued through the built-in HapticImpulsePlayer at the moment an item is observed, confirming successful detection, and a second distinct pulse is issued when the item is grabbed. This haptic confirmation reinforces the sense of physical interaction with the evidence and helps students recognise successful observe and inspect actions without needing to rely solely on visual feedback.

4.6 Investigation System Logic

4.6.1 Evidence Lifecycle State Machine

The progression of each evidence item through the investigation workflow is managed by the EvidenceStateMachine, which implements the item's lifecycle. The set of possible states is defined as a C# enum comprising five values: Hidden, Detected, Marked, Photographed, and Collected. Each evidence object in the scene carries its own EvidenceStateMachine component instance holding the item's current state, allowing every piece of evidence to progress independently through the lifecycle at its own pace as the student moves through the scene.

An item begins in the Hidden state, in which it is present in the scene but has not yet been observed and carries no visual indication of its forensic significance. Once the student approaches within range, hovers over it with the controller ray—which causes the highlight effect—the item transitions to Detected. As the student marks the evidence with the tent marker tool, the item transitions to Marked. The transition to Photographed occurs only once the three-stage photography procedure has been completed for that item, and this is the point at which the item's close-inspection mode is unlocked, permitting the physical examination described earlier. The item reaches Collected — its terminal state — once it has been placed into an appropriate evidence container and tagged with a complete digital evidence tag.

Each transition is guarded by a corresponding condition that must be satisfied before the EvidenceStateMachine permits the state change, directly enforcing the mandatory forensic procedural sequence established in the use case relationships. An enum-driven approach was chosen over an Animator-based state machine, as the evidence lifecycle is a purely logical progression with no associated animation states, and a lightweight C# implementation allows the current state to be queried directly by other systems, such as the scoring and feedback systems described below, without the overhead of an Animator graph. Each forensic action a student performs on an item — marking, photographing, dusting for fingerprints, collecting, and tagging — is recorded by the InvestigationService through its LogAction method, stamped with a UTC timestamp and a correctness flag, for student evaluation.

Additionally, the EvidenceStateMachine stores evidence identity data—evidence type, name, id, and critical status—to maintain evidence data throughtout the investigation session.

Figure 4.8 captures the EvidenceStateMachine component assigned to an evidence item, it displays how the evidence data is stored in the instance in addition to its current state in the investigation.

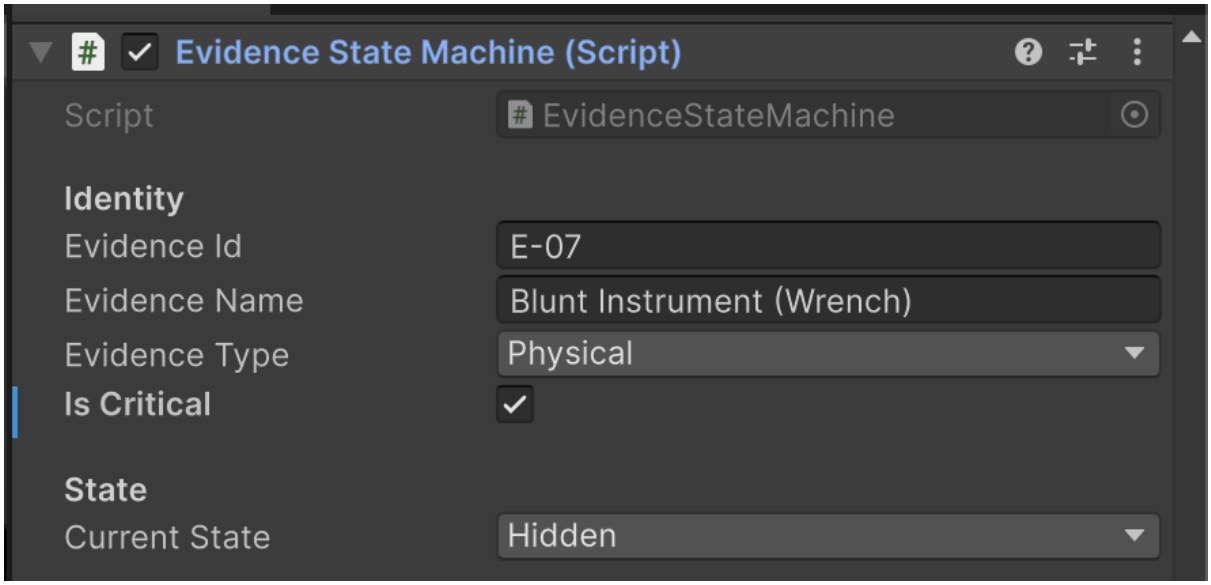


Figure 4.8: EvidenceStateMachine Component on Evidence Object

4.6.2 Dynamic Evidence Placement

To prevent students from relying on memorised evidence locations across repeated sessions, the position of each evidence item is randomised at the start of every investigation using a scene-based spawn group system. Spawn points are authored directly in the Unity scene during development. Each evidence item is represented by multiple pre-positioned variants — one per candidate location — all placed as child GameObjects under a shared parent carrying an EvidenceSpawnGroup component. This keeps the spatial configuration entirely within the scene hierarchy and eliminates the need for a separate database-driven spawn point catalogue.

When an investigation session begins, the EvidenceSpawnGroup sets the active state of every child so that exactly the chosen variant is enabled and all others are disabled.

Once a variant has been activated, the component identifies the EvidenceStateMachine component attached to the chosen child and reads its evidenceId field to produce the session's SpawnedEvidence record. Using the EvidenceStateMachine's own evidenceId, rather than the group object's name, is a deliberate correctness requirement: runtime action logging resolves spawned evidence through InvestigationService.GetSpawnedEvidenceId using this same identifier, so a mismatch between the two would cause every subsequent action — photography, collection, tagging, fingerprint dusting — to be logged with a null spawnedEvidenceId, breaking per-evidence scoring entirely. If no valid EvidenceStateMachine is found on the chosen child, the group's own name is used as a fallback. The resulting SpawnedEvidence record, carrying the evidence identifier, the chosen variant's name, and its index within the candidate list, is returned to the calling session initialisation logic for persistence and evaluation tracking.

Figure 4.9 presents the spawn group parent and all the spawn point candidate children.



Figure 4.9: Candidate Spawn Points as Children to the Same Spawn Group Parent

4.6.3 Procedural Feedback System

The procedural feedback system communicates the consequences of every student action in real time, reinforcing correct procedure and signalling mistakes as they occur. Rather than coupling each tool directly to the interface, scoring, and logging systems, feedback is distributed through an event-driven observer architecture built on C# event delegates: the EvidenceStateMachine and EvidenceRegistry raise OnStateChanged as items advance through their lifecycle, the InvestigationService raises OnActionLogged for each recorded action, the ProcedureViolationTracker raises OnViolationRecorded on a detected error, and the ScoreManager raises OnScoreChanged when the running score updates. Consumers such as the VRHud, the ScoreManager, and the ProcedureViolationTracker subscribe to the events relevant to them when they initialise and detach when torn down, leaving the systems that produce feedback fully decoupled from those that react to it.

Feedback is delivered through three sensory channels at the instant of interaction. Haptic feedback uses the controller's HapticImpulsePlayer, with distinct pulses for observing an item, grabbing it, placing a marker, dusting a surface, and revealing trace evidence under the UV light. Audio feedback plays a confirmation sound at the point of contact for marker placement, fingerprint dusting, and UV detection. Visual feedback is the most prominent: detected evidence is rendered with the emissive highlight described earlier, and the head-locked HUD continuously updates the detected and collected counters and the live score as the corresponding events fire. The system also differentiates correct from incorrect actions, most visibly in the photography workflow, where the VirtualCameraController evaluates each shot against the required framing and state conditions and drives a colour-coded HUD hint — a success state for a valid capture, an error state for a rejected one — alongside one of two matching shutter sounds. Procedural errors such as photographing before marking or sealing evidence in the wrong container are surfaced immediately and additionally recorded by the ProcedureViolationTracker for the scoring process detailed in the following section, so feedback operates on two timescales: an immediate sensory response during the investigation, and a persistent record that informs the final evaluation.

4.6.4 Scoring and Report Generation

Evaluation follows a dual-layer model that separates the at-a-glance feedback a student receives during play from the structured assessment an instructor reviews afterwards. The first layer is a numeric score out of 100, using a weighted task-scoring and penalty approach in which each task contributes in proportion to its significance and incorrect actions are deducted in proportion to their severity. The second layer expresses the same session as the attainment of discrete Learning Objectives on a pass/fail basis, following the competency-based model of Mayne and Green [48], which identifies specific skill gaps more meaningfully than a raw number. The definition of which actions are correct and which are penalised is grounded throughout in the procedural sequence established by the National Institute of Justice [36].

The numeric score combines four weighted components, summarised in Table 4.2.

Component	**Weight**	**How it is computed**
Evidence Score	40%	Weighted found items divided by the weighted total, scaled to 100; critical evidence counts twice as much as non-critical evidence.
Procedure Score	40%	Starts at 100, with a penalty subtracted for each violation of the NIJ sequence — for example collecting an item before photographing it, photographing before a marker is placed, omitting one of the three photography stages, or sealing evidence in the wrong container type.
Tagging Score	10%	Average proportion of the required tag fields completed across all collected items, scaled to 100.
Time Efficiency Bonus	10%	A tiered bonus for finishing within the per-scene benchmark time; a student is never penalised for taking longer.

Table 4.2: The four weighted components of the Layer 1 numeric score.

These component scores are combined into a single weighted sum, from which a fixed deduction of two points is applied for every hint requested from the NPC advisor, with the result constrained to the range 0–100:

$$\text{Total} = \text{clamp}(0.40\,E + 0.40\,P + 0.10\,T + 0.10\,B - 2H,\ 0,\ 100)$$

where E, P, T, and B are the Evidence, Procedure, Tagging, and Time Bonus scores respectively and H is the number of hints requested.

The logic is implemented as a stateless calculator, the RubricScoreCalculator, which gathers live data from the EvidenceRegistry, the ProcedureViolationTracker, the ChainOfCustodyLogger, and the InvestigationService and returns a RubricScoreReport containing both the numeric breakdown and the Learning Objective results; being free of side effects, it can be invoked at any moment to produce the current best estimate of the score. During play the ScoreManager re-runs it on every relevant event — an evidence state change, a logged action, or a recorded violation — and raises OnScoreChanged to keep the live HUD score current. At session end the PerformanceReporter builds the definitive report, automatically when the last item is collected or on demand when the

student submits an incomplete investigation. The report is then surfaced two ways: the ReportPanel subscribes to OnReportReady and presents the breakdown and objective outcomes on a world-space panel inside the headset, while the RubricReportSerializer maps the report onto the Firestore schema defined in the scoring rubric for the InvestigationService to persist as a performance_report sub-document, which the instructor dashboard later reads back for review.

The second layer derives four Learning Objectives from the same metrics. LO-1 (Evidence Identification) passes when the student collects at least 80% of the scene's evidence with no critical item missed; LO-2 (Documentation Sequence) when at least 90% of collected items were photographed before collection and no critical sequence violation occurred; LO-3 (Evidence Tagging) when the required tag fields are completed for at least 80% of collected items; and LO-4 (Container Selection) when the correct container type was used for at least 90% of them. Each is reported as a pass or fail with the underlying figure, such as "7 / 9 items", so the instructor sees not only whether a competency was met but by what margin.

4.7 Forensic Tools

The forensic tools share a common equipment model. Each tool is implemented as a component deriving from a base EquippableTool class, and only one tool is held at a time. The student selects between tools through a world-space radial menu, opened by holding a controller button: the menu appears in front of the player, the thumbstick highlights a slot, and releasing the button equips the highlighted tool through the ToolEquipManager. Equipping and unequipping a tool invokes lifecycle hooks on the component — enabling its visuals and input handling while held and disabling them when put away — and the currently equipped tool is shown on the HUD. The following subsections describe the three primary documentation tools; in each case the controller trigger drives the tool's main action.

4.7.1 Evidence Marker Placement

Once an item has been detected, the student marks it using the evidence marker tool to record its location before any further handling, in keeping with the forensic principle of documenting position first. While the tool is equipped it projects a ray from the tool tip and renders a reticle at the point of intersection with the scene; the reticle is drawn green where placement is permitted and red where it is not, the latter occurring when the ray misses a valid evidence item or strikes a surface whose angle exceeds the allowed threshold. Pressing the trigger casts the ray and, if it hits an evidence item that is eligible to be marked, places a numbered marker on it. The marker style is chosen according to the kind of evidence: ordinary object evidence receives a standing tent card, positioned just past the item's bounding box on the player's side and dropped onto the surface beneath so that it rests cleanly beside the evidence rather than overlapping it, whereas flat surface evidence — such as blood, prints, and impressions rendered as decals — instead receives a low adhesive marker stuck flush against the surface at the exact point the student is aiming. The adhesive form suits evidence that lies on walls, glass, or other steep faces where a free-standing card could not rest, while both forms carry the same sequential number.

Each placement increments a running counter, so markers are numbered sequentially in the order they are placed, and the spawned card displays its number through a MarkerCard component parented to the evidence. Placing a marker transitions the item's state machine to the Marked state, issues a haptic pulse and a confirmation sound, and logs a PlaceMarker action — together with the marker number — through the InvestigationService for evaluation. Figure 4.10 shows the marker tool in use, with the placement reticle and a numbered marker card beside an evidence item.



Figure 4.10: Evidence marker tool: placement reticle and a numbered tent marker card

Figure 4.11 shows the adhesive marker variant applied flush to a piece of flat surface evidence.

Figures/adhesive_marker.png

Figure 4.11: Adhesive marker placed flush on flat surface evidence

4.7.2 Virtual Camera

The virtual camera lets the student photograph evidence following the standard three-stage forensic photography procedure. The tool carries a child camera that renders the scene to a RenderTexture, which is displayed on a screen on the back of the camera body to act as a live viewfinder, so the student frames the shot exactly as they would with a real camera. The tool automatically targets whichever evidence item the lens is pointed at, casting a ray each frame and keeping the last valid target so that brief glances away do not lose progress. Each item must be captured in three successive stages — an Overview shot taken from more than two metres away to establish scene context, a Mid-Range shot taken between roughly half a metre and two metres with the evidence marker in frame, and a Close-Up shot taken from under half a metre with a measurement scale in frame.

Before each shot is accepted, the camera validates a set of conditions appropriate to

the current stage: the evidence must be fully within the frame and not occluded by other geometry, the camera must be within the distance band for that stage, the item must already be in the Marked state, and the stage-specific requirement — marker in frame for the mid-range shot, scale in frame for the close-up — must be satisfied. A valid capture advances the item to the next stage and, once all three stages are complete, transitions the item to the Photographed state, permitting the physical inspection and collection described earlier. An invalid attempt is rejected with a stated reason. Every shutter press, whether accepted or rejected, is logged as a TakePhoto action recording the stage, the capture distance, and the outcome, which the scoring system later uses to detect missing or out-of-sequence photography. Figure 4.12 shows the camera viewfinder while framing an evidence item.

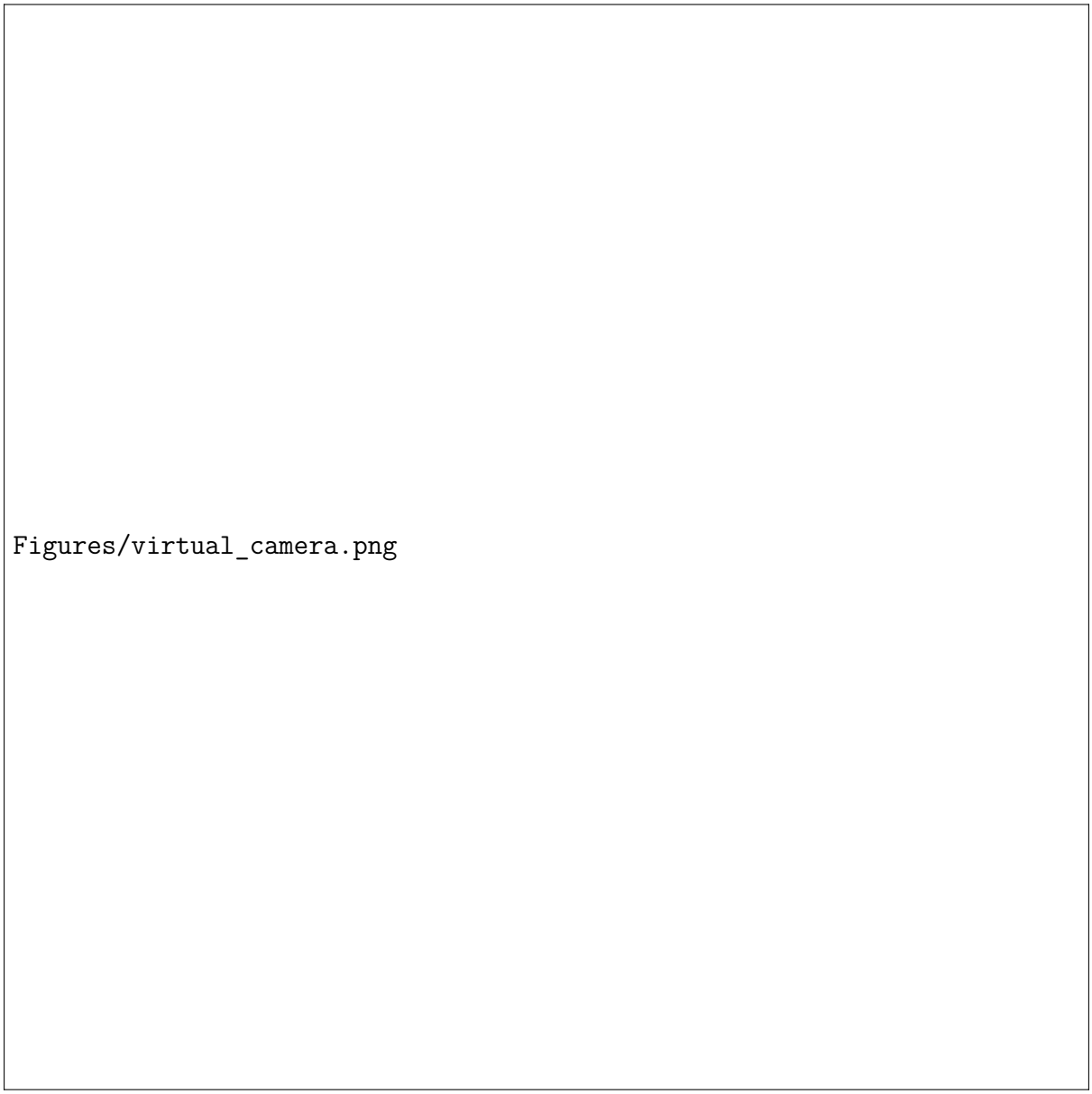


Figure 4.12: Virtual camera with live RenderTexture viewfinder during the photography workflow

Camera UI

Feedback during photography is presented through two surfaces. On the camera body itself, the viewfinder border flashes green when the framed shot currently satisfies all conditions and red when it does not, giving the student an immediate, glanceable indication of shot validity. In parallel, the camera section of the head-locked HUD displays the current photography stage and a contextual hint that names the next requirement to satisfy, for example prompting the student to move closer, bring the marker into frame, or include the scale. And, after a shutter press, briefly shows the success or failure result. Figure 4.13 shows the camera interface with the stage indicator and validity feedback.



Figure 4.13: Camera interface: viewfinder validity border and HUD stage indicator with contextual hint

4.7.3 UV Light

The UV light reveals trace evidence that is not visible under normal lighting, such as biological fluids and latent prints. While the tool is equipped a tinted spotlight is projected from it, configured as a cone with a fixed range and beam angle, so the student physically sweeps the beam across surfaces to search the scene. Each frame the tool gathers nearby colliders within the beam range, filters them to surfaces flagged as UV-reactive, and checks that each candidate lies within the beam's cone angle of the tool's forward direction; the first time a reactive surface falls within the beam, it is revealed. Revealing an item issues a haptic pulse and a confirmation sound and logs a UseUVLight action recording the evidence subtype.

Beam and Reveal Rendering

Each surface that responds to UV light is represented by a ForensicSurfaceEvidence component carrying a subtype — such as a blood pool, smear, or fingerprint — and a pre-placed decal or mesh that depicts the trace. For UV-reactive items this visual is kept hidden until the item is detected, and is enabled the moment the UV beam reveals it by reacting to the item's state machine, so the trace appears to fluoresce as the beam passes over it under the tinted UV spotlight. The reveal is therefore achieved by toggling the pre-authored decal in response to the detection event rather than by a runtime fluorescence shader, which keeps the effect inexpensive and authorable per item. Figure 4.14 shows trace evidence revealed under the UV beam.

Figures/uv_light.png

Figure 4.14: UV light tool revealing trace evidence under the tinted beam

Evidence Reveal System

Detection is gated on both distance and angle. A surface is only revealed when it lies within the beam's range and inside its cone half-angle, so holding the light too far away or pointing it off to the side leaves the evidence hidden, requiring the student to actively sweep the beam across the scene to uncover everything. Each reactive surface is revealed only once: the tool transitions the item from Hidden to Detected through the same state machine used by direct observation, after which the item follows the standard marking, photography, and collection sequence. Because detection routes through the shared state machine, trace evidence uncovered with the UV light counts toward investigation progress and scoring identically to evidence found by direct observation.

4.7.4 Fingerprint Kit

The fingerprint kit recovers latent prints through the standard two-step technique of dusting and lifting. While the tool is equipped it projects a short-range pointer line and reticle from the brush tip, coloured to indicate whether a fingerprint surface is within reach, and a light haptic pulse confirms when the brush first comes over a valid print. Because the kit operates only at very close range, the student must bring the brush right up to the surface rather than pointing at it from across the room, reproducing the deliberate, hands-on nature of the task.

The trigger drives a two-stage process on whichever fingerprint surface the brush is touching. The first press dusts the latent print: this reveals a dusting overlay on the surface, detects the evidence if it was still hidden, and logs a DustFingerprint action. A second press lifts the now-visible print onto tape, logging a LiftFingerprint action. The tool records the dusted and lifted state of each surface, so the two steps must be carried out in order and neither can be repeated, and each step issues a stronger haptic pulse and a confirmation sound to reinforce the sense of performing the physical procedure. Figure 4.15 shows the fingerprint kit being used to dust and lift a latent print.

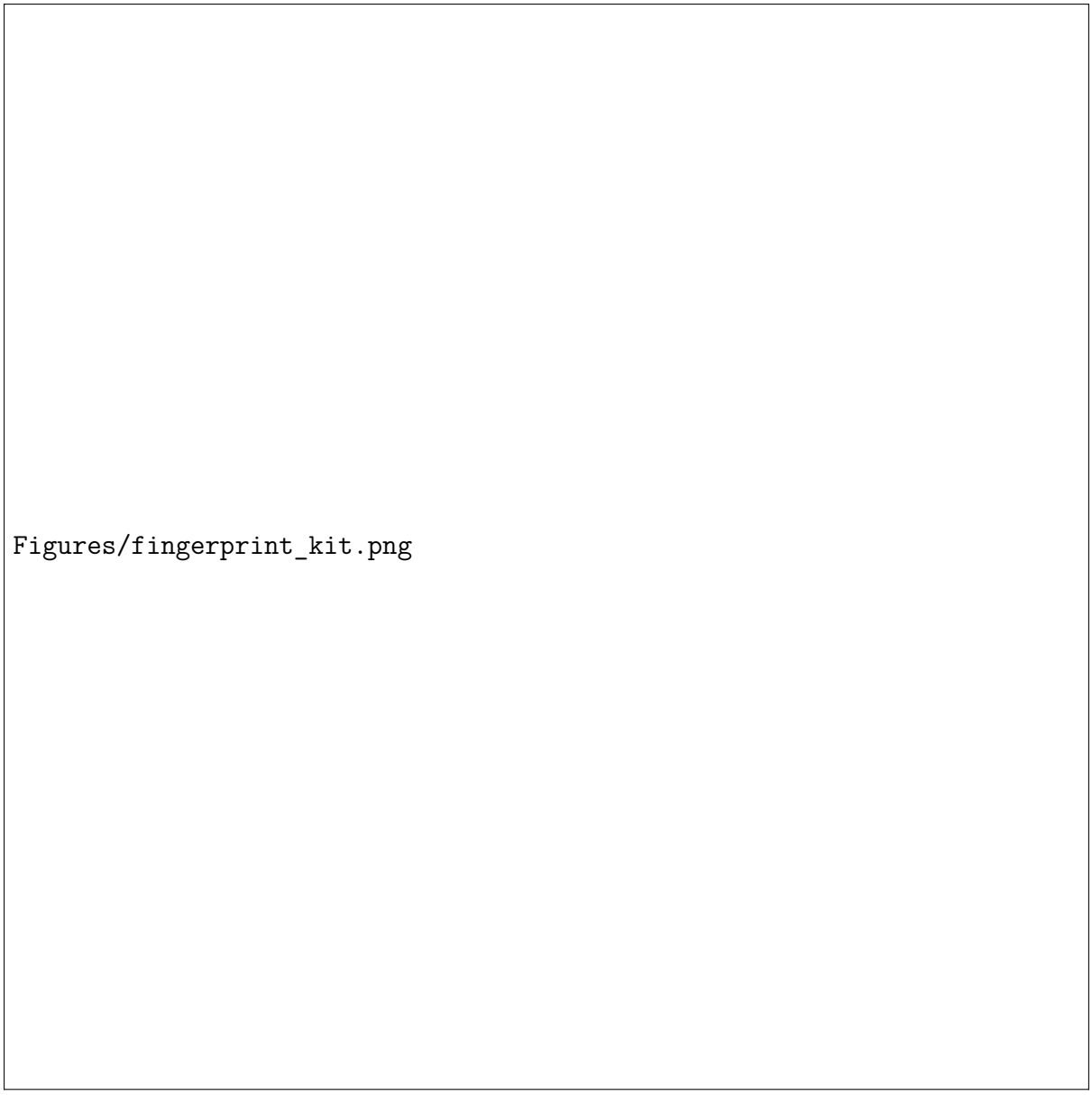


Figure 4.15: Fingerprint kit: dusting a latent print and lifting it onto tape

4.8 Collection and Tagging

4.8.1 Evidence Collection

Evidence may only be collected once it has been fully documented. When the student selects an item that has been photographed, and is therefore eligible for collection — by holding a pick-up item in hand or by pointing the ray at a flat surface item — a prompt floats into view inviting them to squeeze the grip to collect it; the prompt is withheld for items that have not yet reached this stage, enforcing the documentation-before-collection rule. Pressing the collect control opens the tagging interface for that item, through which the item is sealed into a container and transitioned to its terminal Collected state.

Collection is handled by the TaggingSystem, which seals the item, records a chain-of-

custody event, and logs both a TagEvidence and a CollectEvidence action before transitioning the state machine to Collected. Whether the collection is judged correct depends on the container chosen: a container-compatibility table maps each evidence subtype to the containers a technician would accept — blood into a sealed jar or swab kit, a lifted print into an envelope, a hair strand into a sterile bag, and so on — and an incompatible choice still collects the item but is flagged as a wrong-container error for the scoring system. In scenes that do not provide a tagging panel, the system falls back to an automatic seal using the default compatible container so that collection remains possible. Figure 4.16 shows the collection prompt that appears while a fully documented item is held.



Figure 4.16: Collection prompt shown while holding a documented evidence item

4.8.2 Evidence Tagging Interface

When an item is collected, the tagging interface presents the digital evidence tag the system is about to seal. It is implemented as a world-space panel, built at runtime, that displays the tag's fields and lets the student make the one decision that is theirs to make. Most of the tag is populated automatically: a unique identifier is generated for the item, the description defaults to the evidence name, the location is taken from the item's position in the scene, the seal timestamp is set to the moment of collection, and the collector's identity is drawn from the authenticated user. These fields are shown for confirmation rather than manual entry.

The single interactive element is container selection. The panel lists the available container types and highlights the chosen one; confirming with the Collect & Seal button passes the selected container to the TaggingSystem, which seals the tag and logs the action with a correctness flag set according to whether the container suits the evidence subtype, while cancelling leaves the item in the student's hand. Concentrating the student's input on the container choice keeps tagging to a single deliberate decision — the graded chain-of-custody step — while still producing a complete, professionally structured tag. Figure 4.17 shows the tagging panel with its auto-populated fields and the container selection.

Figures/tagging_panel.png

Figure 4.17: Evidence tagging panel: auto-populated tag fields and container selection

4.9 NPC Forensic Advisor

To support students who are unsure how to proceed without simply revealing the answers, the system includes a virtual forensic advisor: a non-player character the student can approach and question in natural language, which replies with contextual guidance generated by a large language model. Consulting the advisor is a graded action — each question carries a small score penalty through the hint deduction described earlier — so the advisor offers a safety net while still encouraging students to reason independently.

4.9.1 NPC Character and Animation

The advisor is a rigged humanoid character positioned within the scene, and the student engages it through the same approach–highlight–prompt–interact idiom used for evidence. When the student aims the controller ray at the advisor from within a set range, the character is outlined with an emissive highlight and a billboarded prompt appears inviting them to select it; selecting opens the conversation panel. While idle, the advisor stays planted but tracks the student naturally through head-and-eye look-at, turning its head, eyes, and a little torso to face the player and re-orienting its body only slowly when the student moves far around to the side, so it appears attentive without walking around the scene.

Its animation is driven by a compact Animator controller with two boolean parameters, one for thinking and one for talking. The character idles by default, plays a thinking animation while a reply is being generated, and a talking animation while the spoken answer is played, returning to idle when finished. Beyond adding a sense of life, these states visually communicate the system's status to the student: the thinking pose signals that the advisor is working on an answer, which masks the latency of the underlying network and model calls. Figure 4.18 shows the forensic advisor character in the scene.

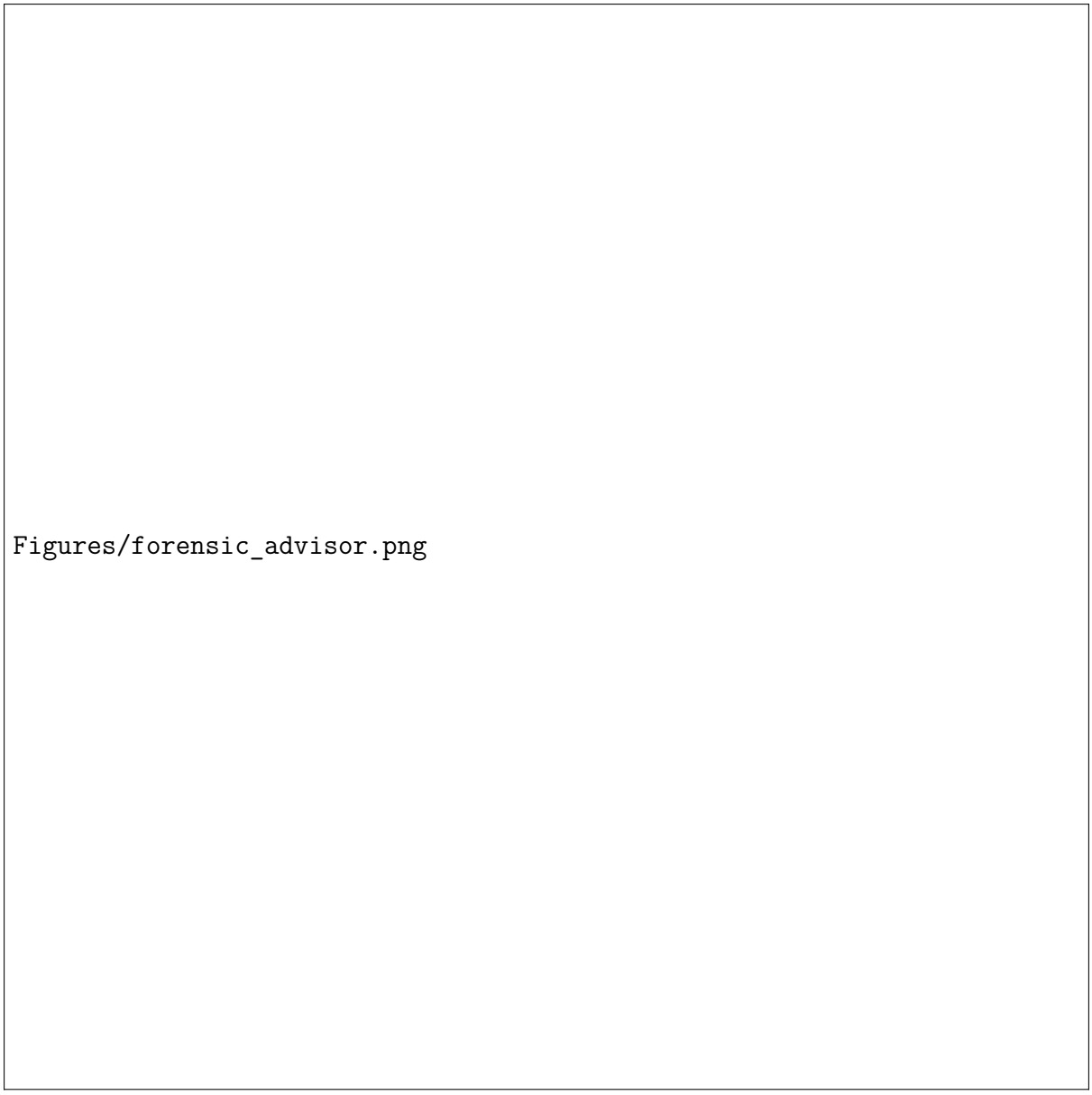


Figure 4.18: Forensic advisor NPC with its engagement prompt

4.9.2 Asynchronous LLM Integration

The advisor's intelligence is provided by a large language model accessed through a separate backend service rather than called directly from the Unity client. When the student submits a question, the client sends it together with a description of the current scene state to the backend, which queries the language model — a Gemini Flash model — and returns a generated hint that the client then displays and speaks aloud. The same backend additionally handles the advisor's two speech functions: it transcribes the student's recorded voice into text and synthesises the advisor's spoken reply from text, so a single service hosts the language, speech-recognition, and speech-synthesis models behind one endpoint. Routing every request through this backend also keeps the models' API credentials on the server rather than embedding them in the distributed build.

The exchange is designed to be non-blocking so the simulation never freezes while

waiting on the network or the model. Each request is issued through Unity's coroutine-based web request mechanism, yielding until the response arrives instead of stalling the main thread, while the advisor's thinking animation plays for the full duration so the wait reads as deliberate consideration rather than a frozen application. Every submitted question is also recorded as a hint-request action, feeding the hint deduction in the scoring rubric.

4.9.3 Input Modalities

The student can pose questions to the advisor in two ways — by speaking or by typing — both surfaced through the conversation panel, a world-space interface that displays the current question, the advisor's answer, and a short status line.

Speech-to-Text

The primary input is voice, using a push-to-talk model. Holding the designated controller button, or pressing the Speak control on the panel, begins recording from the microphone; releasing it stops the recording and submits it for transcription. The captured audio is encoded as a WAV clip and posted to the backend's speech-recognition endpoint, which is backed by a Whisper model and returns the recognised text. The transcription is then revealed on the panel with a typewriter effect and submitted as the question. If nothing intelligible is recognised, the student is prompted to try again or switch to typing.

Virtual Keyboard Text Input

As an alternative to voice — useful in noisy surroundings or when recognition struggles — the panel provides a built-in on-screen keyboard. Selecting the Type control reveals a QWERTY keyboard whose keys are pressed with the controller ray; the keys build up the question text and are accompanied by space, backspace, and enter keys, with enter submitting the typed question through the same path as a spoken one. Offering both modalities keeps the advisor accessible regardless of the student's surroundings or comfort with voice input. Figure 4.19 shows the conversation panel with the on-screen keyboard.

Figures/advisor_panel.png

Figure 4.19: Advisor conversation panel with question, answer, and on-screen keyboard

4.9.4 Scene Context Serialization

For the advisor's answers to be useful they must reflect what is actually happening in the scene, so each question is accompanied by a freshly generated description of the current investigation state. Immediately before a question is sent, the client builds this context from the live evidence registry: it states the setting, summarises overall progress in terms of how many items have been detected and collected out of the total, and lists each evidence item with its current status, name, type, and a human-readable location derived from its object name. It also appends a reminder of the correct handling order — detect, mark, photograph, then collect and tag — so the advisor's guidance stays aligned with the procedure the system teaches.

This context is regenerated for every question rather than stored, so the advisor always reasons over the present state of the investigation, and it is sent alongside the

question to the backend as a single structured request. Each question is therefore treated independently, with the relevant scene state supplied afresh each time, which keeps the request self-contained and the advisor's answers grounded in the student's actual progress.

4.9.5 Text-to-Speech Output

So that the advisor feels like a person being consulted rather than a text box, its answers are delivered as spoken voice in addition to on-screen text. Once the language model's reply has been received and shown on the panel, the client sends the answer text to the backend's text-to-speech endpoint, which is backed by a Piper voice model and returns the synthesised speech as audio. The audio is returned as a WAV stream, which the client decodes at runtime into a Unity AudioClip so it can be played back immediately without writing any intermediate file to disk.

The decoded clip is assigned to the advisor's audio source and played, and the talking animation is enabled for the duration of playback and switched off once the clip finishes, so the character's mouth and gesture animation are synchronised with the spoken answer. Because the audio source sits on the advisor itself, the voice is spatialised to the character's position in the scene, reinforcing the sense that the guidance is coming from the advisor the student is standing in front of. As with the rest of the pipeline, synthesis is performed without blocking the simulation, so the student remains free to look around the scene while the answer is spoken.

4.10 Firebase Integration

The application's backend is built on Firebase, integrated through the Firebase Unity SDK introduced earlier. Rather than scattering Firebase calls across the gameplay code, all backend access is organised into a layered data architecture that separates responsibilities into three tiers. At the lowest level, model classes are plain C# objects that mirror the structure of Firestore documents — users, crime scenes, evidence, investigations, and the actions logged within them — each providing methods to convert itself to and from its Firestore representation. Above them, repository classes encapsulate all direct communication with Firebase, issuing the actual authentication calls and database queries. At the top, service classes expose a clean, task-oriented interface to the rest of the application, so that the user interface and gameplay systems call the services and never touch the Firebase SDK directly.

A single component, FirebaseCore, owns the connection itself: it initialises the SDK at startup as described earlier, holds the shared authentication and Firestore handles, and exposes an awaitable readiness guard that every repository waits on before issuing a call, so no operation runs against an uninitialised connection. This layering keeps the Firebase dependency isolated in one place, leaves the gameplay code agnostic to how data is stored, and made the data layer straightforward to exercise in isolation through the project's test scripts.

4.10.1 Authentication and Role-Based Access

Authentication is handled through Firebase Authentication using email-and-password credentials. The AuthService sits in the service layer and exposes registration, login, logout,

and password-reset operations, delegating the underlying Firebase calls to the UserRepository. Registering a new account performs two steps: it creates the authentication record and then writes a corresponding user profile document to the users collection, storing the username, email, and an assigned role; logging in authenticates the credentials and loads the matching profile. The service also subscribes to Firebase's own authentication-state changes, so that events such as a token refresh or an external sign-out are reflected automatically, and it raises a set of events — signed in, signed out, profile loaded, and error — that the login and menu interfaces subscribe to in order to update what the user sees.

Each user profile carries a role of student, instructor, or administrator, stored as a field on the user document rather than embedded in the credentials, which allows an administrator to change another user's role at runtime. The RoleService reads this role and exposes the simple checks used throughout the application to decide what a user may do. In particular, the role determines which interface a user is presented with on login — the scene-and-mode selection menu for students, the management dashboard for administrators, and the performance-review dashboard for instructors, as described in the user interface section — so a single build serves all three user types from one authenticated entry point.

4.10.2 Cloud Firestore Data Access

Persistent data is stored in Cloud Firestore, Firebase's NoSQL document database, whose hierarchical document-and-collection model suits the project's semi-structured records. The data is organised into top-level collections for users, crime scenes, and investigations, with related records nested beneath them — an investigation document, for instance, owns the sequence of actions the student performed, the set of evidence spawned for that session, the items they collected, and the generated performance report. Each repository owns one of these areas: the SceneRepository reads the available crime scenes and their evidence when a session starts, while the InvestigationRepository creates the investigation record and writes back the session's results.

All database access is asynchronous, using the C# async/await pattern so that queries never block the main thread or stall rendering. The randomised set of evidence spawned for a session is recorded when the investigation begins, while the actions the student performs are accumulated in memory during play and written to Firestore in a single batch when the session ends — alongside the collected-evidence records and the performance report — rather than incurring a separate network write for every action. Reads, such as loading the list of selectable scenes or an instructor retrieving a student's past results, are performed on demand through the corresponding service. Because every document maps cleanly to a model class through its conversion methods, the higher layers work with strongly-typed objects throughout and remain insulated from the details of the Firestore schema.

Chapter 5

Testing and Result

This chapter presents the testing process carried out on the Crime Scene Virtual Reality Simulation system. Three main features were selected for testing, with a total of ten test cases designed and executed across all three features. A structured Test Case Execution Template was followed for each test case, covering the test scenario, description, preconditions, steps, expected results, actual results, and pass/fail status.

5.1 Test Plan

The testing approach followed functional black-box testing, where each feature was tested based on its expected behavior as defined in the system requirements and use case specifications. Test cases were designed to cover positive flows (expected correct behavior), negative flows (incorrect input or out-of-sequence actions), and edge cases (such as network failure). Testing was carried out on the implemented Unity VR application using a Meta Quest headset in Interactive Mode.

5.2 Main Features Selected for Testing

The three main features selected for testing represent the core functional areas of the system: the AI-powered hint system, the evidence handling workflow, and the session initialization flow.

5.2.1 NPC-Based AI Hint System

The NPC-Based AI Hint System allows students to ask a virtual forensic advisor for contextual guidance during an investigation. The system sends the student's question to an LLM service asynchronously, displays a thinking animation while waiting, and presents the response as text and voice. This feature was selected because it is the primary differentiator of the system from existing forensic training tools.

5.2.2 Evidence Collection and Tagging

The Evidence Collection and Tagging feature covers the process of collecting a detected and marked evidence item and automatically generating a complete digital tag containing

a unique identifier, description, location coordinates, timestamp, and chain-of-custody information. This feature was selected because it represents the core investigative workflow that all students must complete in every session.

5.2.3 Mode Selection and Scene Loading

The Mode Selection and Scene Loading feature covers the process by which a student selects either Interactive or Observational mode and loads the crime scene. The mode chosen directly determines whether students can interact with evidence or only observe the scene. This feature was selected because it is the entry point for every investigation session and its correct behavior is a prerequisite for all other features.

5.3 Test Cases Execution

5.3.1 Feature 1: NPC-Based AI Hint System

Table 5.1: TC-01: Approach the NPC and Submit a Question Using Voice Input (Positive Flow)

TEST CASE TITLE	TC-01: Approach the NPC and Submit a Question Using Voice Input (Positive Flow)
TEST DATE	15/06/2026
TEST SCENARIO	Request Hints and Guidance (UC-11) / Generate Contextual Forensic Response (UC-17) — voice input modality
TEST DESIGNED BY	Hanan Naji
TEST DESCRIPTION	Verifying the full NPC interaction flow using voice input: the student aims the controller at the advisor, selects it to open the conversation panel, holds the push-to-talk button to record a question, and receives a spoken and on-screen response while the NPC plays the correct animations throughout.
PRECONDITIONS	• Student is logged in with an active investigation session • Student is in Advanced Interactive Mode within range of the NPC advisor • Internet connection is available so the backend service can be reached • The device microphone is functioning

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Aim the controller ray at the NPC advisor from within interaction range	The advisor is outlined with an emissive highlight and a billboarded prompt appears inviting the student to select it	The advisor glowed with an emissive highlight and the selection prompt appeared	✓	
2	Select the advisor using the controller ray	The conversation panel opens displaying the question area, the advisor's answer area, and a status line	The conversation panel opened with all elements visible	✓	
3	Hold the push-to-talk controller button to record a forensics-related question	Recording begins from the microphone; the panel indicates recording is in progress	Recording started and the panel showed a recording indicator	✓	
4	Release the button to stop recording and submit	The recognised text appears on the panel with a typewriter effect; the NPC begins its thinking animation	The transcribed text appeared and the NPC started its thinking animation	✓	
5	Observe the NPC and simulation while the reply is generated	The thinking animation plays; the simulation stays fully interactive with no freezing	The thinking animation played; the simulation stayed fully interactive	✓	Request runs through a coroutine so the main thread is never stalled

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
6	Wait for the spoken and on-screen response	The answer appears on the panel and is spoken aloud; the NPC switches to talking animation then returns to idle; the response is relevant to the investigation	The answer appeared and was spoken; talking animation played then returned to idle; response was relevant	✓	Voice is spatialised to the advisor's position; talking animation is synchronised with the audio

Table 5.3: TC-02: Submit a Question Using the Virtual Keyboard Text Input (Alternative Modality)

TEST CASE TITLE	TC-02: Submit a Question Using the Virtual Keyboard Text Input (Alternative Modality)
TEST DATE	15/06/2026
TEST SCENARIO	Request Hints and Guidance (UC-11) — text input modality via on-screen keyboard
TEST DESIGNED BY	Hanan Naji
TEST DESCRIPTION	Verifying the alternative text input modality: the student selects the NPC, chooses the Type control on the conversation panel to reveal the on-screen QWERTY keyboard, uses the controller ray to type a question, and receives the same spoken and on-screen response as with voice input.
PRE-CONDITIONS	• Student is logged in with an active investigation session • Student is in Advanced Interactive Mode within range of the NPC advisor • Internet connection is available so the backend service can be reached

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Aim at and select the NPC advisor to open the conversation panel	The conversation panel opens; both a Speak control and a Type control are visible	The conversation panel opened with both input controls visible	✓	
2	Select the Type control on the panel	An on-screen QWERTY keyboard appears; keys can be pressed using the controller ray	The on-screen keyboard appeared and responded to controller ray input	✓	Useful when voice recognition struggles or surroundings are noisy
3	Type a question key by key and press Enter to submit	The typed text builds up on the panel; pressing Enter submits through the same path as voice; the NPC begins its thinking animation	Text built up correctly; Enter submitted the question; thinking animation started	✓	
4	Observe the NPC and simulation while the reply is generated	The thinking animation plays; the simulation remains fully interactive	The thinking animation played; the simulation remained fully interactive	✓	
5	Wait for the spoken and on-screen response	The answer appears on the panel and is spoken aloud; the talking animation plays then returns to idle	The answer appeared, was spoken, and the talking animation was synchronised with the voice	✓	Follows the same pipeline as voice input

Table 5.5: TC-03: NPC Hint Request Does Not Block VR Interaction (Non-Blocking Behavior)

TEST CASE TITLE	TC-03: NPC Hint Request Does Not Block VR Interaction (Non-Blocking Behavior)
TEST DATE	15/06/2026
TEST SCENARIO	Request Hints and Guidance (UC-11) running concurrently with Navigate Crime Scene (UC-2)
TEST DESIGNED BY	Hanan Naji
TEST DESCRIPTION	Verifying that submitting a question to the NPC does not freeze or block the VR simulation: the student can continue navigating the scene and interacting with evidence while the request is being processed asynchronously in the background.
PRECONDITIONS	• Student is logged in with an active investigation session • Student is in Advanced Interactive Mode and has submitted a question to the NPC • The NPC thinking animation is playing and the reply has not yet arrived • Internet connection is available

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Submit a question to the NPC via voice or keyboard	The NPC begins its thinking animation; the request is sent to the backend through a coroutine so the main thread is not blocked	The NPC started its thinking animation and the request was sent in the background	✓	
2	While waiting for the reply, teleport to a different location in the scene	The student can navigate normally using the teleportation arc; no freezing or delay is observed	Navigation worked normally while the NPC was processing; no freezing was observed	✓	

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
3	While still waiting, approach and interact with an evidence item	The student can hover, select, and interact with evidence normally; the evidence highlight and haptic feedback respond as expected	The student interacted with evidence normally; all evidence responses worked as expected	✓	
4	While still waiting, look around the scene freely	Head tracking and view updates respond in real time with no lag or stuttering	Head tracking and view worked normally with no lag	✓	
5	Wait for the NPC response to arrive	The NPC switches from thinking to talking animation; the spoken and on-screen response is delivered; the simulation was fully interactive throughout	The NPC delivered the spoken and on-screen response; the simulation was fully interactive throughout	✓	

Table 5.7: TC-04: NPC Request When the Backend Service Is Unreachable (Negative Flow)

TEST CASE TITLE	TC-04: NPC Request When the Backend Service Is Unreachable (Negative Flow)
TEST DATE	15/06/2026
TEST SCENARIO	Request Hints and Guidance (UC-11) / Generate Contextual Forensic Response (UC-17) — backend service unavailable
TEST DESIGNED BY	Osayed Zaheda
TEST DESCRIPTION	Verifying system behaviour when the backend service cannot be reached: the simulation must remain running, the thinking animation must play during the wait, the student must receive a clear error notification, and no blank or fabricated response must be shown.
PRE-CONDITIONS	• Student is logged in with an active investigation session • Student is in Advanced Interactive Mode and has opened the NPC conversation panel • The device network connection can be disabled to simulate the backend being unreachable

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Disable the device network connection	The simulation continues running normally; no crash or freeze occurs	The simulation kept running normally; no crash occurred	✓	
2	Submit a question via voice or keyboard	The NPC begins its thinking animation as it would for a normal request	The question was submitted and the NPC started its thinking animation	✓	
3	Wait while the system attempts to reach the backend	The NPC stops thinking; a clear error message appears on the panel; no blank or fabricated answer is shown	The error notification appeared on the panel; no fake response was shown; NPC returned to idle	✓	

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
4	Verify the simulation state after the failed request	The VR session remains fully interactive; all other investigation actions still work	The simulation remained fully interactive; all actions continued normally	✓	
5	Re-enable the network and submit the same question again	The request reaches the backend; thinking animation plays; a spoken and on-screen response is returned	The request succeeded; the advisor delivered the spoken and on-screen response	✓	

5.3.2 Feature 2: Evidence Collection and Tagging

Table 5.9: TC-05: Complete Evidence Documentation and Collection — Full Positive Flow

TEST CASE TITLE	TC-05: Complete Evidence Documentation and Collection — Full Positive Flow
TEST DATE	12/06/2026
TEST SCENARIO	Full correct sequence: discover, mark, photograph (3 stages), collect, tag with container selection
TEST DESIGNED BY	Osayed Zaheda
TEST DESCRIPTION	Verifying that following the full correct forensic sequence causes the system to open the tagging panel and generate a complete digital evidence tag after the student selects the correct container.
PRECONDITIONS	• Student is logged in with an active investigation session • Application is in Advanced Interactive Mode • An evidence item is in its initial undiscovered state • Marker Tool, Virtual Camera, and Scale Tool are in the radial menu

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Point the controller at the evidence and touch/select it	The evidence glows yellow and haptic feedback is felt; the item is now discovered	The evidence glowed yellow, haptic feedback was felt, and the item became discoverable	✓	
2	Equip the Marker Tool; point at the evidence and press the trigger	A green aiming indicator appears; a numbered tent-card marker is placed beside the item and haptic feedback fires	The green indicator appeared, the marker was placed, and haptic feedback fired	✓	Marker can only be placed on a discovered item
3	Equip the Virtual Camera; stand more than 2 m away and press the trigger (Shot 1 — Overview)	HUD shows Shot 1/3; viewfinder border turns green at correct distance; shot is captured	HUD showed Shot 1/3, border turned green, and the shot was captured	✓	Must be at least 2 m from the evidence
4	Move to 0.5–2 m, keep the marker in the viewfinder, press trigger (Shot 2 — Mid-Range)	HUD shows Shot 2/3; shot accepted with marker in frame	HUD showed Shot 2/3, marker was in frame, and the shot was accepted	✓	Shot rejected if marker is not in view
5	Place the Scale Tool beside the evidence; move within 0.5 m, keep scale in view, press trigger (Shot 3 — Close-Up)	HUD shows Shot 3/3; shot accepted; HUD confirms documentation complete	HUD showed Shot 3/3, scale in frame, shot accepted, HUD confirmed complete	✓	Shot rejected if scale is not in view

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
6	Grab the evidence and squeeze the left grip to collect it	A collect prompt appears; squeezing opens the Tagging Panel showing auto-filled fields (ID, description, location, timestamp, collector) and a list of container types; selecting the correct container and pressing Collect & Seal removes the item from the scene	The Tagging Panel opened with all auto-filled fields; correct container was selected; Collect & Seal sealed and removed the item	✓	Wrong container still collects but is flagged as an error in the score

Table 5.11: TC-06: Collect Before Photography — Violation Logged, Collection Not Blocked

TEST CASE TITLE	TC-06: Collect Before Photography — Violation Logged, Collection Not Blocked
TEST DATE	12/06/2026
TEST SCENARIO	Evidence collected after marking but without photography — violations recorded and score reduced
TEST DESIGNED BY	Osayed Zaheda
TEST DESCRIPTION	Verifying that skipping photography and collecting right after placing the marker is allowed by the system, but is recorded as procedural violations that reduce the procedure score and cause LO-2 to fail.
PRE-CONDITIONS	• Student is logged in with an active investigation session • Application is in Advanced Interactive Mode • An evidence item has been discovered and marked • No photographs have been taken for this item

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Without taking any photos, grab the marked evidence item	A collect prompt appears; collection is available at this stage	The collect prompt appeared	✓	
2	Squeeze the left grip to collect	The evidence is collected and removed without blocking	The evidence was collected and removed without any blocking	✓	Collection is allowed after marking even without photos
3	Check the score on the HUD right after collection	The procedure score decreases to reflect penalties for skipping photography	The HUD score decreased immediately	✓	
4	End the session and view the performance report	Report lists violations for collecting before photography and for each missing photo stage; LO-2 shows Fail	Report listed all violations correctly and LO-2 showed Fail	✓	

Table 5.13: TC-07: Photograph Before Placing Marker — Camera Rejects the Shot

TEST CASE TITLE	TC-07: Photograph Before Placing Marker — Camera Rejects the Shot
TEST DATE	12/06/2026
TEST SCENARIO	Photography is blocked when no marker is placed; HUD shows error and trigger is rejected
TEST DESIGNED BY	Osayed Zaheda
TEST DESCRIPTION	Verifying that the Virtual Camera rejects all photographs when no marker has been placed on the evidence, showing a clear error on the HUD and enforcing the correct forensic sequence.
PRE-CONDITIONS	• Student is logged in with an active investigation session • Application is in Advanced Interactive Mode • An evidence item has been discovered but no marker has been placed • The Virtual Camera is equipped from the radial menu

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Point the camera at the discovered unmarked evidence	The HUD shows the photo stage and displays a red error message	The HUD showed the error message in red	✓	
2	Stand at a valid overview distance and press the trigger	Viewfinder stays red; trigger is rejected; error sound plays; no photo is taken	Viewfinder stayed red, trigger was rejected, error sound played	✓	Shot rejected because no marker has been placed
3	Place a marker on the evidence then re-equip the camera	The error message disappears; viewfinder turns green at the correct distance	Error disappeared and viewfinder turned green	✓	
4	Press the trigger at a valid overview distance	Shot is accepted; HUD confirms the first photo and shows the next stage	Shot was accepted and HUD advanced to the next stage	✓	Photography works normally after a marker is placed

5.3.3 Feature 3: Mode Selection and Scene Loading

Table 5.15: TC-08: Select Interactive Mode and Load the Crime Scene (Positive Flow)

TEST CASE TITLE	TC-08: Select Interactive Mode and Load the Crime Scene (Positive Flow)
TEST DATE	12/06/2026
TEST SCENARIO	Student selects Interactive mode and the scene loads with evidence interaction enabled
TEST DESIGNED BY	Tasneem Abu Sara
TEST DESCRIPTION	Verifying that selecting Interactive mode loads the crime scene with all evidence items available for grabbing and collection.
PRE-CONDITIONS	• Student is logged in and the mode selection panel is shown • No scene is currently loading

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Press the Interactive button on the mode panel	Interactive mode is selected and the scene selection panel appears	The scene selection panel appeared	✓	
2	Select any scene card	The card is highlighted as selected	The card was highlighted	✓	
3	Press the Start button	The crime scene starts loading	The crime scene started loading	✓	The same default scene loads in the current version
4	Wait for the scene to load and hover over an evidence item	Scene loads fully; evidence items glow on hover and can be grabbed	Scene loaded; evidence items glowed on hover and were grabbable	✓	

Table 5.17: TC-09: Select Observational Mode — Evidence Interaction Disabled (Positive Flow)

TEST CASE TITLE	TC-09: Select Observational Mode — Evidence Interaction Disabled (Positive Flow)
TEST DATE	12/06/2026
TEST SCENARIO	Student selects Observational mode and evidence becomes non-interactive
TEST DESIGNED BY	Tasneem Abu Sara
TEST DESCRIPTION	Verifying that Observational mode loads the scene in a view-only state where evidence cannot be grabbed or collected.
PRE-CONDITIONS	• Student is logged in and the mode selection panel is shown • No scene is currently loading

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Press the Observational button on the mode panel	Observational mode is selected and the scene selection panel appears	The scene selection panel appeared	✓	
2	Press Start and wait for the scene to load	The scene loads in Observational mode	The scene loaded successfully	✓	
3	Hover over and try to grab an evidence item	Evidence does not glow on hover and cannot be grabbed	No glow appeared; the item could not be grabbed	✓	Evidence interaction is disabled in this mode
4	Try to inspect an evidence item	Inspection is not available; no response occurs	Inspection had no effect	✓	
5	Move and look around the scene freely	Navigation and head movement work normally	Movement and view worked normally	✓	Scene navigation is not affected by the mode

Table 5.19: TC-10: Pressing Start Multiple Times During Load Does Not Duplicate the Load (Negative Flow)

TEST CASE TITLE	TC-10: Pressing Start Multiple Times During Load Does Not Duplicate the Load (Negative Flow)
TEST DATE	12/06/2026
TEST SCENARIO	Extra Start button presses during loading are safely ignored
TEST DESIGNED BY	Tasneem Abu Sara
TEST DESCRIPTION	Verifying that pressing Start multiple times while the scene is loading results in only one scene load, with all extra presses being ignored.
PRE-CONDITIONS	• Student is logged in and on the scene selection panel after choosing a mode • No scene is currently loading

Step	Step Description	Expected Results	Actual Results	Pass / Fail	Notes
1	Press the Start button once	The scene starts loading	The scene started loading	✓	
2	Press Start again while loading	The extra press is ignored; no second load starts	The second press had no effect	✓	
3	Keep pressing Start while loading	All extra presses are ignored; loading continues normally	All extra presses were ignored	✓	
4	Wait for the scene to finish loading	The scene loads correctly and becomes ready	The scene loaded correctly	✓	
5	Go back to scene selection and press Start again	A new load starts normally	A new load started normally	✓	

5.4 Test Results Summary

Table 5.21 summarizes the results of all ten test cases.

Table 5.21: Test Results Summary

TC	Title	Feature	Result
TC-01	Approach NPC and Submit Voice Question	NPC Hint System	✓
TC-02	Submit Question via Virtual Keyboard	NPC Hint System	✓
TC-03	NPC Hint Does Not Block VR Interaction	NPC Hint System	✓
TC-04	NPC Request When Backend Unreachable	NPC Hint System	✓
TC-05	Complete Evidence Documentation and Collection	Evidence Collection & Tagging	✓
TC-06	Collect Before Photography – Violation Logged	Evidence Collection & Tagging	✓
TC-07	Photograph Before Marker – Shot Rejected	Evidence Collection & Tagging	✓
TC-08	Select Interactive Mode and Load Scene	Mode Selection & Scene Loading	✓
TC-09	Select Observational Mode – Interaction Disabled	Mode Selection & Scene Loading	✓
TC-10	Repeated Start Press – Load Not Duplicated	Mode Selection & Scene Loading	✓

All ten test cases passed. Each feature behaved as expected based on the system implementation described in Chapter 4.

Chapter 6

Conclusion and Future Works

6.1 Review of the project

This project delivered both the design and a functional implementation of a Virtual Reality-based crime scene simulation intended for forensic science education. Grounded in an analysis of existing forensic VR tools and the requirements derived from them, it produced a working application that combines immersive VR environments with procedural training, gamification, and structured feedback, addressing the difficulties and limitations of traditional forensics training and offering a more interactive, scalable, and risk-free alternative that enhances observational and evidence-handling skills. The core investigative workflow — scene navigation, evidence observation and inspection, marker placement, photography, fingerprint and UV examination, collection and tagging, scoring and report generation, and an NPC-based forensic advisor — was implemented and integrated with a Firebase backend for authentication and data persistence.

6.2 Future Work

The work this semester followed a staged development plan, delivering an implementation of the planned system: the core VR environment, the evidence-interaction and tagging mechanisms, the scoring, feedback, and report-generation components, the controlled evidence randomization, the NPC-based guidance system, the instructor and administrator dashboards, and the base observational mode, together with supporting features such as an onboarding tutorial and an end-of-scene debriefing. With the system implemented, future work will focus on validating and refining it: structured user testing with forensic science students to measure the system's impact on engagement and procedural accuracy, followed by system refinement, performance optimization, creating multiple scene environments and completion of the final project documentation.

Bibliography

[1] Interaction Design Foundation, *What is extended reality (xr)?* Available: https://www.interaction-design.org/literature/topics/extended-reality-xr. Accessed: 12 Dec. 2025, 2025.

[2] Wikipedia Contributors, *Extended reality*, Available: https://en.wikipedia.org/wiki/Extended_reality. Accessed: 13 Dec. 2025, 2025.

[3] Interaction Design Foundation, *What is virtuality continuum?* Available: https://www.interaction-design.org/literature/topics/virtuality-continuum. Accessed: 14 Dec. 2025, Jan. 2022.

[4] Ozemio, *Virtual reality learning: The future of immersive training*, Available: https://ozemio.com/articles/virtual-reality-learning-the-future-of-immersive-training. Accessed: 18 Dec. 2025, 2025.

[5] Wangfred, *Three types of virtual reality (vr): A deep dive into immersive digital worlds*, Available: https://inairspace.com/blogs/learn-with-inair/three-types-of-virtual-reality-vr-a-deep-dive-into-immersive-digital-worlds. Accessed: 16 Dec. 2025, 2025.

[6] Mechatech, *What is a 3dof vs 6dof in vr*, Available: https://www.mechatech.co.uk/journal/what-is-a-3dof-vs-6dof-in-vr. Accessed: 17 Dec. 2025, 2025.

[7] TechTerms, *Hmd - head-mounted display*, Available: https://techterms.com/definition/hmd. Accessed: 19 Dec. 2025, 2025.

[8] Laval Virtual, *Cave - cave automatic virtual environment*, Available: https://blog.laval-virtual.com/en/glossary/cave/. Accessed: 20 Dec. 2025, 2025.

[9] F. Beji, W. d. P. Ferreira, I. Pivotto Dabat, and V. Matias, "Cave automatic virtual environment technology: A patent analysis," *Engineering Proceedings*, vol. 89, no. 1, p. 9, 2025, Available: https://www.mdpi.com/2673-4591/89/1/9. DOI: 10.3390/engproc2025089009

[10] C. Boletsis and D. Chasanidou, "A typology of virtual reality locomotion techniques," *Multimodal Technologies and Interaction*, vol. 6, no. 9, 2022, Available: https://www.researchgate.net/publication/362946674.

[11] D. Lochner and J. Gain, "Vr natural walking in impossible spaces," in *Proceedings of the 27th ACM Symposium on Virtual Reality Software and Technology*, Nov. 2021, pp. 1–9. DOI: 10.1145/3487983.3488305

[12] M.-X. Chen, Y.-T. Chen, C.-J. Hsu, and L.-Y. Cheng, "A survey on the design of virtual reality interaction interfaces," *Sensors*, vol. 24, no. 19, 2024, Available: https://www.mdpi.com/1424-8220/24/19/6204.

[13] J. Jerald, *The VR Book: Human-Centered Design for Virtual Reality*. ACM Books, 2016.

[14] B. Spittle, S. Lips, A. Amresh, and M. Banda, "A review of interaction techniques for immersive environments," *International Journal of Human-Computer Interaction*, 2021, Available: https://www.open-access.bcu.ac.uk/13376/.

[15] Antaeus AR, *Design principles & issues for gaze and pinch interaction*, Available: https://medium.com/antaeus-ar/design-principles-issues-for-gaze-and-pinch-interaction-a95e251169ae. Accessed: 22 Dec. 2025, 2024.

[16] T.-H. Tseng, S. Huron, E. Lecolinet, and B. Gugenheimer, "Fingermapper: Mapping finger motions onto virtual arms," *Proceedings of the ACM CHI Conference on Human Factors in Computing Systems*, 2023, Available: https://arxiv.org/abs/2302.11865.

[17] S. Aufenanger et al., "Immersive virtual reality learning environments for higher education: A student acceptance study," *Computers & Education: X Reality*, vol. 7, 2025, Available: https://doi.org/10.1016/j.cexr.2025.100105.

[18] M. Hamash, P. Tiernan, and G. Young, "Virtual reality in post-primary education research trends from 2013 to 2024," *Computers & Education: X Reality*, vol. 7, 2025, Available: https://doi.org/10.1016/j.cexr.2025.100123.

[19] S. Deterding, D. Dixon, R. Khaled, and L. Nacke, "From game design elements to gamefulness: Defining gamification," *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments*, pp. 9–15, 2011, Available: https://doi.org/10.1145/2181037.2181040.

[20] K. Seaborn and D. I. Fels, "Gamification in theory and action: A survey," *International Journal of Human-Computer Studies*, vol. 74, pp. 14–31, 2015, Available: https://doi.org/10.1016/j.ijhcs.2014.09.006.

[21] E. L. Deci and R. M. Ryan, *The "what" and "why" of goal pursuits: Human needs and the self-determination of behavior*. 2000, vol. 11, pp. 227–268, Available: https://doi.org/10.1207/S15327965PLI1104_01.

[22] M. Sailer, J. U. Hense, S. K. Mayr, and H. Mandl, "How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction," *Computers in Human Behavior*, vol. 69, pp. 371–380, 2017, Available: https://doi.org/10.1016/j.chb.2016.12.033.

[23] D. Dicheva, C. Dichev, G. Agre, and G. Angelova, "Gamification in education: A systematic mapping study," *Educational Technology & Society*, vol. 18, pp. 75–88, Jul. 2015.

[24] A. C. T. Klock, I. Gasparini, M. S. Pimenta, and J. Hamari, "Tailored gamification: A review of literature," *International Journal of Human-Computer Studies*, vol. 144, 2020, Available: https://doi.org/10.1016/j.ijhcs.2020.102495.

[25] G. Makransky and G. B. Petersen, "The cognitive affective model of immersive learning (camil): A theoretical research-based model of learning in immersive virtual reality," *Educational Psychology Review*, vol. 33, pp. 937–958, 2021, Available: https://doi.org/10.1007/s10648-020-09586-2.

[26] I. Gorbanev et al., "A systematic review of serious games in medical education: Quality of evidence and pedagogical strategy," *Medical Education Online*, vol. 23, no. 1, 2018, Available: https://doi.org/10.1080/10872981.2018.1438718.

[27] A. All, E. P. N. Castellar, and J. Van Looy, "Assessing the effectiveness of digital game-based learning: Best practices," *Computers & Education*, vol. 92, pp. 90–103, 2016, Available: https://doi.org/10.1016/j.compedu.2015.10.007.

[28] E. Kasneci et al., "Chatgpt for good? on opportunities and challenges of large language models for education," *Learning and Individual Differences*, vol. 103, 2023, Available: https://doi.org/10.1016/j.lindif.2023.102274.

[29] J. Rickel and W. L. Johnson, "Animated agents for procedural training in virtual reality: Perception, cognition, and motor control," *Applied Artificial Intelligence*, vol. 13, no. 4-5, pp. 343–382, 1999, Available: https://doi.org/10.1080/088395199117315.

[30] Forensics Colleges & Universities, *Forensics colleges*, Available: https://www.forensicscolleges.com. Accessed: 28 Dec. 2025, 2025.

[31] R. Saferstein, *Criminalistics: An Introduction to Forensic Science*, 11th. Pearson, 2015.

[32] B. A. J. Fisher and D. R. Fisher, *Techniques of Crime Scene Investigation*, 8th. CRC Press, 2012.

[33] M. M. Houck and J. A. Siegel, *Fundamentals of Forensic Science*, 2nd. Academic Press, 2010.

[34] E. Locard, "The analysis of dust traces," *American Journal of Police Science*, vol. 1, no. 3, pp. 276–298, 1930.

[35] R. M. Gardner and T. Bevel, *Practical Crime Scene Analysis and Reconstruction*. CRC Press, 2012.

[36] Technical Working Group on Crime Scene Investigation, "Crime scene investigation: A guide for law enforcement," National Institute of Justice, U.S. Department of Justice, Tech. Rep. NCJ 243598, 2013, Available: https://www.ojp.gov/pdffiles1/ncjrs/243598.pdf.

[37] Crime Scene, *Evidence tents with scale numbered 1-20*, Available: https://shop.crimescene.com/product/evidence-tents-with-scale-numbered-1-20/. Accessed: 30 Dec. 2025, 2025.

[38] Crime Scene, *Evidence marking flags*, Available: https://shop.crimescene.com/product/evidence-marking-flags-p-161/. Accessed: 31 Dec. 2025, 2025.

[39] Crime Scene, *Evidence labels*, Available: https://shop.crimescene.com/product/evidence-labels-p-311/. Accessed: 2 Dec. 2025, 2025.

[40] Arrowhead Forensics, *Abfo no. 2 photomacrographic scales*, Available: https://arrowheadforensics.com/a-6200-abfo-no-2-photomacrographic-scales-white-plastic.html. Accessed: 3 Dec. 2025, 2025.

[41] D. S. Byrne, "Learning by doing: Csi comes alive," *Journal of Social Science Research*, vol. 13, 2018, Available: https://www.farmingdale.edu/facultycenter/studentsfirst/pdf/learning-by-doing.pdf.

[42] J. Jani and A. Johnson, "Virtual reality and its transformation in forensic education and research practices," *Journal of Visual Communication in Medicine*, vol. 45, no. 1, pp. 18–25, 2022, Available: https://doi.org/10.1080/17453054.2021.1971516.

[43] J. Wang et al., "Virtual reality and integrated crime scene scanning for immersive and heterogeneous crime scene reconstruction," *Forensic Science International*, vol. 303, 2019, Available: https://doi.org/10.1016/j.forsciint.2019.109943.

[44] W. M. Khalilia, M. Gombár, Z. Palková, M. Palko, M. Harničárová, and J. Valiček, "Using virtual reality as support to the learning process of forensic scenarios," *IEEE Access*, 2022, Available: https://ieeexplore.ieee.org/document/9849677.

[45] E. Raymer, Á. MacDermott, and A. Akinbi, "Virtual reality forensics: Forensic analysis of meta quest 2," *Forensic Science International: Digital Investigation*, vol. 47, 2023, Available: https://www.sciencedirect.com/science/article/pii/S2666281723001774.

[46] S. Koller, L. C. Ebert, R. M. Martinez, and T. Sieberth, "Using virtual reality for forensic examinations of injuries," *Journal of Forensic Sciences*, 2018, Available: https://www.sciencedirect.com/science/article/abs/pii/S0379073818308831.

[47] T. Sieberth, A. Dobay, R. Affolter, and L. Ebert, "A toolbox for the rapid prototyping of crime scene reconstructions in virtual reality," *Journal of Forensic Sciences*, 2019, Available: https://www.sciencedirect.com/science/article/abs/pii/S0379073819304189.

[48] R. Mayne and H. Green, "Virtual reality for teaching and learning in crime scene investigation," *Science & Justice*, vol. 60, no. 5, pp. 466–472, 2020, Available: https://www.sciencedirect.com/science/article/abs/pii/S1355030620300927.

[49] M. Albeedan, H. Kolivand, and R. Hammady, "Designing and evaluation of a mixed reality system for crime scene investigation training: A hybrid approach," *Virtual Reality*, vol. 28, no. 165, 2024, Available: https://link.springer.com/article/10.1007/s10055-024-01018-8.

[50] Staples VR, *Forensic virtual reality training simulator*, Available: https://www.phfscience.nz/news-publications/forensic-virtual-reality-training-simulator/. Accessed: 10 Dec. 2025, 2025.

[51] Vizitech USA, *Vr forensics*, Available: https://ttaweb.com/product/vr-forensics/. Accessed: 12 Dec. 2025, 2025.

[52] VictoryXR, *Csi & forensics (vxrlabs)*, Available: https://www.victoryxr.com/csi-forensics/. Accessed: 15 Dec. 2025, 2025.

[53] Meta, *Home detective*, Available: https://www.meta.com/experiences/home-detective/6123516891014614/. Accessed: 21 Dec. 2025, 2025.

[54] Arab American University, *Virtual crime scene simulation for forensic biology students*, Available: https://www.al-ayyam.ps/ar/Article/382141. Accessed: 8 Dec. 2025, 2025.

[55] Road to VR, *How to tell if your pc is virtual reality ready*, Available: https://www.roadtovr.com/how-to-tell-pc-virtual-reality-vr-oculus-rift-htc-vive-steam-vr-compatibility-tool/. Accessed: 25 Jan. 2026, 2024.

Appendix A

Appendices

A.1 Use Case Specifications

This appendix provides detailed specifications for the main use cases in the system.

A.1.1 UC-1: Select Mode and Scene

Table A.1: UC-1: Select Mode and Scene

Use Case Name	Select Mode and Scene
Actor	Student
Description	This use case enables the students to select the mode and scene they want to investigate. The selection depends on whether they choose the Base Observational mode (allows students to explore and view the scene without interacting with the evidence) or the Advanced Interactive mode (enables students to interact with evidence within the scene and collect it using forensic tools).
Preconditions	• The student must have logged in successfully using their credentials • The scenes must be available and ready in the system for selection • The mode should be available for selection

Event Flow	1. The application displays the mode selection menu with available options: • Base Observational: No interaction with evidence • Advanced Interactive: Ability to interact with evidence 2. The student selects the desired mode 3. The application displays available scenes for selection 4. The student selects the scene 5. The application sends a request to Firebase to load the selected scene with the chosen mode configuration 6. If the mode is Base Observational: The scene is displayed for observation only (either 3D or 360-degree recorded scene) 7. If the mode is Advanced Interactive: The scene is loaded with forensic tools enabled to allow the student to interact with and collect evidence
Alternative Flows	**A1: Login Failure** • If the login data is incorrect: The application displays an error message **A2: Mode or Scene Not Selected** • If the mode or scene is not selected, a warning message is displayed prompting the student to make the required selections
Special Requirements	• The application must be connected to the Firebase for verifying login data • The scene should be loaded quickly and without issues (either 3D environments or 360-degree recorded scenes)
Postconditions	• The student has successfully selected the mode and scene • The scene is loaded successfully based on the student's selections • If the Base Observational Mode is chosen: the scene is displayed for observation only • If the Advanced Interactive Mode is chosen: the tools for interacting with the evidence are enabled

A.1.2 UC-2: Navigate Crime Scene

Table A.2: UC-2: Navigate Crime Scene

Use Case Name	Navigate Crime Scene
Actor	Student
Description	Student explores the crime scene using teleportation-based movement and natural head motion for 360° viewing.
Preconditions	• Student has selected mode and scene • Crime scene is loaded • VR headset and controllers are tracking properly
Event Flow	1. System displays crime scene environment 2. Student uses head movements to look around (360° viewing) 3. Student points controller at desired location 4. Student activates teleportation mechanism 5. System validates target location (accessible, within scene bounds) 6. System transitions student to new location 7. Student can crouch physically or use controller input to lower viewpoint 8. System updates position tracking 9. Steps 2-7 repeat as student explores the scene
Alternative Flows	**A1: Invalid teleportation target** • At step 5, if location is invalid (outside bounds, obstructed), system provides visual feedback (red indicator) • System prevents teleportation • Use case resumes at step 3 **A2: Collision detection** • During step 6, if student would collide with scene objects, system adjusts final position to prevent overlap • Use case continues normally
Special Requirements	• Maintain 90 FPS frame rate during movement • Minimize motion sickness through smooth transitions • Provide clear visual indicators for valid/invalid teleportation targets
Postconditions	• Student has explored the crime scene • Student position and viewing history are tracked

A.1.3 UC-4: Observe Evidence

Table A.3: UC-4: Observe Evidence

Use Case Name	Observe Evidence
Actor	Student
Description	Student visually observes and identifies potential evidence items without physical interaction. This initial observation follows proper forensic protocol of not disturbing evidence before it is properly documented through marking and photography.
Preconditions	• Student is in Advanced Interactive Mode • Student has navigated to an area containing evidence
Event Flow	1. Student approaches a potential evidence item in the scene 2. System highlights the evidence item when controller is within interaction range 3. Student observes the evidence visually without touching 4. Student uses head movements and position changes to view evidence from multiple angles 5. Student identifies characteristics: color, size, texture, markings, position, context 6. Student mentally notes observations 7. Student decides whether evidence requires documentation and collection 8. If evidence requires collection, student proceeds to marker placement (UC-5)
Alternative Flows	**A1: Evidence Not Relevant** 1. At step 7, student determines item is not evidence 2. Student continues navigation to search for other potential evidence 3. Use case ends
Special Requirements	• System must prevent grabbing/touching until evidence is properly documented • Visual highlighting to indicate evidence can be observed • Clear feedback that physical interaction is not yet permitted

Postconditions	• Evidence item remains in original position undisturbed • Observation action is logged with timestamp • Student can proceed to marker placement for documented evidence

A.1.4 UC-4a: Inspect Evidence Physically

Table A.4: UC-4a: Inspect Evidence Physically

Use Case Name	Inspect Evidence Physically
Actor	Student
Description	Student picks up, rotates, and closely examines evidence items using VR controllers after proper documentation has been completed. This physical inspection occurs only after evidence has been marked and photographed following forensic protocols.
Preconditions	• Student is in Advanced Interactive Mode • Evidence has been observed (UC-4) • Evidence marker has been placed (UC-5) • Three-stage photographic documentation has been completed (UC-6) • Optional specialized examinations (UC-7, UC-8) completed if applicable

Event Flow	1. Student approaches the documented evidence item 2. System confirms evidence has been properly documented 3. System enables physical interaction with the evidence 4. Student presses the grip button to grab the evidence 5. System attaches the evidence to the virtual hand 6. Student brings the evidence closer for detailed physical inspection 7. Student rotates the evidence using hand movements to view all angles 8. System renders high-detail textures and surfaces as evidence is examined 9. Student observes additional characteristics through physical manipulation 10. Student releases grip button to place evidence back 11. System returns evidence to its original position with proper orientation
Alternative Flows	**A1: Two-Handed Inspection** 1. At step 5, student grabs evidence with both controllers 2. System enables two-handed manipulation for larger items 3. Student can scale view or rotate more precisely 4. Flow continues from step 6 **A2: Evidence Not Properly Documented** • At step 2, if evidence has not been marked or photographed • System displays message indicating documentation is incomplete • System prevents physical interaction • Student must complete UC-5 and UC-6 first • Use case resumes from step 1 after documentation is complete
Special Requirements	• Haptic feedback when evidence is grabbed • Smooth rotation and scaling without jitter • System must enforce proper documentation before allowing physical interaction

Postconditions	• Evidence item returns to its original position if not collected • Physical inspection action is logged with timestamp and duration • Evidence is ready for collection (UC-9)

A.1.5 UC-10: Tag Evidence

Table A.5: UC-10: Tag Evidence

Use Case Name	Tag Evidence
Actor	Student
Description	Student attaches comprehensive digital evidence tags to collected items immediately upon collection, including unique identifiers, descriptions, location coordinates, timestamps, and chain of custody information. This use case implements professional evidence tagging and documentation procedures following forensic standards.
Preconditions	• Student is in Advanced Interactive Mode • Student has successfully logged in to the application • Evidence has been observed (UC-4) • Evidence marker has been placed adjacent to the evidence (UC-5) • Three-stage photographic documentation has been completed (UC-6) • Evidence has been physically inspected if needed (UC-4a) - optional • Evidence is being collected (UC-9) - tagging occurs during or immediately after collection

Event Flow	1. Student selects the collected evidence item from the virtual evidence container 2. System displays the digital evidence tagging interface 3. Student enters a unique evidence identifier 4. Student provides a detailed description of the evidence item 5. System automatically captures the exact location coordinates where the evidence was found 6. System automatically records the collection timestamp 7. System automatically logs the student's identification for chain of custody tracking 8. Student reviews the evidence tag information for completeness 9. Student confirms and submits the evidence tag 10. System validates that all required fields are completed 11. System attaches the comprehensive digital tag to the collected evidence 12. System provides visual confirmation that tagging is complete
Alternative Flows	**A1: Incomplete Tag Information** • At step 10, if required fields are missing or incomplete: System displays an error message indicating which fields need to be completed • Use case resumes at step 3 **A2: Incorrect Procedural Sequence** • If student attempts to tag evidence before placing marker or completing photography: System provides procedural feedback indicating the correct sequence • System prevents tagging until prerequisites are met • Use case resumes when proper sequence is followed **A3: Tag Modification** • Student selects a previously tagged evidence item • System displays existing tag information • Student can modify description or other editable fields • System logs modification with timestamp and maintains chain of custody • Flow continues from step 9

Special Requirements	• System must ensure all evidence tags include unique identifiers to prevent duplication • Location coordinates must be captured automatically with precision • Chain of custody tracking must be immutable and automatically maintained • System must validate that marker placement and photographic documentation are completed before allowing tagging • Visual and haptic feedback must be provided for procedural errors
Postconditions	• Evidence item has a complete digital tag attached with all required forensic documentation • Evidence tag is stored in the system database linked to the investigation session • Chain of custody record is established and tracked • Tagging action is logged for performance evaluation including correctness and procedural compliance • Evidence is ready for final submission as part of the investigation

A.1.6 UC-13: View Performance Evaluation

Table A.6: UC-13: View Performance Evaluation

Use Case Name	View Performance Evaluation
Actor	Student
Description	Student views a post-investigation performance summary displayed in VR, including score, evidence identification accuracy, tagging correctness, procedural errors, and detailed evaluation metrics.
Preconditions	• Student has successfully logged in to the application • Student has completed an investigation session • Performance report has been generated by the system

Event Flow	1. System displays performance evaluation panel in VR interface 2. Student reviews overall score and performance summary including: • Overall investigation score • Evidence identification accuracy (found vs. missed evidence) • Evidence tagging correctness and completeness • Procedural errors and compliance violations • Time efficiency metrics • Detailed breakdown of actions taken 3. Student navigates through detailed evaluation sections 4. Student views specific errors and missed evidence with explanations 5. Student reviews correct procedures and best practices 6. Student compares performance to previous investigations (if available) 7. Student closes the evaluation panel when finished reviewing
Alternative Flows	**A1: Incomplete Investigation** • If investigation was not properly completed, system displays partial results with notification • System indicates which sections were incomplete • Flow continues from step 4
Special Requirements	• Performance metrics must be calculated accurately based on predefined scoring rubric • Evaluation panel must be clearly readable in VR environment • System must provide constructive feedback highlighting both strengths and areas for improvement • Report data must be persisted for future reference and instructor access
Postconditions	• Student has viewed complete performance evaluation • Performance report is stored in student's investigation history • Report is available for instructor review • Student has gained feedback on investigative performance and procedural accuracy

A.1.7 UC-14: Review Student Performance

Table A.7: UC-14: Review Student Performance

Use Case Name	Review Student Performance
Actor	Instructor
Description	Instructor accesses and reviews detailed performance reports of students' investigation sessions, including scores, evidence handling, procedural compliance, and investigation completion metrics. This enables instructors to assess student learning outcomes, provide targeted feedback, and track progress over time.
Preconditions	• Instructor has successfully logged in to the system with instructor credentials • One or more students have completed investigation sessions • Performance reports have been generated and stored in the system
Event Flow	1. System displays instructor dashboard with list of students 2. Instructor selects a student from the list 3. System displays the selected student's investigation history with available sessions 4. Instructor selects a specific investigation session to review 5. System retrieves and displays comprehensive performance report including: • Overall investigation score and grade • Evidence identification metrics (found, missed, incorrectly identified) • Evidence tagging accuracy and completeness scores • Procedural compliance and errors committed • Time spent on investigation • Detailed timeline of actions taken 6. Instructor reviews performance metrics and detailed breakdowns 7. Instructor examines specific errors or areas of concern 8. Instructor can view comparison with class average or learning objectives 9. Instructor closes the performance report when finished reviewing

Alternative Flows	**A1: No Completed Sessions Available** • At step 3, if selected student has no completed investigation sessions • System displays message indicating no data available for review • Instructor returns to student list to select another student • Flow resumes from step 2 **A2: Report Loading Failure** • At step 5, if system fails to retrieve performance report • System displays error message • Instructor can retry loading the report • Flow resumes from step 4
Special Requirements	• System must protect student data privacy and comply with educational data regulations • Performance metrics must be calculated consistently using the same rubric for all students • Interface must allow easy navigation between multiple student reports
Postconditions	• Instructor has reviewed student performance report • Instructor has gained insight into student learning progress and areas needing improvement